SUBURBAN RAIL LOOP

OFFICIAL

Systems Engineering Management Plan

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SUBURBAN RAIL LOOP AUTHORITY

VICTORIA'S BIG BUILD

VICTORIA State Government

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1 Introduction

Increasing complexity in large-scale projects is fuelled by the need to build and integrate sophisticated and novel products, that deliver enhanced performance, and which must have the flexibility to adapt to users' future needs.

Suburban Rail Loop Authority (SRLA) must have the multidisciplinary expertise needed to apply Systems Engineering (SE) effectively throughout the implementation of infrastructure and technology projects.

From inception to operation, the Systems Engineering procedures being put in place by the Engineering Assurance team will enable SRLA to effectively make decisions both explicitly and justifiably and ensure the envisaged project outcomes are achieved by the final delivered product.

For an organisation to gain the most benefit from the Engineering Assurance team, the activities and principles laid out in this plan must be actively embraced and embedded across the Engineering division of SRLA.

1.1 Purpose

As specified in the Network Integrity Assurance Plan (NIAP) [2] SRLA must produce a 'Systems Engineering and Assurance Management Plan' to present to the Department of Transport Planning (DTP) at each N3 review or as deemed necessary under the guidance of SRLA and DTP.

SRLA has chosen to develop a Systems Engineering Management Plan (SEMP) and a Systems Assurance Management Plan (SAMP) to meet the above requirement and therefore should be read in conjunction with the SRLA Systems Assurance Management Plan to give a complete view of all Systems Engineering & Systems Assurance disciplines, activities and stratagem.

This SEMP defines and documents the top-level Systems Engineering activities to be applied to the Suburban Rail Loop (SRL) Project to assure the successful planning, phasing, migration, integration, and acceptance of the SRL, and its handover to operations and maintenance.

This SEMP is an integral part of documenting an integrated approach to engineering activities, tailoring a typical lifecycle model to the project characteristics and identifying the methods that will be used.

The SEMP will provide:

- An overview/summary of the progressive SE strategy for the SRL project
- An overview/summary of the systems integration strategy for the SRL project, the development of a SRL system architecture to support capture & dissemination of consistent & quality information, and how SE processes relate and support this approach
- A lifecycle model to translate the SE approach contained in ISO/IEC/IEEE 15288:2015 [3] into one suitable for the SRL project, supporting assurance against the Department of Treasure & Finance (DTF) Investment Lifecycle [1] and tying into DTP's assurance reviews as specified in the Network Integrity Assurance Plan (NIAP [2])
- The organisational structure of the SRLA Engineering Assurance Team, their roles and responsibilities and their functional relationships with the rest of the SRLA organisation
- Definitions of the SE processes & activities that SRLA shall undertake during each phase of the lifecycle, including the key outputs to be expected. These processes & activities will change between procurement and delivery phases; therefore this plan must clearly differentiate this change in roles
- Give indication as to what SE processes & activities are expected to be undertaken by Contractors during delivery and their expected outputs.

This SEMP is not to be used for:

- Specifying contractual requirements of Contractors, this shall be conveyed in Project Scope and Technical Requirement (PS&TR) part C requirements.
- Low level detail on how specific processes work, the SEMP must instead draw together top-level activities and leave the detail to be provided in more specific lower level management plans, procedures and guidelines
- Discussion with external stakeholders except for DTP or Office of the National Rail Safety Regulator (ONRSR)
- Work Package contractors to implement/show compliance to. Each Work Package contractor is expected to produce their own suite of management plans, as specified by the PS&TR Part C requirements.

1.2 Audience

This SEMP is intended:

- To be **DEVELOPED** by the Manager, Engineering Assurance of SRLA
- For its activities to be **REVIEWED** & **IMPLEMENTED** by the Engineering Assurance and Engineering Design Teams, Digital Engineering / Asset Information Team, Technical & Commercial Interface Team of SRLA
- To be **ENDORSED** by the Executive General Manager – Rail & Infrastructure Delivery
- To be **APPROVED** for use by the Deputy Director – Engineering & Assurance of SRLA
- To be **READ** by anyone in SRLA to understand the SE approach being applied at SRLA.

1.3 Scope

This revision of the SEMP is applicable to activities conducted to produce a complete set of PS&TRs for all SRL East work packages.

This management plan is written with specific regard to SRL East, however any future stages of SRL should follow the guidance & process outlined in this management plan.

The SEMP is intended to be used as an assurance artefact for DTP's N-Gate assurance reviews to evidence how SRLA is undertaking a systematic approach to developing the SRL East solution.

The SEMP is a live document and will be subject to periodic updates, typically triggered by progressing through different phases of the program lifecycle, but may also be updated due to any significant project event.

This management plan is to be used by SRLA only to organise and manage the SE application to its works. Contractors and Systems Integrator roles are described in this document for the purpose of planning what contractors and a system integrator should typically do and therefore inform what minimum expectations SRLA has of these parties to progressively assure and submit assurance evidence to comply with the NIAP (as per SRLA's assurance responsibilities).

This SEMP has been developed as instructed by the Rail & Infrastructure Delivery Management Plan.

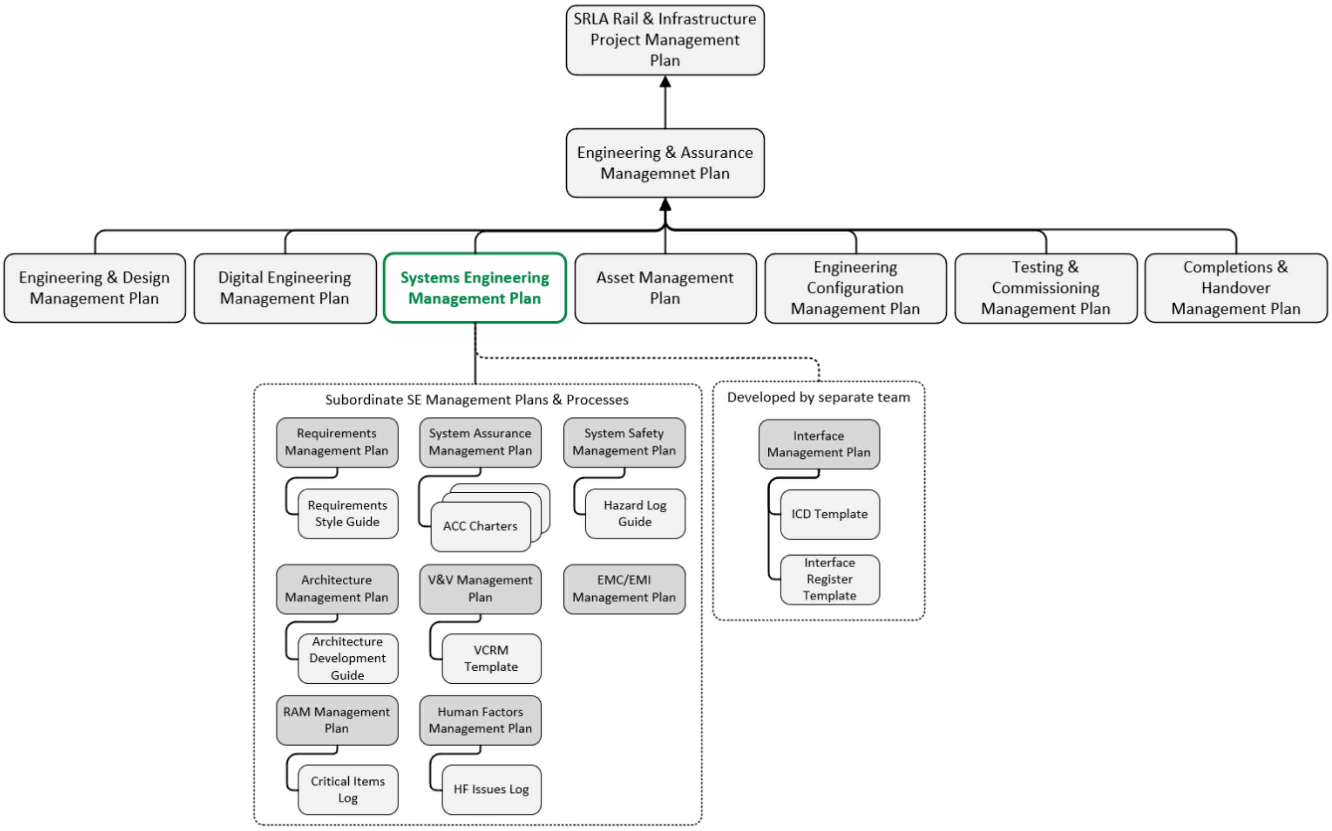


Figure 1 - Management Plan Hierarchy

The SEMP prescribes the development of many subordinate plans to detail how each SE sub-discipline will be conducted. Details on the minimum contents of these management plans can be found in Appendix F, with references to these documents being made throughout this management plan. The sub plans are:

- Requirements Management
- System Architecture
- Interface Management
- System Safety
- RAM
- EMC/EMI
- Human Factors.

2 Engineering Assurance Objectives

The primary objective of the Engineering Assurance Team at SRLA is to provide assurance evidence to SRLA's executive that the works being conducted will meet the directives laid out in the SRL East Client Requirements Documents (CRD) and is safe So Far as Reasonably Practicable (SFAIRP).

DTP's NIAP instructs SRLA to progressively assure there will be no discernible impact to the integrity of DTP's extant Public Transport Network. SRLA must participate in strategic "N Gate" assurance reviews held by DTP where various evidence artefacts have been identified for production. These artefacts include a Systems Engineering & Assurance Management Plan (this plan), a System Requirements Specification (SRS), a system architecture/definition, Requirements Traceability Matrices (RTMs) and Verification Cross Reference Matrices (VCRMs).

To produce and supply these artefacts effectively; a Systems Engineering approach is applied to the definition and management of technical, operational and program integration issues to assure DTP that SRL's benefits will be realised by the end state system.

The SEMP is developed with the following 3 view points in mind:

- Operator, Maintainer & User Integration – ensuring the SRL East system is developed with its end users in mind and that the end user is ready to receive/interact with the system
- Program Delivery Assurance – ensuring that SRL East system is being progressively developed and delivered in line with program scope and constraints
- Systems Integration – ensuring that the SRL East system can be and is being integrated to successfully deliver a reliably performing system.

The above view points will be a central tenement to the SEMP and will provide the structure upon which the team organisation, the planning of activities and the governance arrangements are built upon.

The figure below shows how the viewpoints are covered by activities that are described further in the SRLA Systems Engineering Guideline [Ref. 20].

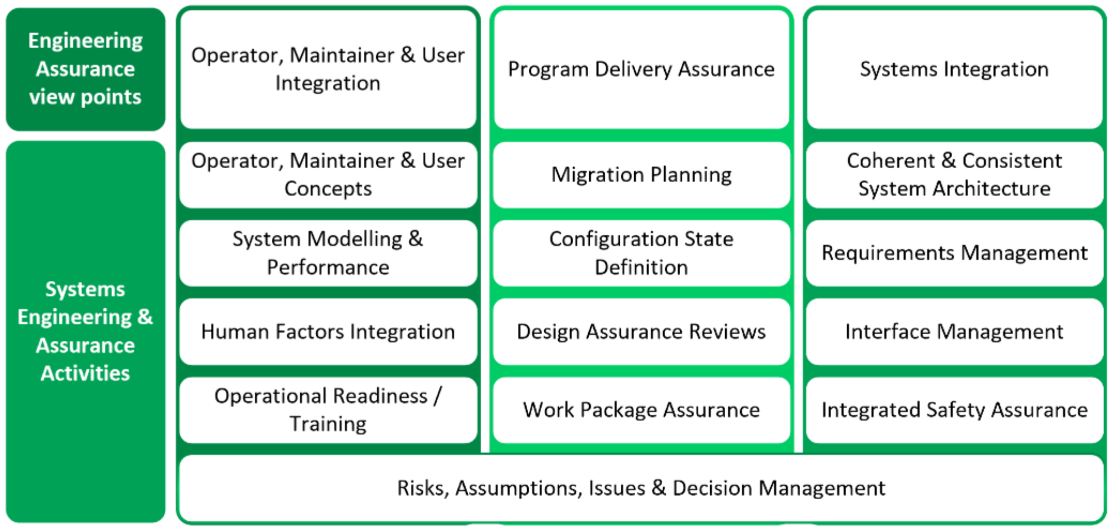


Figure 2 Engineering Assurance view points and activities

SRLA will conduct the following activities for each defined phase of the project lifecycle.

Pre-Procurement Phase (of Linewide Package)

- Be responsible for producing a systematic process for developing a technical railway system solution that meets the Client Requirements Document. The process must coordinate technical expertise in the SRLA Engineering & Assurance division to produce a cohesive and consistent SRL East system solution
- Create an assurance team independent from the work packages, allowing for impartial review of assurance evidence against client requirements
- Produce and maintain a suite of management plans to further define how Requirements Management, System Architecture, Reliability Availability Maintainability (RAM), Human Factors, System Safety, & Systems Assurance disciplines will support the overall systems engineering approach described in this plan
- Establish a team that has appropriate breadth in Systems Engineering & Assurance capability to be able to drive the systems engineering processes necessary to produce the SRL East System solution
- Establish clear roles and responsibilities for all Engineering Assurance Team positions
- Establish mechanisms for the escalation of Engineering Assurance issues to SRLA executive leadership for resolution
- Create a Transition/Migration Plan that will record the critical dependencies between workstreams and establish a set of justifiable configuration points that will be endorsed by the governance forum and be subject to stringent change control.
- Identify any technical risks, issues or assumptions and raise them with the Risk Management team at SRLA
- Work collaboratively with any performance/modelling teams and liaise with both DTP and Digital Engineering teams in the production of clear modelling task remits, control of the baseline data and underlying assumptions.

Post-Procurement Phase (of Linewide Package)

- Provide embedded resources for the interim Systems Integrator in undertaking its duties where necessary
- Support the escalation of Engineering Assurance issues between the work packages up to SRLA governance forums
- Provide guidance on how system engineering tasks (as suggested by this plan) could be implemented by the Work Package teams
- Maintain the SRL East System solution in line with any approved changes
- Maintain & monitor how work packages are keeping to critical dependencies identified in the Augmentation/Migration Plan
- Orchestrate SRLA Systems Assurance reviews (as defined in Table 1) with equal participation from the System Integrator to support the identification and mitigation of risks to SRL East , raise actions for SRLA and the Work Package teams to enact and capture the outcomes in project-wide assurance reports for DTP.

3 SRL East System Overview

The SRL program is planned to be developed and delivered by an integrated set of projects. This overview section intends to give clarity as to when the principles and activities presented in this SEMP must be adhered to.

3.1 System Context & Interfaces

As described in the BRD and subsequent SRLA Program Response; the SRL system can be split geographically into three major transport corridors:

- SRL East – Cheltenham to Box Hill
- SRL North – Box Hill to Melbourne Airport
- SRL West – Melbourne Airport to Werribee.

This revision of the SEMP is applicable to works on SRL East, however can be adapted to apply to SRL North & West should that be necessary.

Furthermore; from a System of Systems perspective, SRL East can be considered as the SRL East System and an SRL East Enabling System:

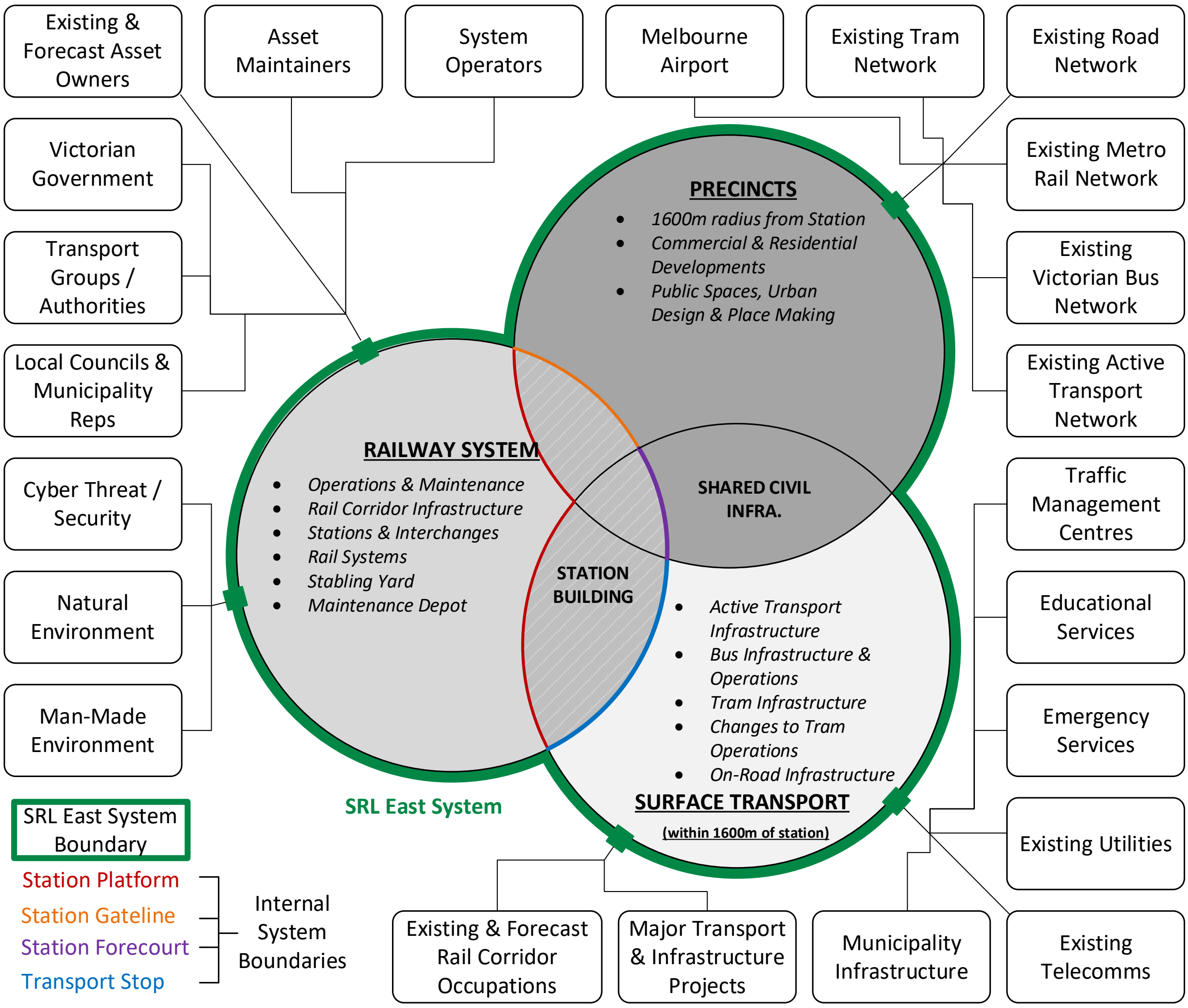


Figure 3 - SRL East Context

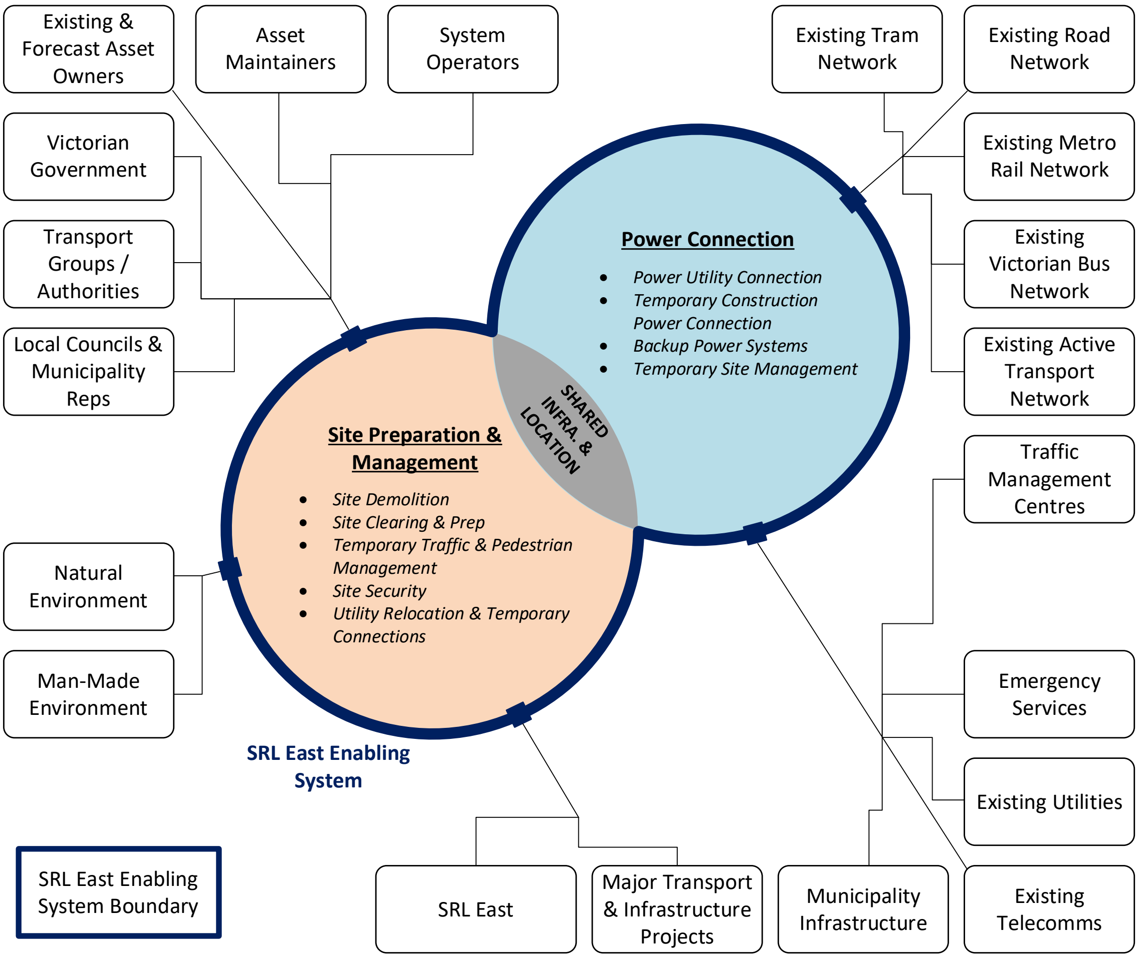


Figure 4 - SRL East Enabling System Context

The Enabling System will support the final implementation of SRL East, as depicted below.

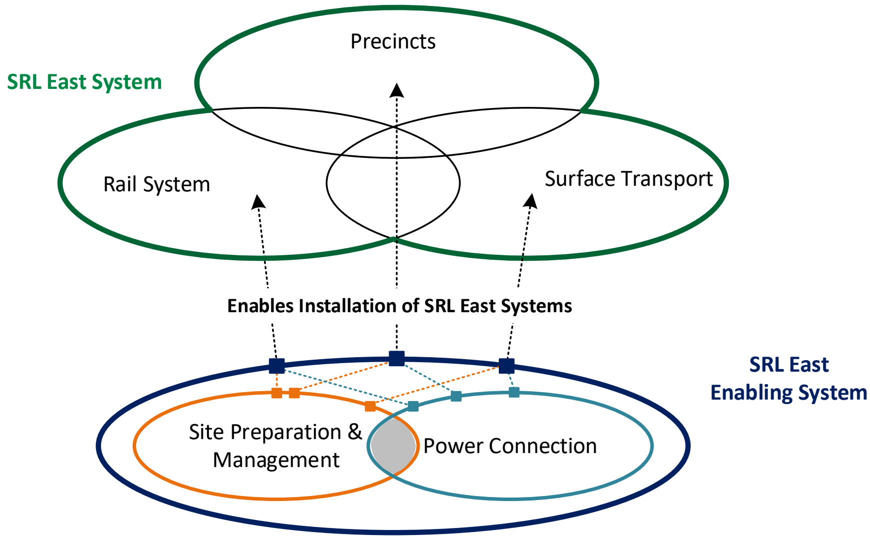


Figure 5 - Relationship between SRL East & Enabling System

This plan is to be used to schedule activities to specify all systems identified (including their interfaces) and support the procurement of those systems through contractual documentation.

The context diagrams provided above are for information only. The definitive definitions of these systems are detailed within the SRL East System Definition Document and Initial and Early Works (IEW) System Definition Document.

With regards to stakeholders, systems, projects and their interfaces, the SEMP and its subordinate plans are written with the following context/relationships in mind:

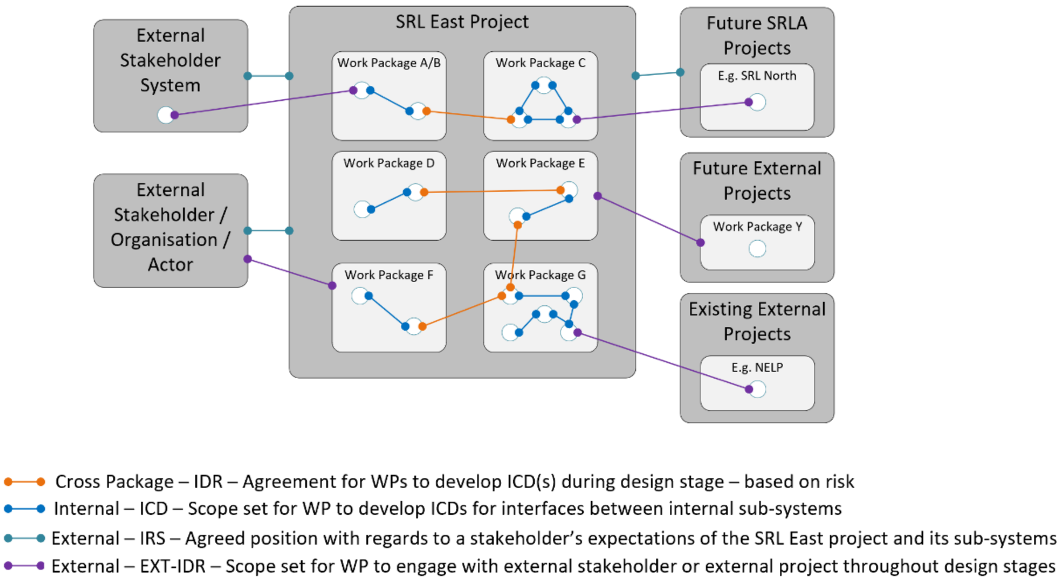


Figure 6 – Interface Types

External Interfaces: Can be both behavioural/technical and/or contractual in nature. They exist between anything within the SRL East boundary and any external project, sub-system or stakeholder.

Cross Package Interfaces: are both behavioural/technical and/or contractual in nature. They exist between two work packages within the SRL East boundary. This can be between sub-systems, or between work package scopes of work to encourage collaboration and an integrated design.

Internal Interfaces: are behavioural/technical in nature. They exist between two sub-systems within a work package.

3.2 SRL Systems Engineering Framework

The SE approach described by this document revolves around utilising a digital model of the 'SRL System Architecture' as the entity that pulls all the engineering information together and acts as the program's single source of truth with regards to:

- The expected usage of the system (as expressed through the eyes of its operator, maintainer & expected user groups)
- How the system needs to function to meet the needs of its users and stakeholders
- What level of performance and other desired characteristics the system must possess
- How the system is composed of multiple interacting physical components, and that they must work together to achieve the overall system functionality and performance
- How the system interfaces with its environment and existing man-made systems
- How the system is expected to be gradually migrated from construction into operational use in concurrent logical stages.

As with all digital models, the model is a tool enabling complex programs to handle and configure data, therefore the success of this approach relies on all SE activities yielding appropriate information that attributes to building the SRL System Architecture. Therefore, the SE activities described in this document must focus on undertaking the necessary activities in line with industry standard and best practice, but more importantly must produce 'good' and 'consistent' information to feed the digital model. Likewise, the digital model must be developed to accept the different types of information likely to be produced as an output of SE activities.

To support best practice and efficient use of the tool, SRLA has chosen to apply the fundamentals of the Arcadia Method [Ref. 17] and tailor the methods for the specific context of SRL East.

The methodology is described in this document throughout the lifecycle stages prior to procurement, with contractors then choosing to either adopt the ARCADIA method for their works, or a similar framework that must interface with SRLA's methodology.

Figure 5 shows how the SRL System Architecture is central to capturing and reproducing data/information for the SRL East program.

Figure 6 shows that when applied, the SE Framework yields a hierarchical set of technical requirements to link top level needs to contracts, via multiple levels of refinement.

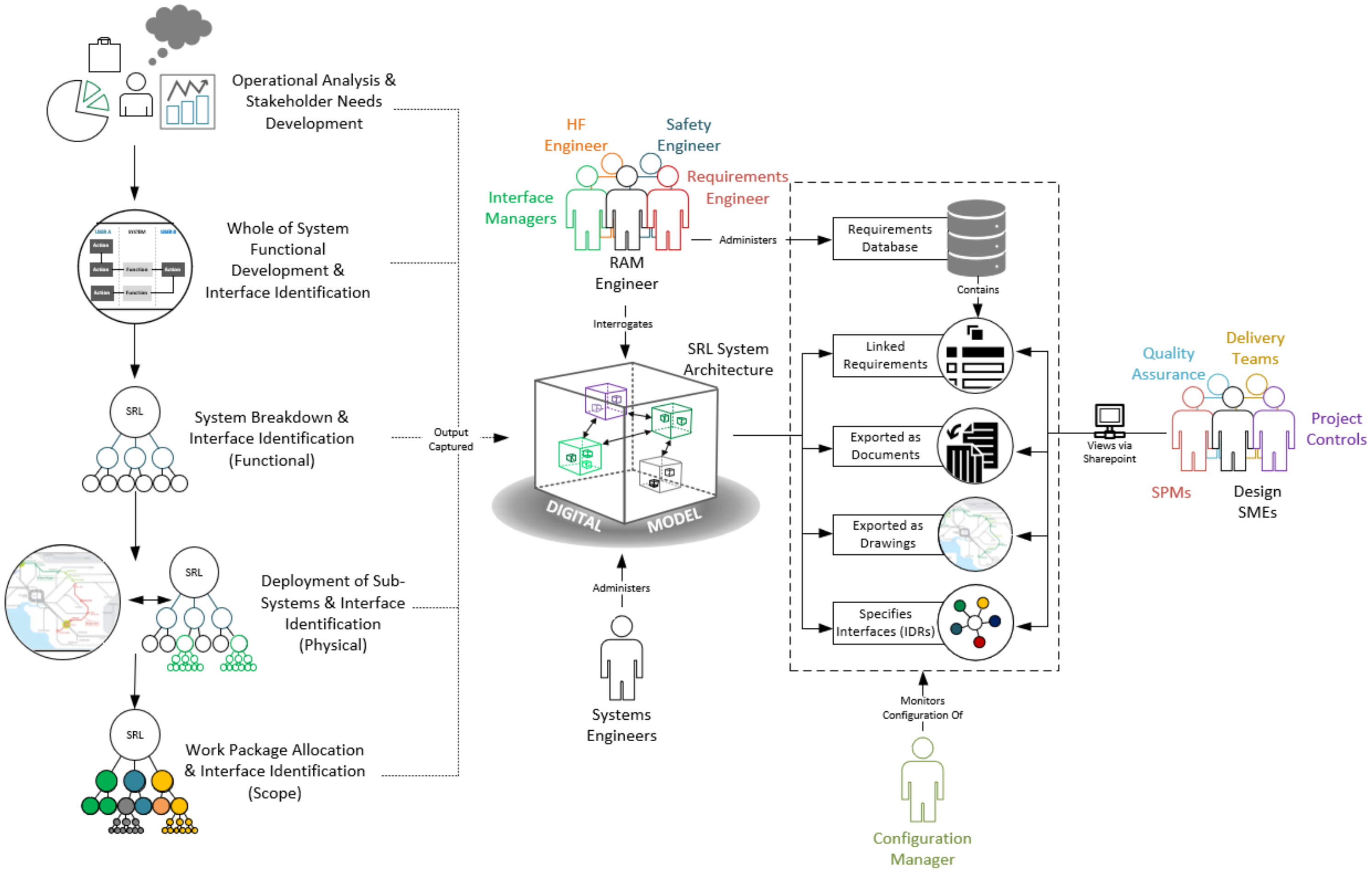


Figure 7 – Architecture Centric Approach

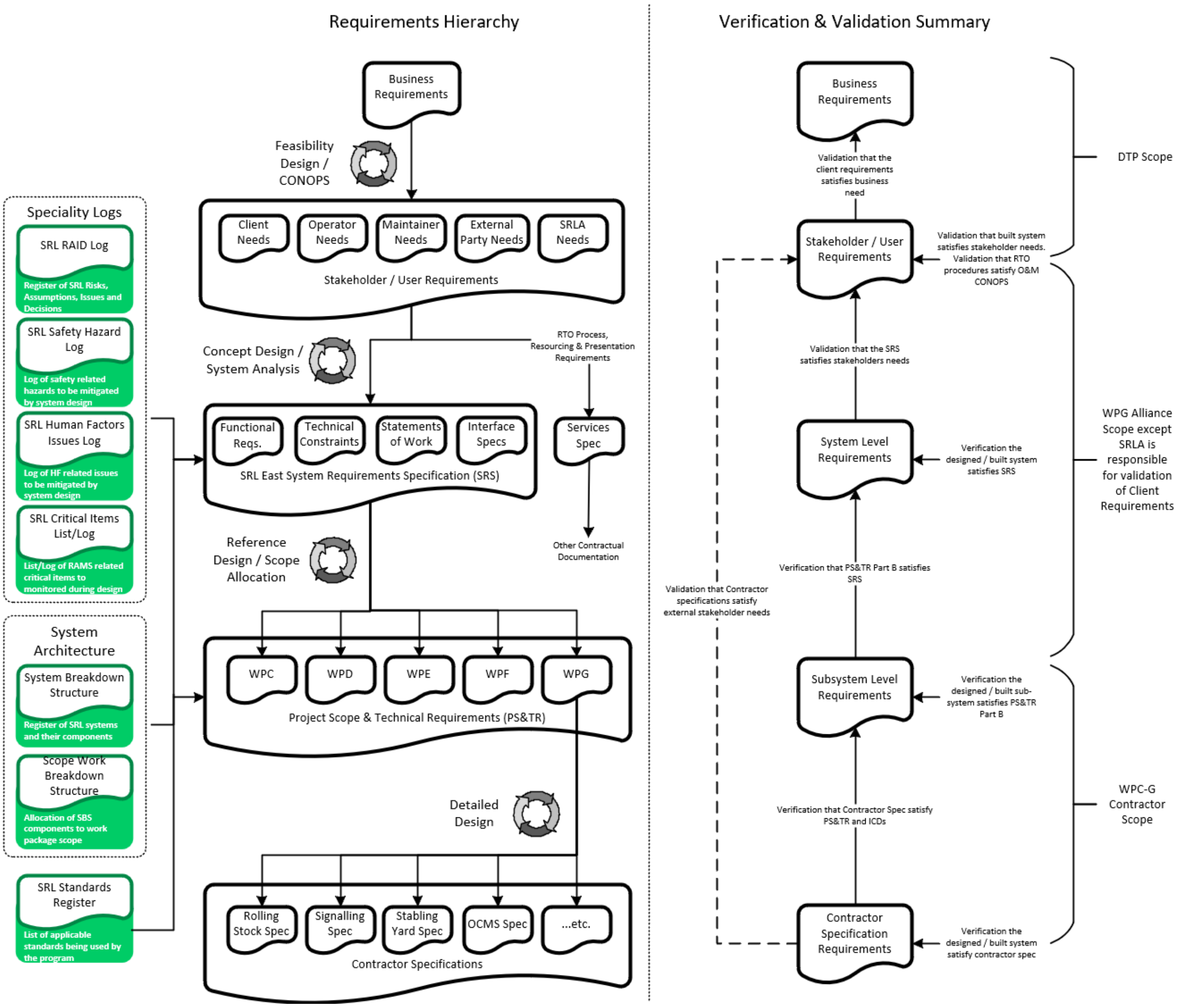


Figure 8 - Requirements Output from applying SE Framework

4 Engineering Assurance & the SRLA Program Lifecycle

It is critical that any works undertaken lead towards producing the necessary products required to pass the DTF Gate Reviews [1], DTP Network Integrity Assurance Review [2] milestones, and the work packages' Assurance & Configuration Committee (ACC) Stage Gate Reviews. To ensure all SE works actively contribute to the smooth passage through each DTF gateway review, this plan explains how the traditional SE lifecycle fits with the DTF Investment Lifecycle and its associated gateway reviews.

4.1 General SE Lifecycle for SRLA Projects

The SEMP is based on tailoring the SE lifecycle and the ISO/IEC/IEEE 15288-2015 [3] system lifecycle processes to meet the development & design needs of the SRLA Project.

The general system V-Lifecycle is shown below in Figure 9 (and defined in the following table) to illustrate the major phases of the SRL East program that this management plan will refer to when defining when particular activities must be applied by SRLA and its Contractors.

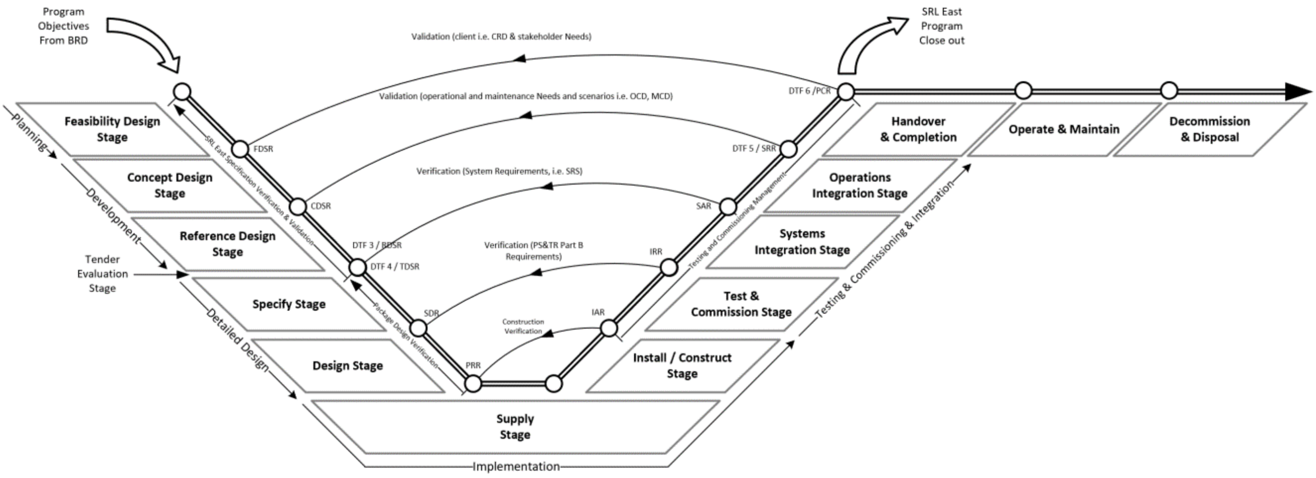


Figure 9 - SRL East System V-Lifecycle

Table 1 - Lifecycle stage definitions

Stage Title	Definition	Applicable WP
Feasibility Design	The initial stage of the project where the project benefits are analysed and defined. Program specific strategies and frameworks are developed, a high level cost estimate is produced based on key decisions being made along with a high level feasibility design being produced. A key deliverable of this stage is a concept of operations (CONOPS) to define the performance objectives of the system and its planned usage/operations/maintenance.	SRLA Only
Concept Design	Work is undertaken to develop a more detailed SRL East System design building of anything defined during the Feasibility Design stage. The Concept of Operations will be developed into a more specific Operations & Maintenance Concept to support the design of the system. The concept design will support detailed decisions to be made, will begin to identify more	SRLA Only

Stage Title	Definition	Applicable WP
	detailed interfaces, and must be supported by simulations/modelling to ensure the emerging solution is fit for purpose.	
Reference Design	Work is undertaken to turn the concept design into a reference design which must prove how the SRL East system requirements along with other decisions and constraints could be met. The reference design should be of the maturity to support de-risking the SRL East System solution with regards to interface issues (clashes), space proofing, constructability issues, and enable a more accurate cost estimate.	SRLA Only
1. Specify	The Specify Stage encapsulates all delivery phase activities prior to the commencement of design, including, but not limited to: • Identification of PS&TRs relevant for Design Packages. • Identification of IDRs relevant for Design Packages • Sourcing and collation of existing condition data (geotechnical, survey, dilapidation, as-in service technical/operation manuals, etc.) • Preparation of Specialist Reports and other equivalent sources of design input specifications, and derivation of associated technical/interface requirements.	All
2. Design	The Design Stage encapsulates the preparation, review, acceptance and certification of all design documentation necessary to detail the Works in accordance with outcomes of the Specification stage. The Design Stage includes all sub-stages at which design documentation is produced and updated, including the Kick-off Workshop stage, Preliminary Design stage, Detailed Design stage, Final Design stage, Issued for Construction (IFC) stage, and any post-IFC revision to IFC designs, including: • Options assessment and single option selection • Iterative identification of derived requirements and deviations/Departures • Production of Design Documentation comprised of, but not limited to design reports, design drawings, design models, O&M manuals, design registers as required • Verification & certification of design against requirements.	All
3. Supply	The Supply Stage covers the procurement of components or assets from manufacturers/suppliers, whether Commercial Off The Shelf (COTS) or bespoke assets procured in accordance with the design. The Manufacturer/supplier manufactures the component/assets/product or provides a service, tests their assets/products in their factory settings, and prepares/assembles/fabricates the assets/product for delivery to the site for installation. Works for this stage are typically performed off-site (not at SRL locations).	All
4. Install / Construct	The Install/Construct Stage covers the necessary works to EITHER receive manufactured/supplied assets and products and installing them on-site/in-situ OR construct infrastructure supporting other assets/products to be installed, all of which is performed in line with the construction schedules and installation schedules developed by owners of the Works. Works for this	All

Stage Title	Definition	Applicable WP
	stage are located on-site (SRL locations). Notwithstanding this, a work package may assemble prefabricated components on-site	
5. Test & Commissioning	The Test & Commission Stage covers the necessary activities required for standalone tests to confirm that the asset/product (that is within scope to deliver) has been proved to be functioning, performing & interfacing as per the specification for that particular asset/product or groups thereof. This includes the production of all necessary completions/as-built documentation and achievement of the necessary certification in preparation for handing over the asset/product to the System Integrator. The majority of the physical verification activities (final verification against the PS&TR Part B requirements) are expected to happen in this stage.	All
6. System Integration	The System Integration Stage covers the necessary activities required to test all received and commissioned SRL East sub-systems against the System Requirements Specification to verify that all system interfaces, functions, and performance is met in preparation of achieving system acceptance by the operator. This also covers the requirement to justify that the delivered SRL East system is safe so far as is reasonably practicable (SFAIRP) and identify any residual risk for transfer for continued ownership by Victoria. The majority of the physical verification activities (final verification) against the SRS requirements are expected to happen in this stage.	WPG
7. Operations Integration	The Operations Integration Stage covers the necessary activities to test the SRL East system against a set of operator and maintenance scenarios (including testing of standard operating procedures) to ensure the SRL East system is fit for purpose and validate that it is an operable and maintainable system. This also covers the requirement to prove that the designated Rail Transport Operator is accredited, staffed & trained appropriately with all procedures and management systems put in place in order to be ready to open up SRL East to the public (for Day 1 services). The majority of the physical validation activities against the needs contained in the Stakeholder and Operational Needs Documents is expected to happen in this stage.	WPG
8. Handover & Completions	The Handover & Completion stage is the period over which the Work Package collates and provides to the Principal/Project Owner, and any applicable accredited party or Returned Asset Owner, all records and demonstration that the Works have been delivered in full. This is in accordance with the contract, requirements and Reference Documents. This is inclusive of activities up to and including all completion milestones, and requires the resolution or accepted transfer of all open issues and risks to the appropriate party for ongoing management.	All
9. Operate & Maintain	The Operate & Maintain stage covers RTO or applicable Returned Asset Owner management of the operations and maintenance of SRL East in line with the agreements laid out in the Services Specification and the RTO's Safety Management System to deliver services to passengers. This includes the sustainment of SRL East through maintenance and renewals. The Operate & Maintain stage also includes the interim operation and maintenance of infrastructure by Works Package contractors, as necessary prior to handover to other SRL East Work Package Contractors, the RTO or Returned Asset Owner.	WPG

Stage Title	Definition	Applicable WP
10. Decommission & Disposal	The Decommission & Disposal Stage covers the necessary activities to safely decommission and dispose of existing or temporary assets/products. Certain existing equipment is typically made redundant by introduction of new systems and is not required to be retained. This equipment is decommissioned and removed from service. The decommissioned equipment is suitably disposed with great consideration to safety and environmental impacts. This may involve the reuse of the equipment elsewhere (where this option is available).	WPG, however others where required

Note that past the development phase, work packages will be responsible for the detailed design, installation and testing commissioning of their own systems. The lifecycles may be slightly different and therefore will be assured in the relevant manner. further expands the work package lifecycles and indicates how they are drawn back together through the Systems Integration and Operations Integration stages.

Systems Engineering Management Plan

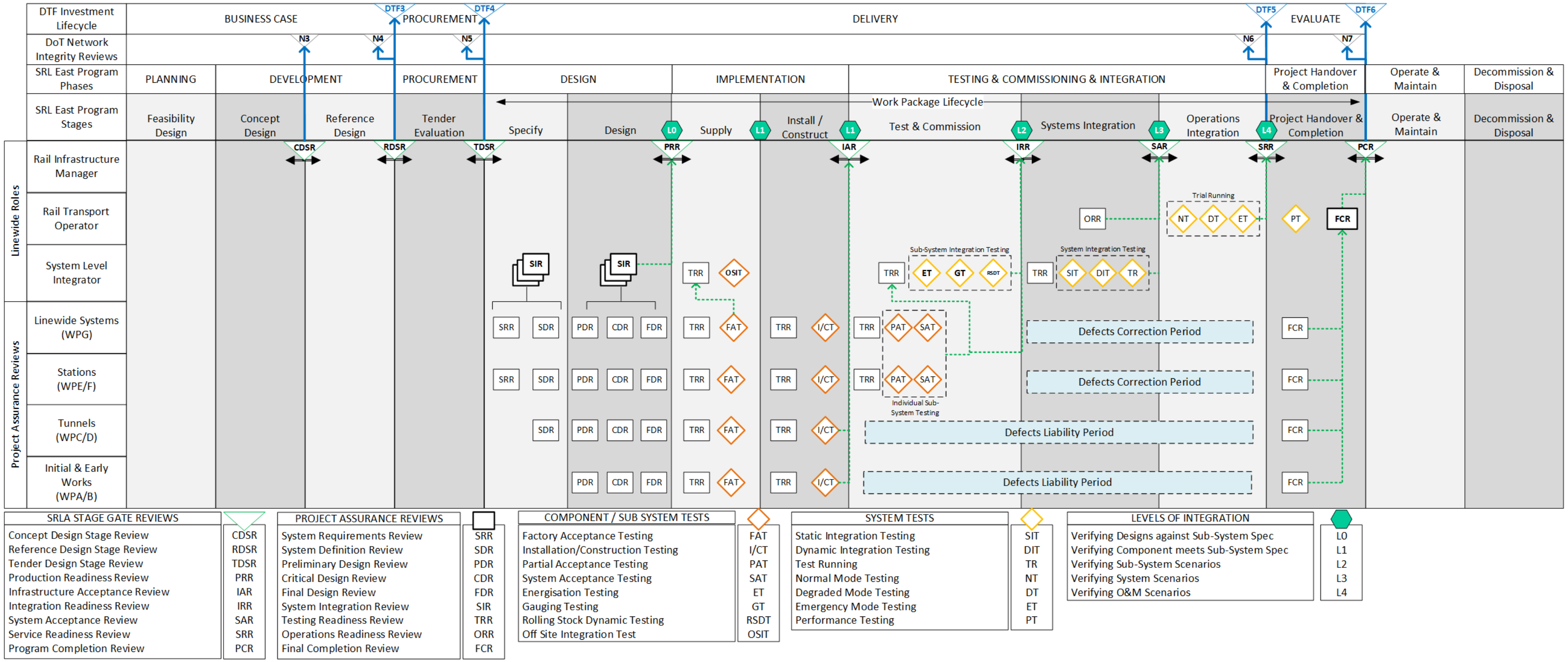


Figure 10 - SRL East System life cycle gates

The following sections in the SEMP cover the various phases (which are applicable to the delivery of the project), as documented in the engineering lifecycle model above.

The classical SE lifecycle applies a set of sequential steps to achieve a single end product/system being delivered by a single entity. A modification to this classical 'V' Model has been made in Appendix A to cater for the phased delivery model described in the SRLA Packaging & Procurement Strategy as well as customising lifecycle stages based upon the type of package.

Appendix A shows that initial and early works & tunnelling work packages have similar lifecycles. This is due to them predominantly being civil infrastructure projects and not requiring complex systems integration.

The line wide package has many more lifecycle stages (and more assurance reviews) as it carries the most risk, as is must be the package physically undertaking integration works of all other packages, and is responsible for validating that SRL East is safe and ready to operate.

Throughout all the package lifecycles, SRLA retains a role of Client and must ensure that all packages are progressing as specified in PS&TRs and other contractual documentation. This is typically done by undertaking gateway reviews and/or periodic assurance reviews. Table 3 describes the minimum expected gateway reviews.

Table 3 Project Assurance Reviews

Project **assurance review**	**System-related Objectives**
Infrastructure Acceptance Review (IAR)	IAR provides assurance the infrastructure is fit to accept by the RIM. It ensures that verification evidence for the sub-system against its PS&TR requirements specification is provided and that all system definition, operation, maintenance and disposal documentation is provided.
Production Readiness Review (PRR)	1. Verify the readiness of the system developer to produce the required number of systems 2. Verify production plans, fabrication, production-enabling products and personnel are in place and ready.
Integration Readiness Review (IRR)	1. Verify an overall integration strategy and plan exist, provide test and integration efficiency and early risk identification and mitigation 2. Ensure the subsystems are on schedule to be integrated into the SRL East 3. Ensure integration facilities, support personnel, and integration plans and procedures are on schedule to support integration. Verify subsystem interfaces are described, stable, have been verified and any change has been implemented by all interface subsystems
System Acceptance Review (SAR)	1. Verify the system complies with its technical baseline 2. Verify the system in its operational environment enables the RTO to perform scenarios identified in the Operations Concept Document (OCD), Maintenance Concept Document (MCD) and Southern Stabling Yard OCD 3. Non-conformances identified and resolved.
Operational Readiness Review (ORR)	Ensure all hardware, software, infrastructure, assets, personnel, procedures and documentation accurately reflect the as-built SRL East and are operationally ready.
Service Readiness Review (SeRR)	Determine the system's readiness for safe and efficient revenue operation, and all personnel, procedures and the SRL East are operationally ready.

Project **assurance review**	**System-related Objectives**
Program Completion Review (PCR)	Ensure that all the works have been completed and documented, and that the SRL system and its operation & maintenance meet the client requirements (CRD) and business requirements (BRD).

For further details on the description of each assurance review including their entry/exit criteria, refer to the SRLA Systems Assurance Management Plan [9].

4.2 SE Framework applied across the SRLA Project Phases & Stages

Each of the lifecycle stages applicable to program delivery are described in sections 1 through 10 of this document. These sections describe the systems engineering approach for the lifecycle stages and are expressed in the following format:

- **Purpose** – specifies what the purpose of the stage in the SRLA project lifecycle is
- **Inputs** – specifies the inputs required to undertake the activities within the stage
- **SE Approach** – defines the activities to be conducted as part of the stage
- **Outputs** - defines the outputs produced as part of the stage activities. These artefacts exist as configuration items to be managed in accordance with the Configuration Management Plan [6].

5 Development Phase – Engineering Assurance Strategy

5.1 Feasibility Design Stage

5.1.1 Purpose

The purpose of the Feasibility Design stage is to determine whether the business needs expressed in the BRD are feasible in the physical, time and cost related constraints that SRLA will face. By passing through this stage SRLA will be able to determine:

- An indicative cost estimate of the program
- Indicative dates for completion and any intermediate staging that may be required to gradually roll out the program
- The risks associated with proceeding to the next stage of development (concept design) based on the information available to SRLA at the time
- The various engineering & program strategies that will need to be put in place to ensure successful achievement of the program.

A Feasibility Design Stage Review (FDSR) concludes and assures the Feasibility Design stage.

5.1.2 Inputs

Title	Description
Business Requirements Document	The purpose of this business requirements document is to provide a clear specification of the envelope within which the SRLA project must deliver, and to what level of service/performance.
Rail Operations Plan	The Rail Operations Plan provides the SRLA with a concept of how the 'future' railway is predicted to be operated and maintained.
Precincts & Planning Plan	The Precincts & Planning Plan provides the SRLA with a concept of how the precincts surrounding each station are predicted to be developed from current day to end state.
Surface Transport Plan	The Surface Transport Plan provides the SRLA with a concept of how amendments to the existing surface transport network are predicted to be made over time to integrate with the new railway.
Key Decisions Register	The Key Decision Register provides SRLA with constraints that must be factored into any development work.

5.1.3 SE Approach

5.1.3.1 Engineering Assurance strategic planning & development of Systems Engineering & assurance related management plans

The Manager, Engineering Assurance is responsible for developing the SEMP & SAMP to set the strategy as to how Engineering Assurance activities are to be conducted throughout the lifecycle phases of the SRL East project. It is imperative that time is taken to strategise and accurately plan EA activities into any master schedules from the start of any program.

The subordinate Engineering Assurance disciplines require their own management plans to be developed. This must be done to structure any Engineering Assurance activities prior to advancing into the next stage of the program.

The subordinate management plans to be created are:

Management Plan	Developed By & Owner
Requirements Management Plan	Principal Engineer – Requirements Management
Verification and Validation Plan	Principal Engineer – Requirements Management
Architecture Management Plan	Principal Engineer – System Architecture
System Safety Management Plan	Principal Engineer – System Safety
Interface Management Plan	Deputy Director, Commercial & Technical Interface
Testing & Commissioning Plan	Deputy Package Director – Linewide & Operations
Human Factors Integration Plan	Principal Engineer – System Safety
EMC Management Plan	Deputy Package Director – Linewide & Operations
RAM Management Plan	Principal Engineer – System Safety

5.1.3.2 Perform Business Needs Analysis of BRD & Rail Operating Plan (ROP) to define Problem Space & characterise the Solution Space

The Engineering Assurance Team must analyse the source documentation issued by DTP to aid in defining the ‘problem space’ for SRLA. The output of which will be a SRLA System Context diagram that defines all external interfaces, stakeholders, and environment that SRLA impacts and will need to interface with. This activity helps to characterise the ‘solution space’ that will be further developed in the Concept Design stage.

This business needs analysis work must support the formation of an Investment Logic Map (ILM) for SRLA to determine the Objectives of the program, the Key Performance Indicators (KPIs) that the program must meet, and the benefits the program is expected to achieve.

5.1.3.3 Define SRL Program Staging Strategy

The SRL East system will be delivered by several projects (work packages). Therefore, a ‘program’ must be drawn together to detail how multiple projects can be logically staged to progressively build SRL overtime.

This program strategy will yield the numerous stages that SRLA must specify within CRDs (per stage) to be put forward for approval by the DTP client and eventual investment by DTF.

This initial alignment of ‘Program’ and ‘Problem Space’ (from the business analysis) is pivotal in enabling effective program integration, predominantly between Project Management, Engineering & Procurement disciplines.

5.1.3.4 Produce SRLA Staged CRDs

To align with DTF gateway reviews, the SRLA program will need a CRD to be developed to support every investment submission. The CRD must align with the business case and other investment submission documents, such as the SRL Stage One Scope Summary Document [7].

The SRLA are responsible for working with DTP to produce the Client Requirements Document that will clearly layout the objectives of what SRLA must deliver to, the stakeholders needs to be met by the SRLA solution and any other constraints. The CRD v1.0 is not the final document. It is an interim stage that requires clear objectives to be agreed, show traceability to the BRD, all stakeholders identified, and any high-level constraints defined.

The requirements of the CRD must be managed within SRLA’s requirements management database. All requirements represented in the CRD must be endorsed by the Requirements Working Group. Further detail can be found in the RWG Terms of Reference [Ref.15].

5.1.4 Feasibility Design Outputs

Title
Engineering Assurance Management Plans
Key Decisions Register
SRLA System Context
SRLA Investment Logic Map
SRLA Client Requirements Document v1.0

Title
SRL Stage One Scope Summary Document

5.2 Concept Design Stage

5.2.1 Purpose

The purpose of the Concept Design stage is to refine the problem and solution space further into a concept system with functionality and characteristics that will enable users of the system to conduct their business. This will be complemented by the production of a Concept Design.

By passing through this stage, SRLA will be able to determine:

- A more refined cost estimate of the program (utilising the concept design)
- Validated dates for completion and any intermediate staging that may be required to gradually roll out the program
- An approved scope for the program
- The risks associated with proceeding to the next stage of development (reference design) based on the information available to SRLA at the time.

A Concept Design Stage Review (CDSR) concludes and assures the Concept Design stage.

5.2.2 Inputs

Title	Description
Systems Engineering & Assurance Management Plans	The various Systems Engineering & Assurance Management Plans (such as Requirements Management Plans) provide guidance on how to undertake EA activities throughout the program lifecycle phases.
Key Decisions Register	The Key Decision Register provides SRLA with constraints that must be factored into any development work and updated as development progresses.
Scope Definition & Staging Document	This document gives high level details on the planned ‘migration’ of the SRLA project to gradually design, build and integrate the system.
SRLA Client Requirements Document v1.0	The CRD is to be developed further during this stage of the program.

5.2.3 SE Approach

5.2.3.1 Perform Operational Analysis and develop SRLA Operations Concept Document

Perform operational analysis of the CRD objectives and develop the SRLA OCD with Rail Infrastructure SMEs, Specialist Engineering SMEs (RAM, Safety, HF), and Operations, Maintenance & Passenger representatives.

The OCD will conceptually define how users of the system (operators, maintainers and passengers) are likely to interact with particular components of the SRLA solution.

The OCD does not define exactly ‘how’ the solution must work, but it gives rise to what parts of the system need to function in a particular way to meet the needs of the users.

The OCD must be developed by performing operational analysis in line with the process provided in Appendix C. A set of user scenarios that cover normal, degraded and emergency modes of the system must be considered as a minimum. These scenarios will form the initial test cases for operator integration testing.

5.2.3.2 Capture User Needs & Other Stakeholder Requirements

Using the user activities generated during the operational analysis, requirements engineers must elicit a set of 'user needs'. These 'user needs' represent what the operator, maintainer and passengers need the SRLA solution to provide them so that they are enabled to do their job/activities.

The user needs must be captured in the SRLA requirements management tool and are treated as additional stakeholder requirements to the CRD. The user needs must be endorsed for use by the SRLA Operational Analysis Working Group, details of which can be found in the OAWG Terms of Reference [Ref. 14].

Further to users, there are many other stakeholders of the system. A stakeholder requirements process must be implemented to manage the engagement of stakeholders outside of the operator, DTP & DTF. The output of this process will be a set of stakeholder needs, statements of work (for SRLA) and/or interface requirements that must be recorded within the SRLA requirements management tool. Once again, these stakeholder needs are considered as additional to the CRD.

The stakeholder requirements must have decisions made against them as to whether SRLA will include them either wholly, partially or not at all within the program scope. This position must be made clear to both interfacing parties, with any risks or issues captured and managed throughout development.

The stakeholder requirements can invoke technical constraints on the system or invoke works that SRLA must undertake as an organisation and report upon, all of which must be progressively assured.

Traceability to requirements sources, such as hazard logs, assumptions, correspondence or decisions must be documented and managed.

5.2.3.3 Establish a System Architecture & Perform System Level Analysis

The System Architecture acts as a configuration managed source of truth for the program with regards to characteristics, performance parameters, functionality and external interfaces/exchanges for the system.

Using the OCD/MCD, User needs and stakeholder requirements, perform System Level Analysis to determine the System Capabilities. These system capabilities should be applied to various system scenarios to enable formation of specific system functionality and sets of functional chains.

The functions should be hierarchically arranged in a functional breakdown structure. The FBS is important to be established early on as it will be used to structure part of the System Requirements Specification.

At this point in time the System Breakdown Structure can begin to be established so that functionality can be apportioned to top level systems. The Level 1 components of the system must be defined, with Level 2 components to be defined in further development phases.

The Level 0 and Level 1 architecture components, the System Capabilities, their associated functions, and the scenarios in which they are required shall be communicated to the program through a System Definition Document. For further details of architecture development, refer to the SRLA Architecture Management Plan [Ref 11].

The requirements output from the System Level Analysis must be captured in the System Requirements Specification (SRS). The SRS must specify all functional requirements, performance requirements, non-functional requirements, and constraints on the SRL East system. The technical details of external interfaces must be published in system level interface requirements specifications. It is expected that a singular IRS be produced for each external interface to detail the functional exchanges and any constraints gathered during engagement with stakeholders.

5.2.3.4 Perform Specialist Engineering Analysis

Whilst performing the system level analysis, Specialist Engineers must undertake targeted analysis to determine the RAM, safety, security, EMC & HF impacts of the proposed system, its inherent risks, and formulate any mitigating actions that must be performed by the project or engineered out during further design stages.

The predominant products that need to be produced as part of these activities are:

- Preliminary Hazard Analysis (PHA) and associated system hazard log
- Failure Modes Effects and Criticality Analysis (FMECA) or another Fault Tree Analysis (FTA) method and its associated critical items list/log
- Early Human Factors Analysis/User Hazard Analysis and its associated Human Factors Issues Log (HFIL)
- Fire & Life Safety strategy based upon the system safety analysis that guides further design
- Interface security/threat analysis (both physical and cyber) and its associated mitigations
- EMC Strategy to limit chance of EMI and to guide further design in how to apply principles of the strategy
- Earthing & Bonding strategy and associated guide to aid an end to end E&B system to be designed in an integrated way
- Noise & Vibration strategy and associated guide to aid design and ensure system level N&V thresholds are not exceeded.

5.2.3.5 Capture System Requirements

The document that holds all the system requirements is referred to as the System Requirements Specification (SRS). The system requirements define the functional and non-functional characteristics that the SRL-East system must exhibit..

Using the FBS the ‘Functional’ section of the SRS can be structured to show the reader how the system behaves/functions to enable System Capabilities justified by the scenarios developed as part of the OCD.

Traceability to requirements sources (such as hazard logs, assumptions, CRD, OCD) must be documented and managed at all times. The requirements in the SRS must be traced to their source(s) of justification, be that an overarching constraint, a control identified in SE logs, parent requirements, or other documentation. For example, a system requirement developed to meet a user need (identified during operational analysis) will be linked to that user need. The traceability must be implemented during requirements development to support progressive requirements traceability matrix production to progressively report satisfaction arguments to stakeholders (where required).

The scenario led analysis described above will not yield 100% of the SRS requirements. Therefore, the EA team must factor in activities to ensure the following are covered:

- Define Customer Expectations
- Define Project & Enterprise Constraints
- Define External Constraints
- Define Operational Scenarios
- Define Measures of Effectiveness
- Define System Boundaries
- Define Interfaces
- Define Utilisation Environments
- Define Life Cycle Concepts
- Define Functional Requirements
- Define Performance Requirements
- Define Modes of Operation.

Once nominated, the System Integrator will own and manage the SRS along with the operation and maintenance needs, and will perform verification against the SRS. SRLA will remain responsible for validating the CRD. SRS changes will be managed under SRLA's management of change process to ensure the delivered system will meet the CRD requirements.

5.2.3.6 Define the system migration or augmentation strategy

Whilst developing the OCD/MCD, the system migration or augmentation strategy must be formed. The migration or augmentation strategy defines consecutive, logically arranged configurations for the system to change from and to. Each configuration state may also have intermediate milestones; all of which need defining so that the program knows when to produce particular baselines. For example, what order infrastructure systems need to be installed to enable the installation of linewide systems; or what facilities is the operator expecting to be functional and when, for the purpose of operational readiness and training ahead of running a live service. Additional considerations include whether the railway system will run at full capacity from Day 1, or whether there will be a state where the railway is run at a lower level of performance.

A migration map must be produced out of the analysis performed to produce a picture of critical dependencies between projects/systems throughout the lifecycle of SRL East. The picture is used as a communication aid to workshop the dependencies further. The dependencies themselves should be documented and defined in an associated interface register.

These critical dependencies and milestones must be scheduled into each work package's program and reported against by project controls.

5.2.4 Concept Design Outputs

Title
CRD v2.0
SRLA OCD/MCD
User Needs
Stakeholder Requirements
SRS v1.0

Title
Safety Hazard Log
System Definition Document
Human Factors Issues Log
Critical Items List/Log
System Migration Map & Configuration States
Concept Design Assurance Report

5.3 Reference Design Stage

5.3.1 Purpose

One of the principal objectives of the Reference Design is to develop the specifications and requirements that will be used during procurement of the works. This is done through the incorporation of the program's System Requirements Specification (SRS) as it details the specifications, the minimum deliverables and the applicable standards against which SRLA 'as a whole' is to be delivered.

Throughout the Reference Design stage, the SRL East System Architecture is developed to a more granular level of detail to determine the physical and functional interfaces that must be specified in IDRs. The Reference Design must be developed by the AJM technical advisor to demonstrate a practical solution can be designed, built, and simulated to meet the requirements of the SRS.

This level of verification closes out the Reference Design stage and yields sufficient evidence to support an N3 NIAP review with DTP.

By the end of the Reference Design stage, SRLA will have:

- A system wide design (based on the components of the SBS) that meets the functionality and performance measures specified in the SRS
- The SRS allocated into smaller Project Scope & Technical Requirements (PS&TR) documents (normally per level 1 or 2 system in the SBS)
- Each PS&TR having associated reference design packs and specialist reports to enable SRLA to release information into the market for tender.

A Reference Design Stage Review (RDSR) concludes and assures the Reference Design stage.

5.3.2 Inputs

Title	Description
SRS v1.0	The SRS is to be developed further during this stage of the program in line with business case development and option selection.
System Architecture	The existing system architecture must continue to be developed to act as the central source of truth for the SRLA solution.
System Configuration States	The configuration states may require updating with more detailed milestones as the System Architecture and SRS are developed further.
Safety Hazard Log	Any previously raised hazard must continue to be designed out during system design, and any emerging hazards should be logged.

Title	Description
Critical Items List/Log	The previous RAM analysis will have given rise to a critical items list. This list must be used when designing the system further to ensure the number of critical items is minimised and managed appropriately.
Human Factors Issues Log	The HFIL provides designers with issues likely to be encountered if designs do not change or cater for human and machine interactions. This log must be continually monitored and updated through this development stage.

5.3.3 SE Approach

5.3.3.1 Iterate SRS, System Architecture, and Specialist Engineering Analysis

This Reference Design stage takes the outputs of the Concept Design stage and refines them based upon further iteration of the previous SE activities and design work.

Once development of the SRS has been completed, the System Architecture shall be published in document form as the System Definition Document (SDD). The SDD defines all system components within the architecture, and shows the links between operational activities, system functions, and component interfaces.

5.3.3.2 Publish the Functional Baseline

The Functional Baseline is an aligned set of configuration items that define the functionality and performance of the SRL East System, including external interface characteristics, and the verification required to meet said functionality, performance and interface characteristics.

For SRL East, the main CIs that form the Functional Baseline are:

- SRL East System Requirements Specification (including IRSs)
- SRL East System Definition Document
- SRL East VCRM
- SRL East SRS to CRD RTM.

SRLA may choose to include other configuration items within the Functional Baseline that provide greater context as to how the functionality, performance & interface characteristics have been justified.

The control, release and monitoring of configuration information is further defined in the SRLA Configuration Management Plan [Ref. 6].

5.3.3.3 Perform Sub-System Level Analysis (where appropriate) and apportion SRS to PS&TRs

Utilising the output of the system analysis, a sub-system analysis can be subsequently conducted. This enables the formation of a more detailed SBS, showing major components of the Rail System.

Similar to System Level Analysis, the Sub-System Level Analysis will yield definitions of sub-system capabilities, scenarios and functionality, along with specifying technical exchanges of information across sub-system interfaces. The technical detail of these interfaces will be captured within IDRs where the interface exists between two sub-systems being delivered by different parties.

The Engineering Assurance team will need to work with design disciplines to determine any technical constraints the sub-system layer must operate within.

Specialist Engineers (RAM, HF, Safety, Security, etc.) must perform further analysis on the output of the sub-system analysis to further refine their technical constraints and ensure any system level hazard mitigations or performance measures have been appropriately apportioned to the sub-systems.

The framework for developing the sub-system layer are completed in the following three steps:

1. Development of a logical architecture, its output being a System Breakdown Structure and functional requirements for each sub-system.
2. Deployment the logical architecture into the physical realm to create a physical architecture, representing what sub-system goes where and identifying more detailed interfaces.
3. Allocation of all elements from the physical architecture to the appropriate work package in line with SRLA's procurement strategy. The allocation must cover the lifecycle stages as specified in this SEMP to indicate clearly what party is responsible for the specification of a particular sub-system, who is responsible for the sub-system design, who is responsible for the supply and installation of the sub-system and who is responsible for the testing & commissioning of the sub-system.

All system integration and operations integration activities are the responsibility of the Systems Integrator.

Once this is completed, SRLA will be responsible for ensuring each PS&TR being complete and consistent. This 'scope' related detail is captured within the System Architecture Model and presented graphically as a Scope Work Breakdown Structure (SWBS). The Scope Work Breakdown Structure should be used to structure the PS&TRs with regards to what scope is described in Part A, what sub-systems are specified in Part B, and what process requirements must exist between PS&TRs to ensure appropriate technical information is provided between parties at the key lifecycle stages.

A Sub-System Definition Document must be produced (using the System Architecture Model) for each work package to define the sub-systems, their functions, and cross package interfaces.

5.3.3.4 Produce PS&TR Part B Requirements

The output of the sub-system analysis will result in a collection of technical requirements allocated to sub-systems. These requirements now need to be allocated to the appropriate PS&TR. These requirements are used to form Part B of each PS&TR. It is important to ensure that any interface requirements are clearly specified throughout this lifecycle phase for collation in PS&TR Part E and their associated IDR.

It is important that all requirements developed have a clear justification and traced to the SRL East SRS, traceability to a control from a specialty engineering log (e.g. Hazard Log), or traceability to a design decision made by SRLA. Without this traceability, the Part B requirement will be considered unjustified and should have the decision made as to whether it should remain or be removed from the PS&TR. Where it remains, the justification needs to be sought (by tracing to a decision) and its traceability updated.

A check against the SWBS must be conducted to ensure the requirements specified in Part B are aligned to the scope split decisions made by SRLA and that the Part B does not put requirements onto sub-systems that the work package is not responsible for delivering (therefore outside of their scope). This check should also create alignment with Part A of the PS&TR.

5.3.3.5 Produce IDRs & PS&TR Part E Requirements

To support clarification of interfaces, Interface Definition Requirements shall be developed for each cross package interface and critical external interfaces, and referred to in the requirements. The IDRs contain statements of what information/services must be exchanged across the interface, what performance levels the exchange must have, and any constraints the interface has (such as standard fixings). For further information on IDRs, refer to the SRLA Interface Management Plan [Ref 12].

5.3.3.6 Produce PS&TR Part A Requirements

The E&A Senior Engineering Managers must support the SRLA Work Package teams in coordinating the development of project scope related statements to be used in Part A of their respective PS&TR. The information provided should be produced using the SWBS to ensure consistency of information and ensure the PS&TR does not include any scope that has not been agreed to be delivered by each work package. If gaps are identified, the Senior Engineering Manager should feed back these gaps to the Senior Systems Engineer for the work package to be incorporated into SWBS updates using the change control process.

Once complete, comparison should be conducted to confirm that the scope described in Part A is represented by sufficient technical requirements in Part B, has the necessary statements of work in Part C and has the appropriate interfaces defined in Part E.

5.3.3.7 Produce PS&TR Part C Requirements

Individual teams within SRLA are responsible for producing any Part C requirements that require Work Packages to perform specific activities or processes commensurate with the scope of their respective PS&TRs.

The E&A Senior Engineering Managers are responsible for ensuring all engineering related teams have provided suitable Part C requirements for each PS&TR.

Other teams outside engineering, such as Land Planning, Environment and Sustainability (LPES) must also ensure their needs are met by a set of Part C requirements that must be included within each PS&TR. They must provide their requirements sets to the SRLA Work Package teams for inclusion within each PS&TR. The E&A Senior Engineering Manager should review the additional Part C requirements and ensure there are no conflicts with the other engineering related Part C requirements.

The Engineering Assurance team must elicit a set of Systems Engineering and Systems Assurance related requirements , typically referred to as “Statement of Work” requirements that must be captured in Part C of each PS&TR, and subsequently executed by the Work Packages.

If established at this point, the Systems Integrator must work with the Engineering Assurance team to ensure any Systems Integrator needs are captured within the Systems Engineering and Systems Assurance related requirements. The Interim Systems Integrator should elicit further Part C requirements where necessary to set up the System Integrator for success and enable their processes.

5.3.3.8 Verify SRL East System Requirements Specification

During the Reference Design stage, the E&A team should progressively assess the satisfaction of the SRL East SRS as each PS&TR is produced.

This will be completed by analysis of the top-down requirements traceability and any bottom up traceability provided for each PS&TR.

The E&A team may also utilise artefacts such as the reference design and simulations (e.g. RailSys) to evidence that SRL East is fit for purpose and represents a feasible system solution to stakeholder requirements.

The output of this exercise should be captured within a SRS Verification Report.

5.3.3.9 Publish Allocated Baselines (per PS&TR)

An Allocated Baseline will exist for each PS&TR and is an aligned set of configuration items detailing how system function and performance requirements are allocated across lower-level configuration items (hence the term allocated baseline). It includes all functional and interface characteristics allocated from the SRL East SRS or higher-level configuration items, derived requirements, interface requirements with other configuration items, design constraints, and the verification required to demonstrate the traceability and achievement of specified functional, performance, and interface characteristics.

For SRL East, the main CIs forming the Allocated Baseline are:

- Project Scope & Technical Requirements (including IDRs)
- Work Package specific Sub-System Definition Document (including SWBS)
- Work Package specific procurement drawings
- SRL East SRS to PS&TR RTM
- PS&TR initial VCRM.

SRLA may choose to include other configuration items within the Allocated Baseline that provide greater context as to how the functionality, performance & interface characteristics have been justified, along with any justification for the allocation between work packages.

The control, release and monitoring of configuration information is further defined in the SRLA Configuration Management Plan [Ref. 6].

5.3.3.10 Establish an interim Systems Integrator and processes

During the Reference Design stage, SRLA must prepare for managing the integration of multiple work packages. To do this, it is proposed that an interim Systems Integrator be established, supported by a process to identify, manage and resolve integration issues, assure system wide performance is met, and to deconflict cross package interfaces. The interim Systems Integrator must be responsible for providing recommendations on integration related topics to SRLA who shall decide, based on the proposal, and instruct any resulting actions to be undertaken by the work packages.

The System Integrator will setup working groups as necessary to dispose their duty/function as the integrator.

Each working group must have Terms of Reference developed defining the members, responsibilities, functions, and governance of the working group.

For further detail on the roles and responsibilities of the interim Systems Integrator (and supporting working groups), refer to the Integration Management Team Terms of Reference [Ref. 13].

The minimum processes for the interim Systems Integrator must be to support:

- Impact assessment of change requests (that impact the PS&TR, SRS or Operational Concept)
- Monitoring interface definition, design & conflict resolution with proposals being formally made to SRLA for enactment
- Integration test planning and implementation
- Progressive assurance of work packages against the SRL East SRS (via System Integration Reviews) to ensure the SRL East railway will be fit for purpose and operational by Day 1.

5.3.3.11 Undertake Reference Design Stage Review

The SRLA Assurance and Configuration Committee (ACC) will undertake an internal stage gate review to ensure the SRL East Assurance Case is still achievable following the finalised Reference Design.

The Engineering Assurance Team must compile the outcomes of the RDSR in a RDSR Assurance Report.

5.3.4 Reference Design Outputs

Title
SRS v2.0
SRS Verification Report
Sub-System Definition Document (per Work Package)
Interface Definition Requirements (external and cross package)
System Configuration States
Safety Hazard Log
Critical Items List/Log
Human Factors Issues Log
Project Scope & Technical Requirements Documents (per Work Package)
SRL East Reference Design
Functional Baseline
Allocated Baselines (per Work Package)
Systems Integrator ToR (and working group ToRs)
RDSR Assurance Report

6 Procurement Phase – Engineering Assurance Strategy

6.1 Tender Evaluation Stage

6.1.1 Purpose

Between passing through the RDSR and approaching the Tender Design Stage Review (TDSR), the program must review tender responses from the market to determine:

- What technical scope the market can commit to delivering
- What parties from the market have the necessary skills to deliver the program solution
- What timescales and budget can be committed to
- If there should be any changes to the reference designs and associated PS&TRs due to the market's response.

Based on the market's response to tenders and the support and guidance of the procurement procedures, SRLA will procure delivery entities to adopt the PS&TRs and start the design of their allocated system components.

A Tender Design Stage Review (TDSR) concludes and assures the Tender Evaluation stage.

A TDSR must be done for each work package, with the Tender Design and associated PS&TRs being baselined following each review.

The Tender Evaluation stage may take a multi-staged approach throughout which the Engineering Assurance Team will support workshops, reviews and conclusions regarding the suitability of the submitted tenders pertaining to systems engineering and assurance.

It is expected that each work package will go through an expression of interest phase to shortlist several proponents from the market, a request for proposal phase to assess the strongest proposal from the shortlist of proponents, and a request for tender phase to fine tune the tender design provided by the preferred proponent and award the contract based on this final agreement.

6.1.2 Inputs

Title	Description
Project Scope & Technical Requirements Documents	The existing PS&TRs that have been approved at the preceding RSDR and that have been released to the market
Expression of Interest Responses	Initial responses from the market typically come in the form of responses to an Expression of Interest (EOI).

6.1.3 SE Approach per tender evaluation

6.1.3.1 Test the interim Systems Integrator and its working groups

Based on the processes and working group Terms of References (ToR) established in the previous stage, the Systems Integrator should begin to be populated with applicable employees from SRLA.

Appropriate members with the necessary skills must be enabled to dedicate a proportion of their effort to supporting the functions of the interim Systems Integrator to stress test processes and governance arrangements based on analysis of the SRL Project in its current form.

Any process improvements must be captured and implemented going forward.

As a minimum, a System Integration Review should be undertaken per work package to understand the integration risks to SRLA with regards to SRL Project maturity, proposed PS&TR changes, and proposed schedule changes ahead of contract award.

6.1.3.2 Review Responses to Expression of Interest

Collectively, SRLA must review the responses received from the market during the EOI phase and use decision criteria to shortlist several tenderers, who will be invited to provide a more detailed proposal as to how they would implement the PS&TR.

More specifically, the Engineering & Assurance Team will provide support by reviewing EOI material regarding how it matches SRLA's expectations for systems engineering and assurance practices.

6.1.3.3 Review Responses to Request for Proposal (System Design Review)

Collectively, SRLA must review the responses received from the shortlisted proponents during the RFP phase and use decision criteria to determine the preferred proponent who will be invited to provide a final tender as to how they would implement the PS&TR.

More specifically, the Engineering & Assurance Team will provide support by reviewing RFP material regarding how it matches SRLA's expectations for systems engineering and assurance practices.

Collectively, SRLA must review any responses to the RFT to determine whether the requirements within the PS&TR parts B & C (and the impact to the SRS) are achievable by the tenderer.

Where there are necessary changes or non-compliances to PS&TR requirements, the Engineering Assurance and Work Package teams must determine the impacts the change or non-compliance has not only on the PS&TR in question, but also upon all other developed or in-flight SRL East PS&TRs. The Engineering Assurance team are responsible for identifying impacts these changes have to the CRD and other identified stakeholder requirements.

The interim Systems Integrator must determine the impacts the change or non-compliance has across SRL East allocated baselines and impacts to the SRL East functional baseline, in so much to protect the integrity of the operations & maintenance concept, its associated needs, and the SRL East SRS to ensure a fit for purpose and operational railway can be delivered on Day 1.

6.1.3.4 Amend Allocated Baseline and publish ahead of Contract Award

If the tenderer has requested any amendments to the PS&TRs and SRLA accept these changes through the engineering change management procedure, the Engineering Assurance team must update the relevant PS&TRs (and the SRS and/or needs requirements based upon the impacts of the change) in the requirements management tool and baseline them ahead of contract award.

Non-compliance & technical departures may be proposed at this time, however SRLA will not formally accept them at this stage.

Once the PS&TR and its associated CIs are all updated, a new Allocated Baseline will be published and issued to the proponent to begin the specify stage of the project lifecycle.

6.1.3.5 Amend Functional Baseline and publish ahead of Contract Award

If the tenderer has requested any amendments to the PS&TRs that alter the information contained within the Functional Baseline (System Architecture, SRL East SRS etc.), and these changes have been accepted, the interim Systems Integrator must update the System Architecture, produce the SAM artefacts (SDD, SBS, Interface Register, etc.) and baseline them ahead of contract award. The Engineering Assurance Team will update the SRL East SRS based on the approved changes.

Following completion of the changes, a new Functional Baseline will be published and issued to all necessary SRLA stakeholders.

6.1.3.6 Undertake Tender Design Stage Review

The SRLA ACC will undertake an internal stage gate review to ensure the SRL East Assurance Case is still achievable following the agreements laid out in the awarded contract. This must be done for each work package PS&TR.

The Engineering Assurance Team must compile the outcomes of the TDSR in a TDSR Assurance Report.

6.1.4 Tender Design Outputs

Title
Project Scope & Technical Requirements Documents (per work package)
Systems Requirements Specification & System Definition Document (updated where necessary)
Sub-System Definition Documents (per work package & updated where necessary)
Updated Allocated Baseline (per work package)
Updated Functional Baseline (updated where necessary)
Interface Definition Requirements (updated where necessary)
Tender Response Reviews
RFT Tender Design Baseline
Contract Award Tender Design Baseline
TDSR Assurance Report

7 Design Phase – Engineering Assurance Strategy

7.1 Specify Stage (SRR & SDR)

7.1.1 Purpose

Following the procurement process and successful award/engagement of a Delivery Agency, the next phase focuses on design (commonly known as detailed design) on the scope and technical detail specified in the allocated PS&TR.

Before delving into technical design works, it is important to establish that the work package has firstly understood the scope and technical requirements of the PS&TR, have setup plans and processes to conduct detailed design, have the necessary skills and accreditations to undertake the works, and have had their chance to feed back on system level designs, and to develop a technical specification (where necessary) to design to. This is the 'Specify Stage' of the project.

7.1.2 Inputs

Title
Project Scope & Technical Requirements Documents
Sub-System Definition Document
Interface Definition Requirements
SRLA Management Guidelines & Supporting Documentation
Tender Design reference material & drawings

7.1.3 SE Approach per work package

7.1.3.1 Work Package Tasks

The Engineering teams from each work package is expected to undertake the following (or technical equivalent)

- Establish a Systems Engineering Management Plan specific to the context and complexity of the work package
- Agree an approach with SRLA on how compliance to PS&TRs can be Verified
- Agree an approach with SRLA on how changes to PS&TRs must be undertaken
- Agree an approach with SRLA on how risks with significant technical, safety, cost and schedule impacts are identified, mitigated or escalated to SRLA
- Agree an approach with SRLA how cross package interfaces are to be refined from Interface Definition Requirements to Interface Control Documents and managed by lead work packages
- Demonstrate how the PS&TR for the specific work package are consolidated into the Work Package's objectives and their feasibility
- Support the formation of a cost and schedule for the work package
- Identify and hire the required systems engineering resource for the work package or procure skills where necessary
- Agree with the Systems Integrator on work package resources to participate in Systems Integrator working groups
- Undertake agreed SE processes to specify the work package system solution and accompanying hazard logs, technical reports and decision registers:
    - Detailed analysis of the various system components that are specified to determine how they should be designed
    - Where further decomposition (from the PS&TRs) is required, the work package must undertake further sub-system level analysis to output one of many sub-system requirements specifications, each of which must show traceability to the technical requirements of the PS&TR.
    - Human Factors analysis to determine the Human/System interface and how it should be designed based on its intended usage. This activity must feed into the appropriate working group of the Systems Integrator to determine how the system level HFIL is impacted by each work package and whether there are any emergent human factors issues to be managed.
    - Detailed RAM analysis to ensure the component parts are designed with the appropriate level of reliability, availability or maintainability. This activity must feed into the appropriate working group of the Systems Integrator to determine impacts on system level RAM targets.
    - Detailed Safety analysis (following SRLA SiD procedure) to ensure that the components being designed are safe, both individually but also when joined back together as a larger system. This activity must feed into the appropriate working group of the Systems Integrator to determine how system level risks are being appropriately mitigated by each work package and whether there are any emergent risks to be mitigated.
    - Definition and specification of technical interfaces both internal and external (as identified in IDRs) to the work package, the result of which will form ICDs. Should there be any conflicts or issues during design of the interface, this must be escalated to the Systems Integrator to aid deconfliction. Further information on ICD development can be found in the SRLA IMP [Ref. 12].
- Prepare for project assurance reviews and publish assurance artefacts, as specified in the work package ACC Charter.
- Participate in a System Requirements Review (SRR) and a System Definition Review (SDR), as detailed in the SRLA SAMP.
- Close out any actions from the SRR prior to the SDR.

7.1.3.2 SRLA E&A Tasks

The SRLA E&A Team must undertake the following:

- Receive and review all management plans, raise any comments, resolve the comments with the Work Packages and then finally accept the management plans. The review of the management plans is an entry criteria for the System Requirements Review
- Be on hand to field any request for information from the Work Packages. Provide guidance consistent with the SEMP and is to be applied consistently across all Work Packages
- maintain the SRL East System architecture and assess any changes with regards to impacts to the SRL East System solution outside of the Systems Integrator's remit (i.e., road infrastructure)
- Maintain all SRLA engineering identified configuration items throughout the change process and publish the appropriate baselines as necessary.

The embedded Senior Engineering Managers for each work package, supported by embedded Assurance Engineers must orchestrate SRLAs assurance reviews for each work package (SRR and SDR) and compile the SRL East assurance reports. The core E&A team must review the outcomes of the work package assurance reports and assess whether the SRL East assurance case is being upheld based on the evidence received.

- A System Requirements Review must be evaluated by SRLA per work package to understand the impacts of the work package's progression or design has on the SRL East System solution and how the Work Package is upkeeping their obligations, as specified by their PS&TR
- Any actions required as a result of the SRR must be actioned by the appropriate product owner. For example, if the SRS requires updating, the work package must lodge a change request. Following change request approval, the SRLA EA Team must amend the SRS, publish a new baseline, and distribute the updated baseline to all affected parties
- A System Definition Review must be held per work package to evaluate the credibility and responsiveness of the work package to the PS&TRs and SRL East schedule, as well as the work packages specification(s) and maturity of definition of interfaces with other packages. The SDR ensures that the work package is ready to begin design works, being prepared with integrated processes and having necessary skilled employees. Any risks the work package sees to the SRL East program must have been identified at this stage, with mitigation strategies formed

7.1.3.3 Systems Integrator Tasks

The Engineering Assurance Team is expected to provide the work package with all SRL East system contextual information developed during the reference design phase. Similarly, the Systems Integrator will provide system-level analysis for work package change requests.

During this stage, the Systems Integrator must support work packages to evaluate their involvement in system wide disciplines (such as human factors, earthing & bonding, safety & security, etc.) and agree on memberships of the Systems Integrator's working groups.

Each work package is expected to undertake an SRR and SDR unless stated otherwise in the PS&TR.

A System Requirements Review must be evaluated by the Systems Integrator per review per work package to understand the impacts the work package's progression or design has on system wide functionality & constraints.

The Systems Integrator must evaluate the SDR(s) to determine whether the SRS is being satisfied by the work package's specification(s) and note any actions or recommendations.

7.1.4 Engineering Assurance Outputs

Title
Work Package specific Management Plans
Work Package functional & non-functional specifications
Technical Reports
Interface Control Documents
Hazard Logs & Risk/Issues Registers
System Requirements Review Assurance Report
System Definition Review Assurance Report
Work Package Design Decisions Register

7.2 Design Stage (PDR, CDR & FDR)

7.2.1 Purpose

Following the Specify Stage, the Design Stage focuses on delivering design based on the scope and technical detail specified in the allocated PS&TR.

The assurance reviews that each work package must undertake are Preliminary Design, Critical Design & Final Design at a point in time agreed between SRLA and the work package.

However, the Work Package may develop their design in multiple ‘design packs’ or ‘lots’.

A Preliminary Design Review (PDR) concludes and assures the Preliminary Design sub-stage.

7.2.2 Inputs

Title	Description
Work Package Functional & Non-Functional Specifications	The technical specifications that describe the functional characteristics, performance levels and constraints of the work package’s System of Interest (SOI). May be one or many specifications.
Interface Control Documents	Documents that define the interface characteristics both physically and functionally that exist between systems or components, that are to be developed jointly by both interfacing parties.
Hazard Logs and Risks/Issues Registers	The logs and registers that document the output of specialty analysis that must be considered during detailed design.
System Definition Review Assurance Report	Detailing any actions that must be undertaken by the Work Package prior to performing the PDR.

7.2.3 SE Approach per work package

7.2.3.1 Work Package Tasks

The Engineering teams from each work package are expected to undertake the following (or technical equivalent), the extent of which will vary between work packages and will be indicated within PS&TR Part C requirements:

- Detailed analysis of the various system components that are specified to determine how they should be designed
- Where further decomposition (from the Work Package Specifications) is required, the work package must undertake further sub-system level analysis to output one of many sub-system requirements specifications, each of which must show eventual traceability to the technical requirements of the PS&TR
- Undertake Safety in Design workshops to determine that risks are managed SFAIRP and implemented through designs
- Develop solutions in line with the work package specifications and in line with processes defined in the SRLA Engineering & Design Management Plan [Ref. 4]
- Verification of any designs produced to ensure they meet the work package specifications and the PS&TR. This should be in the form of a VCRM
- Demonstrate that prelim designs have undergone proof engineering to ensure the adequacy, consistency, and integrity against technical specifications
- Develop and finalise all ICDs
- Eventual validation of the system safety argument based upon the final design
- Development of test schedules and procedures to demonstrate that FATs, I/CTs, SATs, etc. are planned, responsibilities for testing has been determined, and the testing & commissioning effort is well understood. Typically documented in a Testing & Commissioning Plan
- Demonstrate that final designs have been certified
- Identification and resolution or SRLA acceptance of any non-conformances against PS&TRs
- Propose changes to SRLA utilising the SRLA engineering change control procedure where necessary.
- Participate in a Preliminary Design Review (PDR), Critical Design Review (CDR) and a Final Design Review (FDR), as defined in the SRLA SAMP.

7.2.3.2 SRLA E&A Tasks

The Engineering Assurance Team is expected to provide the work package with all necessary information regarding the system solution.

Each work package is expected to undertake a PDR, CDR and FDR unless stated otherwise in the PS&TR.

A Preliminary Design Review must be held per work package to evaluate the initial detailed design of the system against the requirements specifying its functionality and characteristics and that it does not unnecessarily induce any safety hazards. The review must present any associated risk not meeting any cost or schedule constraints. Any actions required as a result of the PDR must be actioned by the appropriate product owner prior to the Critical Design Review.

A Critical Design Review must be held per work package to demonstrate that the maturity of the design is appropriate to support proceeding with full-scale fabrication, assembly, integration, and test. This review examines whether the technical effort is on track to complete the detailed designs while meeting functional and performance requirements within the identified cost and schedule constraints at an acceptable risk.

A Final Design Review must be held per work package to examine whether the detailed designs are certified, comply with requirements, and that safety hazards, RAM issues, HF issues and safety arguments have all been resolved prior to acquiring subsystems. It ensures that the production plans, fabrication, assembly, integration enabling products, operational support, and personnel are in place and ready to begin production.

7.2.3.3 Systems Integrator Tasks

During design, the Systems Integrator must support work packages to define sub-system performance in line with system wide performance measures. This is to be enacted by the Systems Integrator working groups to coordinate the analysis of sub-system & system level issues and/or deconflict cross package interfaces. The System Integrator will escalate issues to SRLA for action where the issue cannot be resolved by the Work Packages.

The Systems Integrator must assess the impact of any engineering change requests raised by work packages and report its findings to SRLA to support decision making.

The Systems Integrator must develop operational test procedures based upon understanding of detailed designs from work packages, and to validate the Operator Needs.

The Systems Integrator must support the development of standard operating procedures that detail how the end state operator must undertake its day to day running (and maintenance) of SRL East.

The Systems Integrator must plan for System Integration tests based on the schedules of work packages and delivery of their sub-systems being reported on a timescale in line with contract agreements.

Periodic Systems Integration Reviews are to be run by the Systems Integrator at a frequency to be agreed between the work packages, the Systems Integrator and SRLA. The purpose of the SIR is to progressively ensure that work packages are meeting time & schedule commitments as well as technical development progress required to pass through to implementation. The Systems Integrator will raise any conflicts or issues (along with proposed mitigations) at each ACC review where the System Integrator and the Client will jointly provide formal feedback to the Work Packages. The SIR is an intermediary review to keep touch with work packages and understand their progression between PDR, CDR & FDR, as deemed necessary by the Systems Integrator. It is also to ensure that the Systems Integrator does not lose configuration of the System Architecture and can receive more frequent information from work packages in order to update the system architecture and analyse the impacts of the changes.

The System Integrator must be in attendance at all Work Package ACC reviews to ensure assurance evidences required for WPG's assurance case are requested, reported on, received and any issues or actions are raised and captured in the ACC reports.

7.2.4 Engineering Assurance Outputs

Title
Project Scope & Technical Requirements Traceability Matrix
Preliminary Design Review Assurance Report
Preliminary Design Baseline (Drawings, Specifications, Management Plans, Change Request, Non-compliances etc.)
Critical Design Review Assurance Report
Critical Design Baseline (Drawings, Specifications, Management Plans, Change Request, Non-compliances etc.)
Final Design Review Assurance Report
Final Design Baseline (Drawings, Specifications, Management Plans, Change Request, Non-compliances etc.)
System Integration Review Assurance Reports
Test plans/procedures and initial schedules
Interface Control Documents
Work Package Technical Specifications
Standard Operating Procedures

Title
Design Certification Documents
SRL East System Architecture updated

7.3 Production Readiness Review (PRR)

7.3.1 Purpose

In order to judge the readiness of construction partners or system developers for production of the required infrastructure materials or system, an overall Production Readiness Review must be conducted. This review ensures that the production plans, fabrication, assembly, integration enabling products, operational support, and personnel are in place and ready to begin production. It also ensures that one work package is not advancing further than another work package without the interface risks being appropriately identified and mitigated.

The Product Readiness Review is a stage gate review for SRLA to judge the risk of allowing a singular (or multiple) work package to progress into the implementation phase, and to make a decision on what mitigations or actions must be recorded going forward. Details of the entry/exit criteria for the PRR can be found in the SAMP.

7.3.2 Inputs

Title	Description
Project Scope & Technical Requirements Documents	The existing PS&TRs that the Work Package is being held accountable to
Prelim, Critical & Final Design Assurance Reports from all work packages	Assurance reports collected by SRLA during PDR, CDR & FDR reviews with multiple work packages
System Integration Reviews	Collection of system integration reviews and their associated reports undertaken throughout the Design phase

7.3.3 SE Approach

7.3.3.1 SRLA E&A Tasks

SRLA E&A must undertake a Production Readiness Review to inform SRLA about the associated risks with allowing individual projects to progress into construction or manufacture stages of their lifecycle.

SRLA must assess the maturity of all in flight projects at this time so as to classify the risk to SRL East as a whole.

SRLA E&A must capture the output of their analysis in a PRR assurance report, which must also detail the resultant recommendations or necessary actions for SRLA to consider in its decisions.

7.3.3.2 Systems Integrator Tasks

The Systems Integrator must be in attendance in each PRR ACC review to ensure assurance evidences required for WPG's assurance case are requested, reported on, received and any issues or actions are raised and captured in the ACC reports.

The Systems Integrator must update their risk register to capture any emergent risks from individual work packages proceeding past this stage that may impact integration activities.

7.3.4 Engineering Assurance Outputs

Title
Production Readiness Review assurance report

8 Implementation Phase – Engineering Assurance Strategy

Once a Work Package has commenced manufacturing/supply of its assets, the Implementation Phase begins. This phase covers the procurement of, manufacture of, factory testing of, shipping of, installation of and final acceptance testing of individual assets and infrastructure.

SRLA must be kept updated on progress of manufacturing and testing. SRLA must also be assure the assets being supplied are verified as meeting the technical specifications, not only in the factory, but also on site after shipping and after final installation/construction.

The infrastructure built during this phase must eventually be handed over to and accepted by the Rail Infrastructure Manager.

Any assets delivered or installed during this phase must be tested standalone and Verified as meeting its functional, performance and interface requirements (via simulation) before any sub-system integration may take place.

8.1 Supply Stage (TRR & FATs)

8.1.1 Purpose

During the manufacturing and building of each asset/product, it is necessary to test that each product meets the specification of the final designs prior to being shipped to site and being installed. This is typically known as Factory Acceptance Testing to ensure the product is fit for purpose before leaving its place of manufacture with any defects or corrections being recorded and fixed before the product can be shipped to site. The product will be tested by itself to ensure its own characteristics and behaviour meets the performance and functionality specified. This must be done to limit the risk of causing harm to a larger system should this singular component experience any failure or unexpected/emergent behaviour on site.

Prior to testing, it is expected that each work package undertakes Test Readiness Reviews to determine whether the product itself is ready for factory testing, that all parties are aware of the scope and procedures for testing, and that all associated documentation and engineering activities are at an agreed state.

The Systems Integrator must review all TRRs received to ensure Work Packages are prepared for testing, sticking to schedule, and that there are no unmitigated interface risks (as far as is reasonably practicable).

Once a work package successfully passes a TRR, they may begin undertaking factory tests.

The results of the factory testing must be captured by the work packages, with any defects or non-conformances being recorded and resolved. The acceptance of any product by a work package must be documented as part of the Factory Acceptance Test report.

8.2 Install/Construct Stage (TRR, I/CT)

8.2.1 Purpose

Having undertaken factory tests to verify that the product meets its specification, the product must then be transported to the location of its installation, unloaded, tested to ensure its integrity remains after transportation, installed in place (or permanently assembled/constructed), tested to ensure the standalone product still meets its specification once installed, and (with regards to infrastructure) is accepted by the RIM. Therefore, the Install/Construct Stage covers the necessary activities to do this.

As with the Supply Stage, the work package must undertake a Test Readiness Review prior to beginning any onsite tests.

The Systems Integrator must review all TRRs received to ensure Work Packages are prepared for testing, sticking to schedule, and that there are no unmitigated interface risks (as far as is reasonably practicable).

Once a work package successfully passes a TRR, they may begin undertaking site based tests.

SRLA has not specified that there needs to be a Site Acceptance Test, however it is recommended that work packages specify SATs be performed to ensure the product delivered has the same integrity as the product certified in the factory.

SRLA has specified that work packages must undertake Installation/Construction Tests. These tests are to ensure that once a product is installed (or permanently assembled or infrastructure built) it functions and performs as specified in situ.

For any products commissioned following this stage; the product must come with a guarantee it meets the performance and functionality specified, whilst also being certified as safe. Therefore, any certifications and type approvals must be provided as evidence that the product is suitable to be used.

Any defects and derogations identified during this review must be captured in a defects rectification report and managed in line with the Defects Rectification Procedure. Any defects or derogations of a product that will be put to use following this stage must be captured within a management system, such as a Failure Reporting and Analysis and Corrective Actions System (FRACAS) to list future actions that will correct the defects prior to acceptance.

Following the practical completion of construction; an Infrastructure Acceptance Review (IAR) will be undertaken to conclude this phase for the Site Prep & Tunnel Boring type projects. The Station Fit Out & Linewide Systems projects shall continue through further integration & testing stages.

The System Integrator, Linewide Systems and Station Fit Out work packages must be involved in the IAR held by SRLA to ensure they are accepting of the infrastructure that they will be receiving in order to conduct their fit out work upon.

At this stage, there is opportunity for the Linewide Work Package to undertake some off site integration testing. This means that the Linewide Work Package can strategically integrate some sub-systems in a dedicated facility before transporting the sub-systems to site for wider integration. An example of this would be OSIT of rollingstock, track and signalling subsystems. A dedicated test facility can be built where these critical interfaces between sub-systems can be trialled and tested separate from the site environment so as to limit the risk of major integration issues in the future. For further detail about OSIT opportunities, refer to the SRLA Testing & Commissioning Management Plan.

8.3 Infrastructure Acceptance Review (IAR)

8.3.1 Purpose

To gauge whether the built infrastructure of work packages is acceptable to be used (for hardware fitment by other work packages), SRLA must review the works and 'accept' the infrastructure as the Rail Infrastructure Manager (RIM). This is for WPC and WPA work packages only, with an Infrastructure Acceptance Review being the stage gate performed.

This review must include the necessary activities to confirm practical completion of the work package assets, after which the defects liability period will commence.

This review ensures there is evidence the built infrastructure meets the specification of the PS&TRs (Verification) and is fit for purpose to be used by other work packages.

It is also necessary to have evidence that the installed infrastructure is deemed to have been designed to mitigate safety hazards, was installed safely, and is therefore a 'safe' system. This forms part of the safety case to be proposed to ONRSR for SRLA to maintain its RIM Accreditation.

The Infrastructure Acceptance Review is a stage gate review for SRLA to mark the formal acceptance of civil infrastructure built by work packages. This signifies the practical completion of the work packages and marks the point in which risk transfer from the work packages to SRLA whilst retaining the work packages for a defects and liability period. Details of the entry/exit criteria for the IAR can be found in the SAMP.

8.3.1.1 SRLA E&A Tasks

The SRLA Engineering Assurance Team must undertake the IAR assurance review with the first tunnelling Work Package, as this is the only work package being delivered under SRLA's accreditation as RIM. The evidence from the assurance review will be used to support SRLA to formally accept the handover of the tunnel infrastructure, and will inform the linewide RTO what they will be accepting once the tunnel infrastructure gets handed over to them for operations.

8.3.1.2 Work Package tasks

Both Tunnel Boring type projects and Site Preparation type projects must collate the necessary evidence to submit to SRLA in order for SRLA to undertake an IAR and determine whether the completed infrastructure can be accepted as part of SRLA's role as the RIM.

The details of the necessary evidence is contained in the SAMP.

The projects must also ensure activities relating to practical completions and project close out are monitored and completed ahead of the IAR. Where activities cannot be completed, a plan must be made and agreed with SRLA for the closure of these outstanding actions after the IAR. The work packages must then report on the closure of these outstanding activities with SRLA. All activities must be confirmed as closed before the work package can be deemed 'complete.' Further details of these activities can be found in the Completions & Handover Management Plan [Ref. 18].

8.3.1.3 Systems Integrator Tasks

The Systems Integrator must review the evidence provided by work packages for their IAR. The Systems Integrator must provide feedback and recommendations to SRLA on any issues that may be outstanding or have been identified during the review that will impact/impede the development of the remaining work packages.

9 Testing & Commissioning & Integration – Assurance Strategy

Testing & Commissioning (and overall integration of SRL East) will be conducted based on 'integration levels' as specified in the System Integration Strategy with the work package successfully verifying that the component, sub-system, or system is proved to meet the requirements of the appropriate specification and/or scenarios specified during Reference Design.

9.1 Test & Commission Stage (TRR WPE-G, PAT, SAT, TRR SI, ET, GT, RSDT, IRR)

9.1.1 Purpose

After the product has been tested standalone and confirmed it performs as specified, multiple products are tested together. This is referred to as sub-system integration, as multiple products tested together comprise a more prominent sub-system in the system breakdown structure (such as the Rolling Stock). The multiple, interfacing components will be tested to ensure they behave and meet the performance and functionality specified without failure or any unexpected/emergent behaviours. The testing must be made against sub-system specifications. It is recognised at this time that not all products may be available for complete sub-system integration, therefore partial acceptance testing can also be utilised to support a progressive testing and commissioning strategy.

There may be multiple types of tests that work packages may undertake during this stage. Further details are provided in the SRLA Testing & Commissioning Management Plan, but for the purposes of this SEMP, SRLA specifies two types: Partial Acceptance Tests and System Acceptance Tests.

As per previous stages, the Work Packages are expected to undertake Test Readiness Reviews prior to commencing any sub-system integration testing.

The Systems Integrator must review all TRRs received to ensure Work Packages are prepared for testing, sticking to schedule, and that there are no unmitigated interface and integration risks (as far as is reasonably practicable).

Once a work package successfully passes a TRR, they may begin undertaking sub-system integration tests.

The tests undertaken must be conducted to verify that the sub-systems, when integrated together, meet the technical specifications of the sub-systems and their internal interfaces. External interfaces must be simulated during these tests to partially satisfy ICDs.

The outcome of the PATS & SAT must be collated into testing reports to be used as evidence for the Integration Readiness Revie (IRR).

An IRR concludes and assures the Test & Commission stage and judges whether the System Integrator is prepared to undertake system integration tests, and that all sub-systems have been verified as being built, installed and performing correctly.

During this stage, the System Integrator may choose to undertake strategic system level tests of integrated sub-systems as they become verified and pass SATs. These tests typically include: electrification testing, gauge testing, and rolling stock dynamic testing. The results of these tests can be used to evaluate the readiness of the System Integrator to pass the IRR gate.

9.1.2 Inputs

Title	Description
Commissioning Plan & Commissioning Event Plans	The overarching plans that specify how commissioning must take place and key configurations that must be tested progressively
Approved Test Plans, Procedures, Detailed Interface Test Plans & checklists	Detailed documentations that specify how individual tests must be undertaken, recorded and published
Interface Control Documents	Interface specific documents that identify the properties of the interface under testing and which must be verified by the tests specified in the test plans
VCRM	Record of a test report against requirements providing evidence that they have been verified by the performed tests
Test Schedule	Master schedule of all tests to be undertaken

9.1.3 SE Approach

9.1.3.1 Work Packages E/F & G Activities

Each Work Package is expected to individually test their own delivered assets/components to ensure they meet the performance and functionality specified in their associated asset/component specification without failure or any unexpected/emergent behaviours.
To achieve this, each work package must conduct the following activities for their delivered assets/components only:

- Confirm schedules, work force readiness and site preparedness to undertake sub-system tests by passing a TRR
- Conduct system or partial acceptance tests specified in Detailed Interface Test Plans with both interfacing work packages. These tests should verify the interface is functioning as expected and that both components meet the ICD. Where one side of the interface is not available to be tested, then partial acceptance testing of the interface may be conducted simulating interface exchanges. However, once the reciprocating component is available, the interface must be tested again by both parties in order to 'close' the interface
- Escalation of any interface issues to the Systems Integrator for resolution
- Conduct system or partial acceptance tests specified in the approved test plans by following the necessary test procedures to verify test cases are either partially or fully satisfied. Where partial satisfactions cannot be resolved, a non-compliance must be raised to the System Integrator for assessment
- Recording the outcomes of each test in test reports, with any defects being added to a defects list (or equivalent) and rectification plans being produced
- Provide results of all tests, defects, redline markups of drawings and commissioning certificates to the Systems Integrator to undertake an Integration Readiness Review
- Undertake a Functional Configuration Audit & Physical Configuration Audit of installed components.
- Further detail can be found in the SRLA Testing & Commissioning Management Plan and SRLA Interface Management Plans.

9.1.3.2 Systems Integrator Activities

During the sub-system tests, the Systems Integrator:

- Will support resolution of any interface issues that may arise (such as scheduling)

- Will review any non-compliance or defect to support work package resolutions with 'best for program' outcomes.
- Will perform witness testing of activities performed by other Work Packages

Once all sub-system tests have been completed, drawings finalised, commissioning certificates signed, etc. the Systems Integrator will conduct an Integration Readiness Review.

The results of the Integration Readiness Review must be captured in an assurance report and submitted to the SRLA ACC for approval.

The details of the IRR and its associated checklist can be found in the SRLA System Assurance Management Plan.

9.1.4 Engineering Assurance Outputs

Title
Sub-System Test Reports
Closed ICDs (supported by Detailed Interface Test Results)
Redline Drawings
Commissioning Certificates
Sub-System FCAs
Sub-System PCAs
VCRMS
IRR Assurance Report

9.2 Systems Integration Stage (SIT, DIT, TR, ORR & SAR)

9.2.1 Purpose

This stage of testing accomplishes complete integration and verification of the SRL East System as a whole after subsystems integration is complete.

This stage will be led by the System Integrator who will test the SRL East system against the SRS to verify that all functional interfaces, functions and performance requirements are met in preparation of achieving system acceptance by the operator.

The focus of these tests is to ensure the procured system behaves and performs as specified in the System Definition Document and SRS.

The System Integrator will determine the strategy in which integration testing will be applied, however SRLA has classified three typical types of tests typically performed when integrating railways: Static Integration Tests (SIT), Dynamic Integration Tests (DIT) and Test Running (TR).

During this stage, the RTO will prepare itself for undertaking trail running. This means that all standard operating procedures must be documented, staff hired and trained appropriately to run services, RTO accreditations from ONRSR have been achieved, and all other organisational support systems established.

The RTO will confirm it is ready to begin trial running by performing an ORR and submitting the results to the System Integrator.

A System Acceptance Review (SAR) concludes and assures the System Acceptance phase.

9.2.2 Inputs

Title	Description
System Integration Test Plans, Procedures, & Checklists, including Acceptance Verification Procedures	All the necessary test plans that have been produced prior to commencing tests that detail what tests must be run, by whom and what evidence must be collected and documented
Closed ICDs	All closed ICDs from sub-system integration testing. The remaining ICDs will be for external system interfaces.
SRS and System Scenarios	The technical specification for SRL East and the scenarios that must be tested against to prove the procured system meets the specification and expected use cases
Sub-System Test Reports	Test reports documenting the outcomes of all sub-system integration tests
IRR assurance reports	The Integration Readiness Review assurance report that would highlight any integration risks that the Systems Integrator must be aware of before commencing testing
System Integration Test Schedule	A schedule of all the system integration tests to be conducted

9.2.3 SE Approach

9.2.3.1 Systems Integrator Activities

Key activities expected to be conducted are:

- System Integration Readiness Reviews
- Verification of the SRL System as a whole and all functions, performance parameters, functional interfaces
- System Integration Test Procedures/Acceptance Verification Procedures & Reports, Test Plans.

Key integration areas/interfaces to be Verified are:

- Integration of Rolling stock, CBTC, PSD, OHW, Traction Power in Static Mode and Dynamic Mode
- Controlling and Monitoring of rail systems, automatic, manual and emergency PA/PIDS announcements to selected stations
- Control and monitoring of Tunnel Ventilation System in Normal, Abnormal and Congestion modes within Tunnels with interfaces from Fire Systems, Station HVAC, Vertical Transport system
- Tunnel Services, Lighting, Cross Passage, Dynamic Signage, Evacuation
- SRL system as a whole is verified with all functions and performance tested
- Ensure all records provide an accurate representation of the system integration tests conducted and operation across agreed interfaces is achieved
- Verify requirements traceability and closure of SRS requirements as applicable
- Integrated SRL system functions and performs as expected
- Functional Configuration Audits completed and all issues resolved.

The Systems Integrator must conduct a Systems Acceptance Review following the conclusion of all system integration tests. Subsequently, the RTO, SRLA & the Systems Integrator will jointly judge whether the SRL East system is ready to commence trial running.

9.2.3.2 RTO Activities

To demonstrate operational readiness, the RTO will have to at least demonstrate:

- Standard Operating Procedures have been developed from Service Specification
- Competent staff are available and trained to conduct the standard operating procedures
- Operational risks have been assessed based upon testing reports from system integration activities
- Maintenance plans and staff rosters have been developed
- Engagement with stakeholders/end users to prepare them for SRL East launch
- Accreditation as Authorised Rail Transport Operator (ARTO) has been granted by ONRSR to the RTO.

Most of the activities can be undertaken ahead of the stage input's availability.

9.2.4 Engineering Assurance Outputs

Title
System Integration Test Report/Acceptance Verification Reports
Functional Configuration Audits
SRS VCRM
SAR Assurance Report
ORR Assurance Report

9.3 Operations Integration Stage (Trial Running & SRR)

9.3.1 Purpose

Once all program works have been completed, the SRL East system has been successfully integrated, all staff have been hired and trained, and all relevant information for assets has been collated and exchanged between SRLA and DTP, the SRL East system is ready to be trialled in operations before the service is ready to go live. This stage is to verify everything has been put in place and that operationally the RTO is ready to take control of SRL services and open up the line to the public (Day 1 services). This is the stage where most of the operations and maintenance needs are validated.

A Service Readiness Review (SRR) concludes and assures the Operations Integration Stage.

9.3.2 Inputs

Title	Description
SAR & ORR Assurance Reports	The reports of the pervious assurance reviews will identify any specific actions still to be closed out ahead of operational testing, or any advisory notices that the operator should be aware of.
Standard Operating Procedures	These Standard Operating Procedures detail how the operator expects to operate their railway in normal, degraded & emergency scenarios.
OCD/MCD & Scenarios	These are the original scenarios defined during the operations and maintenance concepts. These scenarios must be validated during operations testing prior to opening the railway.

Title	Description
Trial Running Test Plans/System Validation Procedures	The outcomes of these tests will inform the operator how the system is performing as an integrated set of systems and whether there are any specific safety conditions that need to be considered when undertaking operational tests.

9.3.3 SE Approach

9.3.3.1 RTO Activities

Key activities to be conducted are:

- Undertake Normal, Degraded & Emergency trials as defined in trial running plans and procedures

9.3.4 Engineering Assurance Outputs

Title
Trial Running Reports/System Validation Reports
Operations and maintenance Needs VCRMs
SRR Assurance Report

10 Project Handover & Completion Phase – Engineering Assurance Strategy

10.1 Project Handover & Completion Stage – Work Package Final Completions

10.1.1 Purpose

Before a Work Package can be deemed as complete, SRLA must ensure all documentation is completed, drawings are uploaded into a document management system, and any outstanding issues either mitigated or can be transferred, etc.

SRLA must verify all of this has been done and close out the work package a Final Completion Review (FCR).

Collection of completion evidence will be undertaken by the Completions, Integration & Commissioning Team.

The Final Completion Review (FCR) concludes this stage for each work package and assures the work package has practically completed.

The details for activities in this stage can be found in the SRLA Handover & Completions Management Plan.

10.2 Project Handover & Completion Stage – SRL East Completion

10.2.1 Purpose

The technical system has undergone the necessary level of operational testing/trials to ensure it is ready to be used in service, however there is more to completing a project than just delivering a working system. This stage focuses on measuring whether the project has been completed in line with the objectives and stakeholder needs set in the client requirements documents and eventually the intent of the BRD (once all stages of SRL have been delivered).

Collection of assurance evidence will be coordinated by the Engineering Assurance team. The evidence will be assessed against the System Assurance Case to judge whether SRL East meets the requirements set out in the CRD. They must also validate the stakeholder requirements and user needs.

It is then necessary to judge whether the outcomes of the business requirements document have been met for the whole of SRL. This phase focuses on measuring whether the project has been completed in line with the objectives and stakeholder needs set in the client requirements documents.

A Program Completion Review (PCR) concludes and assures the Program Completion phase.

10.2.2 Inputs

Title	Description
SRR Assurance Report	The assurance report produced for the Service Readiness Review gate. It will provide a program wide status prior to Day 1 operations commencing.
All Work Package FCR Assurance Reports	All FCR assurance reports detailing the status of each Work Package upon "project completion."
FRACAS	The FRACAS details any open actions with regards to rectifications that must be undertaken and closed out

10.2.3 SE Approach

10.2.3.1 Assess SRL East Performance

As soon as Day 1 operations commence, the collation and recording of service performance data will begin. The RTO will be held responsible for recording the data and providing requested service-related data to the Franchise Manager for assessment as per the Franchise Agreement.

The data gathered must be sufficient and specific enough so that SRLA can pass through the information to DTP to be used as evidence to determine whether the service performance parameters (and wider benefits) of the BRD are being met.

The Franchise will undertake user and service performance related tests/surveys to pass over focussed data to SRLA to be used to assure DTP that the CRD and BRD objectives are being upheld post Day 1.

10.2.3.2 Collate assurance evidence and provide to client

The Engineering Assurance Team is responsible to gather the outcome of all performance testing undertaken post Day 1 operations. They must assess the test results against the SRL East CRD and external stakeholder requirements and determine whether the requirements have been validated.

The EA Team must document the results of their analysis in the Program Completion Review assurance report. This PCR assurance report must be reviewed by the SRL ACC. Upon approval of the PCR, the assurance report will then be submitted to DTP as assurance evidence for the N7 review.

10.2.4 Engineering Assurance Outputs

Title
CRD, EPR, stakeholder needs VCRMs
Program Completion Review assurance report

11 Roles & Responsibilities

11.1 Pre Linewide Package Contract Award

11.1.1 SRLA

The Engineering Assurance team structure is shown below in Figure 4. While the Manager, Engineering Assurance reports to the Deputy Director, Engineering & Assurance, the Manager, Engineering Assurance interfaces with all other Deputy Directors within Rail & Infrastructure to ensure the SE processes captured within this document are being implemented appropriately across all areas of delivery and that the Engineering Assurance team members are supporting the Deputy Directors (and their direct reports) sufficiently.

For the purpose of this document, the ‘Engineering Assurance Team’ will consist of all roles that sit under (and including) the Manager, Engineering Assurance.

The Engineering Assurance Team’s main responsibility is to strategise, plan and implement a set of integrated processes & procedures (based upon SE best practice) that enable the clear specification of a functional railway system, based upon meeting user & business needs and program constraints. The Engineering Assurance Team must educate, train, and support other teams in the wider organisation to understand the need for these SE processes, provide the tools and knowledge to undertake such processes, with the overall outcome of improving SRLA’s maturity with regards to SE.

Creating the SEMP and its maintenance is the responsibility of the Manager, Engineering Assurance and it shall guide what all members of the Engineering Assurance Team will do.

Engineering Assurance Team

The Engineering Assurance team must:

1. Plan the SRLA engineering assurance team strategy and subsequent Systems Engineering and assurance strategies for the whole of SRL East & North works
2. Prepare the following management plans (as a minimum) in line with minimum requirements of the SRLA Systems Engineering Guideline document:
   - Systems Engineering Management Plan (this document)
   - Systems Assurance Management Plan
   - Requirements Management Plan

- Interface Management Plan
- System Architecture Management Plan
- Verification and Validation Management Plan
- System Safety Management Plan
- Reliability, Availability and Maintainability Management Plan
- Human Factors Integration Plan

3. Compose and manage the SRL East System definition and its configuration throughout lifecycle phases through implementing an engineering change control process defined in the Configuration Management Plan
4. Support Rail Operations Planning in defining 'Day In The Life Of' (DiTLO) studies for the expected normal, degraded and emergency scenarios of the SRL East System
5. Establish a User Needs Analysis process to elicit a set of Operator & Maintainer needs from the DiTLO workstream, to complement the CRD, the output of which must be appended to the OCD/MCD for review and verification from the future RTO
6. Establish a System Requirements Analysis process to establish a set of system capabilities based upon achieving Operator & Maintainer needs and stakeholder constraints. Once established, the SRA must be utilised by the Systems Integrator to elicit system functional and non-functional requirements to be captured in the System Requirements Specification
7. Establish a Sub-System or Work Package allocation process to enable the appropriate allocation of system requirements to particular work packages for use in PS&TRs
8. Establish an SRL East system architecture to function as the source of truth for all operational, functional, logical & physical analysis conducted by the organisation and its contractors
9. Procure and manage software tools for the following applications:
    a. Model Based Systems Engineering
    b. Requirements Management Database
    c. Configuration Management Database
    d. Formal documentation production/word processing from MBSE software
    e. Collation of Objective Quality Evidence and VCRMs during assurance reviews
10. Establish Requirements Endorsement processes and hold responsibility for the administration of a Requirements Management database.
11. Undertake specialist engineering analysis for all works conducted by SRLA during development, to ensure a safe and appropriately performing system is specified, that will be able to be operated, maintained and used. Specialist Engineering activities covers RAM, Human Factors, System Safety & EMC
12. Establish a Verification and Validation program and processes to ensure verification and validation of needs and requirements at all levels of the requirements hierarchy is assessed both during development, design & testing & commissioning
13. Establish an assurance board that is responsible for assessing and making determination on assurance artefacts that demonstrate compliance to requirements during gateway reviews
14. Maintain regular engagement with key internal stakeholders (Rail Safety Team, Commercial & Technical Interface Team, etc.) to communicate early identification of any issues unearthed during progressive assurance activities

15. Establish a Systems Integration strategy for SRL East and support its implementation throughout the organisation
16. Collaborate with the Package Directors for each Work Package to ensure they provide input to the initial migration plan and continue to update and track their packages against it during delivery
17. Collaborate with Program Controls to ensure Engineering Assurance activities are included within SRLA's master program
18. Identify Engineering & Assurance related risks and escalate them to the Deputy Director, Engineering & Assurance for acknowledgement and action.

Design Management Teams (Civil Infrastructure & Rail Systems)

In relation to the activities undertaken by the Engineering Assurance Team as described within this SEMP; the Design Management Teams must:

1. Provide subject matter expertise during the application of any operational, functional, logical or physical analysis undertaken by the Engineering Assurance team
2. Identify the correct SMEs to be owners of appropriate System Capabilities
3. Ensure any requirements developed by the Design Management Team have been done so in line with the requirements development processes defined by the Engineering Assurance Team
4. Undertake Design Verification using Verification and Validation (V&V) processes established by the Engineering Assurance Team
5. Support the Engineering Assurance Team in identifying, classifying and managing design safety risks & potential human factors issues
6. Produce technical reports to inform functional & non-functional requirements for the SRS such as Sustainability, Noise & Vibration, EMI/EMC, Earthing & Bonding, etc.
7. Support impact assessments of change requests
8. Establish a Standards baseline and manage its configuration in accordance with processes defined in the Configuration Management Plan
9. Inform the Engineering Assurance Team of any expected support as part of undertaking activities defined in the Design Management Plan.

Commercial & Technical Interface Team

In relation to the activities undertaken by the Engineering Assurance Team as described within this SEMP; the Commercial & Technical Interface Team must:

1. Collaborate in developing the Interface Management Plan to ensure both teams are aligned in what processes must be put in place for effective interface management
2. Adhere to the processes laid out in the Interface Management Plan, Requirements Management Plan & System Architecture Management Plan
3. Collaborate with the Principal Engineer – Systems Architecture to ensure that all technical interfaces are identified and recorded within the SRL East system architecture (at system & sub-system levels)
4. Produce Interface Definition Requirements for external & cross package interfaces, based upon information contained within the SRL East system architecture and to be referenced by PS&TRs

5. Undertake continual analysis of the Interface Register & Interface Matrix produced from the SRL East system architecture for updates, errors, and conflicts. Any changes to these artefacts must be communicated to the Principal Engineer – Systems Architecture
6. Coordinate the production and administration of PS&TR Parts E
7. Inform the Engineering Assurance Team of any expected support as part of undertaking commercial & technical interface activities.
8. Manage/support (either directly or indirectly) the engagement of external stakeholders to understand their needs and constraints for SRL East, and any interface requirements, should SRL East interface with their systems. This involves administering the stakeholder engagement process, developing needs and requirements (where necessary) based upon stakeholder inputs, identifying stakeholder risks, and eliciting needs, statements of work or interface requirements for use in the System and Sub-System Level Analysis
9. Collaboration with the Principal Engineer - Assurance in producing assurance artefacts for SRLA assurance reviews and DTP gateway reviews, with particular regard to stakeholder needs and interface definition
10. Notify the Principal Engineer – Assurance of any DTP N-Gate reviews and consult with regards to deliverability of assurance artefacts to support the review.

Configuration Management Team

In relation to the activities undertaken by the Engineering Assurance Team as described within this SEMP; the Configuration Management Team must:

1. Collaborate in developing the Configuration Management Plan with the Engineering Assurance Team
2. Support SRLA in identifying CIs and managing their configuration information within a CI Register
3. Facilitate the usage of the SRL Change Management Process in the Engineering Workflow Management tool (EWM) and monitor CIs produced by SRLA
4. Liaise with the Engineering Assurance Team in the procurement of configuration management database software/tools to enable streamlining of software or at the very least integration of different platforms (requirements and change management)
5. Report on the outcome of status accounting and audits of the configuration management database to highlight any configuration issues with SRLA CIs
6. Coordinate with the Engineering Assurance Team to maintain the integrity of requirements sets in DOORS, including minor revisions of requirement sets
7. Support the Engineering Assurance Team to produce and manage configuration baselines ahead of each gateway review (such as functional baseline, allocated baseline etc.).

Project Controls

In relation to the activities undertaken by the Engineering Assurance Team as described within this SEMP; the Project Controls Team must:

1. Manage all changes in accordance with the change management plan and procedure [21].

Digital Engineering/Asset Information Team

In relation to the activities undertaken by the Engineering Assurance Team as described within this SEMP; the Digital Engineering/Asset Information Team must:

1. Inform the Engineering Assurance Team of the Digital Engineering Strategy for SRLA and request any digital engineering needs from the team to support their activities and inform information sharing mechanisms between SRLA and delivery agencies
2. Provide the Engineering Assurance Team with a procured Digital Environment to share SRL East system definition information onto
3. Collaborate with the Engineering Assurance Team on developing the Asset Management Strategy for SRLA to ensure Project lifecycles models, SE lifecycle models and asset management lifecycle models are being considered and tailored to be complementary

Deputy Director – Engineering & Assurance

In relation to activities undertaken by the Engineering Assurance Team as described within this SEMP, the Deputy Director, Engineering & Assurance must:

1. Review all SRLA Engineering Assurance management plans and approve them for implementation
2. Acknowledge any engineering & assurance related risk items identified by the Engineering Assurance Team and action them accordingly
3. Review and approve/deny resource requests from the Engineering Assurance Team for AJM support and internal hiring of permanent SRLA staff
4. Support the Engineering Assurance Team in implementing the processes outlined in the SEMP (and its subordinate plans) by influencing and educating the other Deputy Directors, Directors & General Managers of SRLA
5. Provide direction to the Engineering Assurance Team to prioritise their activities to best benefit SRLA at any given time
6. Communicate top-down governance decisions to the Engineering Assurance Team that affect their products, processes and plans.

Director – Rail & Infrastructure Delivery

In relation to activities undertaken by the Engineering Assurance Team as described within this SEMP, the Director. Rail & Infrastructure Delivery must:

1. Endorse all SRLA Engineering Assurance management plans and approve them for implementation
2. Endorse the System Requirements Specification & System Definition Document
3. Escalate issues to the executive leadership team received from the Engineering Assurance Team or Head of interim Systems Integrator.

Deputy Director – Rail Operations

In relation to activities undertaken by the Engineering Assurance Team as described within this SEMP, the Deputy Director, Rail Operations must:

1. Endorse all Operational Analysis processes developed by the Engineering Assurance Team
2. Endorse and own the Operator & Maintenance needs developed by the Engineering Assurance Team
3. Sign off on the Validation evidence that Operator & Maintenance needs have been met by the System Requirements Specification

4. Collaborate with the RAM team to determine the operational performance and system performance to be specified for SRL East.

Deputy Director – Quality Assurance

1. Accept this management plan (and subordinate management plans) via the Plans & Procedures Committee and retain record of its configuration.
2. Audit delivery agencies against management plans specified within this document (as per PS&TR Part C requirements) and notify the Engineering Assurance Team where non-compliances have been identified.

Package Director – Linewide & Operations

In relation to the activities undertaken by the Engineering Assurance Team as described within this SEMP; the Package Director – Linewide & Operations must:

1. Support the application of the SE framework to produce the PS&TR for their specific contract and agree to provide Linewide & Operations resources to collaborate under the SE framework
2. Be accountable for the final output of the sub-system level analysis allocated to the linewide & operations work package
3. Take ownership of the information contained within the Linewide & Operations Sub-System Definition Document and report any proposed changes to the Engineering Assurance Team
4. Head up the interim Systems Integrator function and instruct its working groups to undertake tasks.
5. Develop a Testing & Commissioning Management Plan to define what processes must be put in place for effective testing and commissioning.
6. Manage an integrated program of works that results in the output of a linewide & operations PS&TR, its associated assurance evidences and other supporting documentation necessary to gain endorsement from DTF to award contracts
7. Raise system integration issues to the Director, Rail & Infrastructure Delivery for escalation
8. Participate in the Reference Design Stage Gate Review (RDSR) & Tender Design Stage Gate Review (TDSR) prior to contract award.

Package Director – Tunnels & Civils

In relation to the activities undertaken by the Engineering Assurance Team as described within this SEMP; the Package Director – Tunnels & Civil must:

1. Understand how the SE Framework interacts with their PS&TR and identify any issues with the current approach
2. Be accountable for the final allocation of sub-systems to the Tunnels & Civils work packages and its associated scope (that must be specified in the PS&TR)
3. Take ownership of the information contained within the Tunnels & Civils Sub-System Definition Documents and report any proposed changes to the Engineering Assurance Team
4. Work with the systems Integrator to ensure cross package interfaces are identified, specified and understood between package directors, as well as package allocation

5. Work with the Engineering Assurance Team to form an assurance case and other supporting documentation necessary to gain endorsement from DTF to award contracts
6. Work with the Engineering Assurance Team to produce a Tunnel Safety Case necessary to prove to ONRSR that SRLA has done all it can to procure a tunnel that will be safe SFAIRP
7. Raise interface/package allocation conflicts/issues to the interim systems Integrator for resolution or escalation
8. Participate in the Reference Design Stage Gate Review (RDSR) & Tender Design Stage Gate Review (TDSR) prior to contract award.

Package Director – Stations

In relation to the activities undertaken by the Engineering Assurance Team as described within this SEMP; the Package Director – Stations must:

1. Understand how the SE Framework interacts with their PS&TR and identify any issues with the current approach
2. Be accountable for the final output of the sub-system level analysis allocated to the station work packages
3. Take ownership of the information contained within the Stations Sub-System Definition Document and report any proposed changes to the Engineering Assurance Team
4. Work with the Systems Integrator to ensure cross package interfaces are identified, specified and understood between package directors, as well as package allocation
5. Is responsible for provision of each station work package assurance case and must coordinate with the Engineering Assurance Team to form an assurance case and other supporting documentation necessary to gain endorsement from DTF to award contracts
6. Raise interface/package allocation conflicts/issues to the interim Systems Integrator for resolution or escalation
7. Participate in the Reference Design Stage Gate Review (RDSR) & Tender Design Stage Gate Review (TDSR) prior to contract award.

11.1.2 Technical Advisor

Systems Engineering & Systems Assurance Specialists

The Technical Advisor (TA) must provide appropriately skilled Systems Engineering & Safety Assurance resources to support the SRLA Engineering Assurance Team to undertake tasks described in the Engineering Assurance management plans.

Whilst supporting the SRLA Engineering Assurance Team, the TA resource will perform tasks as scoped and monitored by the Manager, Engineering Assurance and/or one of the Manager's Principal Engineers.

Any resource put forward must be assessed as fit for purpose by the Manager, Engineering Assurance before being put forward for sign off by the Deputy Director, Engineering & Assurance.

Should the TA wish to alter any resource whilst they are performing tasks for SRLA, the TA's Project Manager must notify the Manager, Engineering Assurance prior to the resource being altered and provide immediate and appropriate replacement.

Design Leads (Subject Matter Experts)

The TA design leads will be requested to provide subject matter expertise for particular operational, functional, logical and physical analysis undertaken by the Engineering Assurance Team. This may be on a regular or infrequent basis but shall be conducted under approved Works Orders and through a resource forecast and agreement between SLRA & AJM, approved by the Deputy Director, Engineering & Assurance.

Design Authority

The TA must utilise a designated Design Authority accountable for signing off the technical works undertaken by the TA and attest to their safety and quality. SRLA must have an equivalent Design Authority to countersign the works done by the TA and attest to them being in line with the scope of works laid out in Work Orders.

11.2 Post Work Package Contract Award

11.2.1 SRLA

SRLA will oversee the systems engineering and assurance approaches being implemented by each Work Package.

This overseeing will be conducted as formal assurance reviews, as detailed in the SAMP.

Upon request from a Work Package, SRLA must provide guidance/advice/information following the established governance arrangements.

Any top down change requests proposed by SRLA that impact Work Packages, must be done so in accordance with the SRLA Configuration Management Plan [Ref. 6].

A core E&A capability of systems engineers, specialty engineers, requirements engineers, design coordinators, and technical SMEs will be centrally retained by SRLA to be able to give technical engineering advice and guidance to the Executive General Manager R&ID that is independent of the work packages.

11.2.1.1 Deputy Director – Engineering & Assurance

The Deputy Director – Engineering & Assurance is accountable for:

- Defining the process and evidence required to manage requirements.
- Ensuring SME participation in Design review workshops
- Scope of requirements and assurance processes for the packages.
- Reviewing compliance of assurance processes and associated evidence
- Review of delivery partner management plans
- Quality audits, co-ordinated with the Quality team
- Reviewing and endorsing developed rail safety artefacts that support safe decision-making
- Contribute as SME to risk assessments across all levels of the project (together with the SEM)
- Escalation point for matters pertaining to rail safety (together with the SEM)
- Technical Review of change for endorsement (including process definition, criteria and SME engagement) (together with the Principal Project Specialist Lead Design Interface)

- Defining different types of baselines relevant to SRLA CIs and how they are internally managed (together with the Configuration Manager)

Similar capability will be embedded within the Work Packages, however these resources will be coordinated and managed day to day by the respective Senior Engineering Managers. With specific regards to assurance, these embedded resources will support the formation of Work Package assurance reports that will be reviewed and assessed by the core E&A capability.

11.2.2 System Integrator

The System Integrator (SI) for the SRL East Project manages the delivery and implementation of the SRL East Railway System from specify phase through to system acceptance and handover phase of the project lifecycle. This includes integrating subsystems, maintaining the allocated systems from the SRL East architecture, managing the associated requirements and configuration baselines, and developing/monitoring the system integration strategy and stage definition.

In addition, the SI maintains the Line-wide integration program, coordinating activities in relation to SRL East Railway System integration across work packages to ensure alignment with the project goals. This involves managing change impacts, ensuring consistency in specialty engineering disciplines, and promoting cost-effective integration solutions. Regular progress meetings and reporting are conducted to monitor task status and address risks and delays.

Moreover, technical risk management is also a key responsibility, involving the identification, assessment, and mitigation of risks impacting the SRL East Railway System . The SI develops and implements risk management plans, monitors potential issues, and ensures appropriate mitigation and/or contingency measures are in place, contributing to the overall safety, functionality, and efficiency of the SRL East Railway System delivery.

Operational Integration, readiness for operations and maintenance, and digital integration also fall under the remit of the System Integrator.

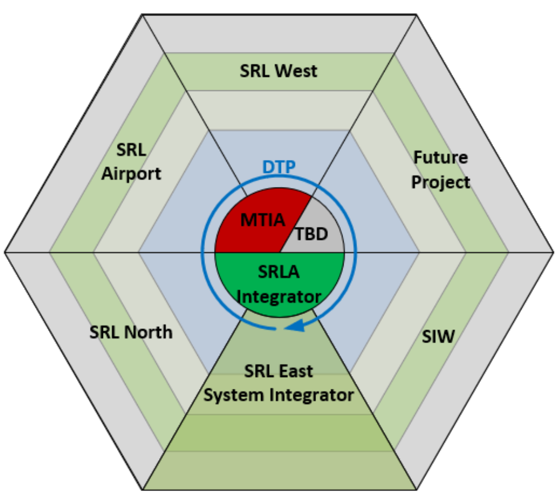


Figure 12 - Integration of SRL programs

As shown in Figure 12; SRLA must also provide interface information to the Systems Integrator regarding external interfaces SRL East has with other SRLA projects, such as SRL North and Station Interchange Works.

SRLA will be the interface with DTP and **Victorian Infrastructure Delivery Authority** on all other suburban rail loop projects, as specified by the SRL Business Requirements Document (BRD).

11.2.3 Work Package Roles & Responsibilities

Line Wide Work Package

The major responsibility of the Work Package is to undertake the works as defined in their PS&TR (and SRS in the case of the Linewide & Operations Work Package) and Commercial/Legal contracts.

The Line Wide Delivery Agency is to work collaboratively with SRLA and given the responsibility of being the "Systems Integrator" during delivery and the "Operator & Maintainer" for post day 1 operations of the SRL East Railway System .

The Line Wide Package Delivery Agency must ensure they employ a skilled workforce capable of undertaking the systems engineering processes. The systems engineering capabilities of the workforce will be judged as suitable by SRLA through submission and assessment of systems engineering related management plans as well as demonstrated competency of individuals in line with the competency management framework.

Stations Work Packages

The major responsibility of the Work Package is to undertake the works as defined in their PS&TR and Commercial/Legal contracts.

It is expected that any stations packages will:

1. Review and acknowledge the program lifecycle and its associated gateway reviews being enforced by SRLA
2. As a minimum, provide the expected assurance artefacts in line with the Project Assurance Guideline prior to each gateway or assurance review
3. Identify and notify SRLA of any risks or issues with regards to any SE discipline defined within any received documents
4. Identify and notify SRLA of any change requests to be assessed that has impact on the Site Prep Package meeting one or many of their PS&TR requirements
5. Develop a requirements management plan and a V&V management plan (or equivalent) to inform SRLA how they are going to manage the PS&TR and its associated V&V activities.
6. Work collaboratively with the System Integrator to apply and achieve consistent, efficient SE outcomes that support project objectives

Tunnels & Civils Work Packages

The major responsibility of the Work Package is to undertake the works as defined in their PS&TR and Commercial/Legal contracts.

It is expected that any tunnels & civils packages will:

1. Review and acknowledge the program lifecycle and its associated gateway reviews being enforced by SRLA
2. As a minimum, provide the expected assurance artefacts in line with the Project Assurance Guideline prior to each gateway or assurance review

3. Identify and notify SRLA of any risks or issues with regards to any SE discipline defined within any received documents
4. Identify and notify SRLA of any change requests to be assessed that has impact on the Tunnels & Civil work package meeting one or many of their PS&TR requirements
5. Develop a requirements management plan and a V&V management plan (or equivalent) to inform SRLA how they are going to manage the PS&TR and its associated V&V activities.
6. Work collaboratively with the System Integrator to apply and achieve consistent, efficient SE outcomes that support project objectives

Initial & Early Works Packages

The major responsibility of the Work Package is to undertake the works as defined in their PS&TR and Commercial/Legal contracts.

It is expected that any site preparation packages will:

1. Review and acknowledge the program lifecycle and its associated gateway reviews being enforced by SRLA
2. As a minimum, provide the expected assurance artefacts in line with the Project Assurance Guideline prior to each gateway or assurance review
3. Identify and notify SRLA of any risks or issues with regards to any SE discipline defined within this document
4. Identify and notify SRLA of any change requests to be assessed impacting on the Site Prep Package meeting one or many of their PS&TR requirements
5. Develop a requirements management plan and a V&V management plan (or equivalent) to inform SRLA how they are going to manage the PS&TR and its associated V&V activities.

11.3 Governance Arrangements

This management plan must follow the governance arrangements as specified in the SRLA Governance Management Framework [20]

Concerning decision making and establishing an overall authority, DTP, SRLA and its engaged Delivery Agents will be expected to work using the following breakdown of scope and responsibility for key Configurable Items shown in Figure 10.

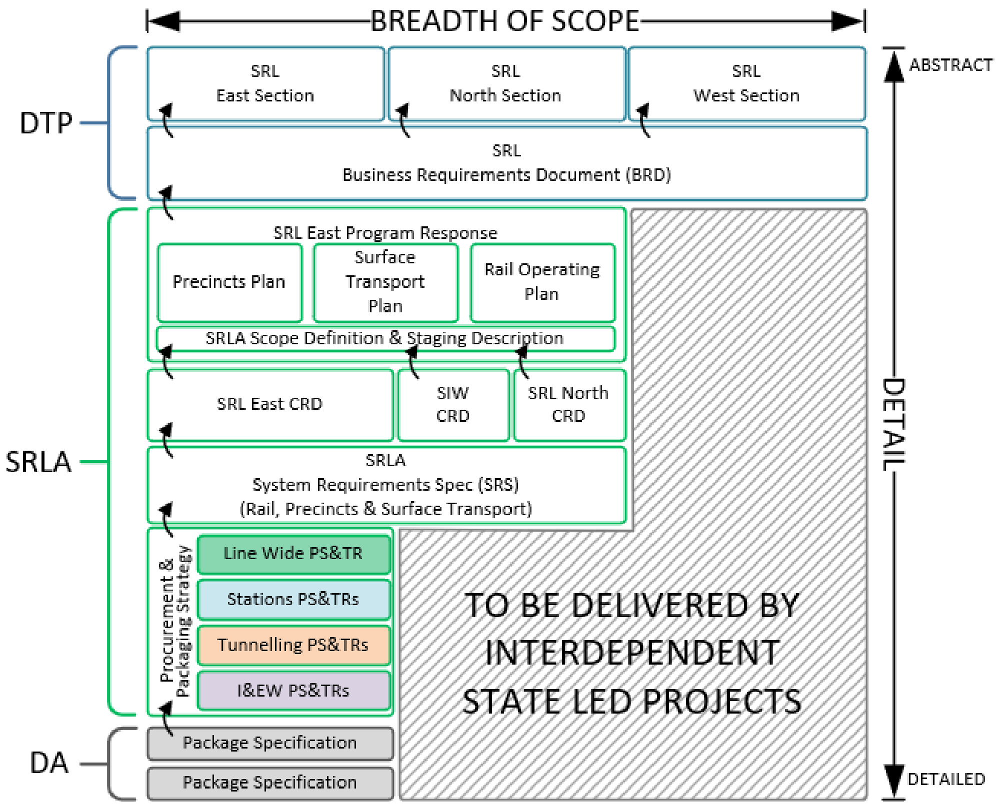


Figure 13 Break-down of scope and responsibility of key Configurable Items

This gives rise to different levels of authority for different organisations with regards to decisions being made and the impacts these decisions have on the key CIs.

The Suburban Rail Loop Act 2021 empowers SRLA to determine the best approach on how to stage, design and deliver the Eastern & Northern sections of the SRL Program (as deposed onto them by the BRD).

It is SRLA's responsibility to ensure that the SRL East Program Response is endorsed by DTP and DTF and that it satisfies the BRD and remains within the investment case.

This is to be met by SRLA providing timely evidence to DTP and DTF at key program stages, as laid out in the Network Integrity Assurance Plan and the Investment Management Lifecycle (described further in this document).

A Project Steering Committee (PSC) has been established to act as the highest level of governing authority for the project. Therefore, any decision or change being proposed by SRLA or Work Packages that impacts CIs that sit above the SRL East Program Response; must be escalated to the relevant DTP Governance Body for endorsement/rejection.

Should the change impact the ability for the SRL East Program Response to satisfy the BRD, the decision or change must be escalated to SRL PSC as described above.

Systems Engineering Management Plan

Whilst being responsible for satisfying upwards, SRLA is also responsible for managing a set of contracts/packages that should progressively design, construct, test and integrate the SRL East System.

To make this happen efficiently a Systems Integrator will be procured/established to govern the design, construction, testing, integrating and handover stages across each work package as part of SRL East.

SRLA retains the role as Client whilst the Systems Integrator must ensure successful implementation of a running railway for SRL East. SRLA in its role as Client must therefore ensure that SRL East is:

- Supported by a feasible and safe technical system solution and O&M concept
- Future proofed for future operations with SRL North
- Has its external interfaces defined that support future integration with other planned work packages
- Has suitable processes in place to continually assure DTP that the SRL East objectives will be met
- Has suitable processes in place to ensure integrity of the extant Public Transport Network is maintained throughout delivery of the SRL East works
- Meets the performance requirements laid out in the Environmental Effects Statement
- Meets external stakeholder requirements for which SRLA is responsible for validating.

Collating, assessing, and reporting on technical assurance evidence for the above is the responsibility of the Engineering & Assurance branch of SRLA.

SRLA will progressively assure SRL East using the following breakdown of assurance cases:

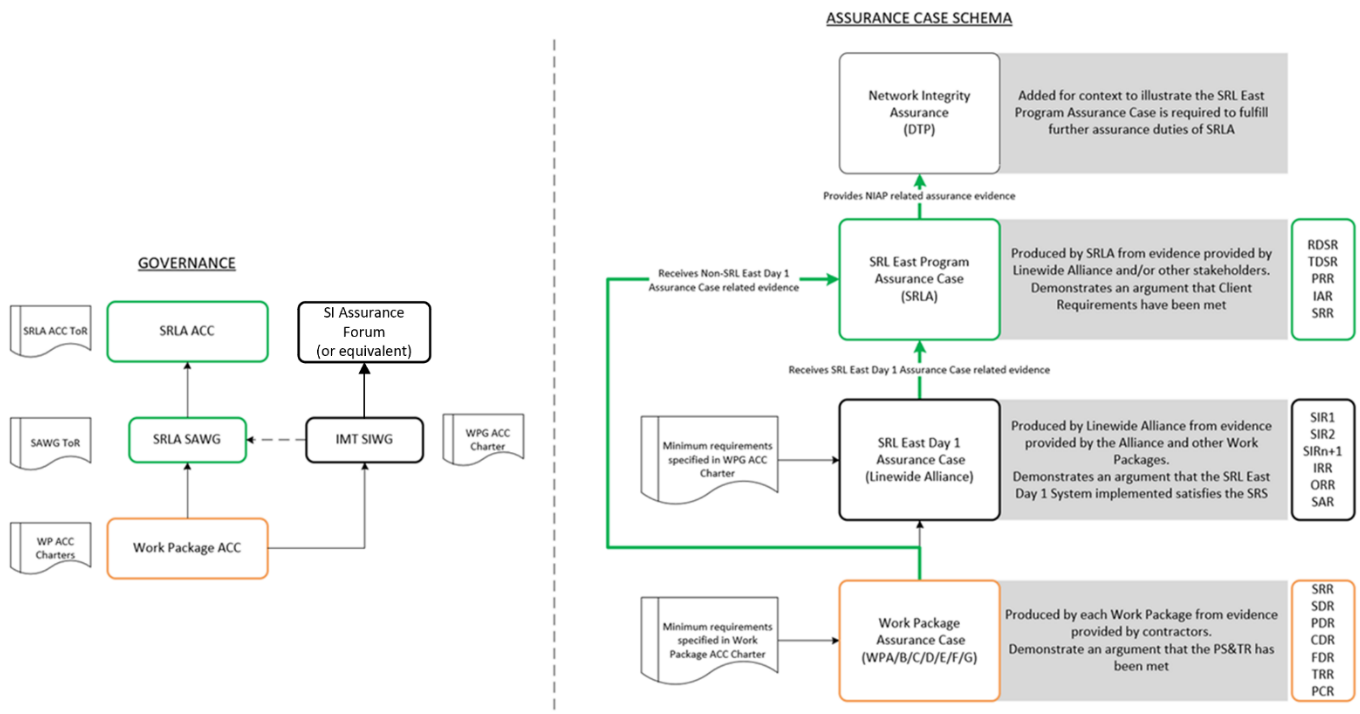


Figure 14 Assurance Case

For further detail on how the assurance cases are structure and governed, refer to the Systems Assurance Management Plan [Ref.9].

12 Quality Management and Continuous Improvement

This SEMP is to be reviewed by the Manager, Engineering Assurance on an 18 monthly basis to assess its:

- Satisfaction of the Rail & Infrastructure Management Plan
- Applicability & relevance to SRLA practices
- Definition of responsible parties/roles, etc.

For further details on the quality assurance (and expected gradual maturity of this plan) refer to the Quality Management Plan [8].

This SEMP is expected to be updated prior to any procurement decisions to ensure there is sufficient directive included to guide external parties on what Systems Engineering & assurance activities are expected of them.

13 Glossary

Abbreviation	Definition
ACC	Assurance Configuration Committee
BRD	Business Requirements Document
CI	Configuration Item
CMDB	Configuration Management Database
CMP	Configuration Management Plan
CONOPS	Concept of Operations
CRD	Client Requirements Document
DiTLO	Day in The Life Of
DoT	Department of Transport (DTP after 2022)
DTF	Department of Treasure & Finance
DTP	Department of Transport and Planning (formerly DoT)
EMC/EMI	Electro-Magnetic Compatibility/Interference
EOI	Expression of Interest
FCA	Functional Configuration Audit
FMECA	Failure Modes Effects and Criticality Analysis
FRACAS	Failure Reporting, Analysis, and Corrective Action System
FTA	Fault Tree Analysis
GSN	Goal Structured Notation
HF	Human Factors
HFIL	Human Factors Issues Log
HFIP	Human Factors Integration Plan
ICD	Interface Control Document
IEW	Initial & Early Works
ILM	Investment Logic Map
INCOSE	International Council of Systems Engineers

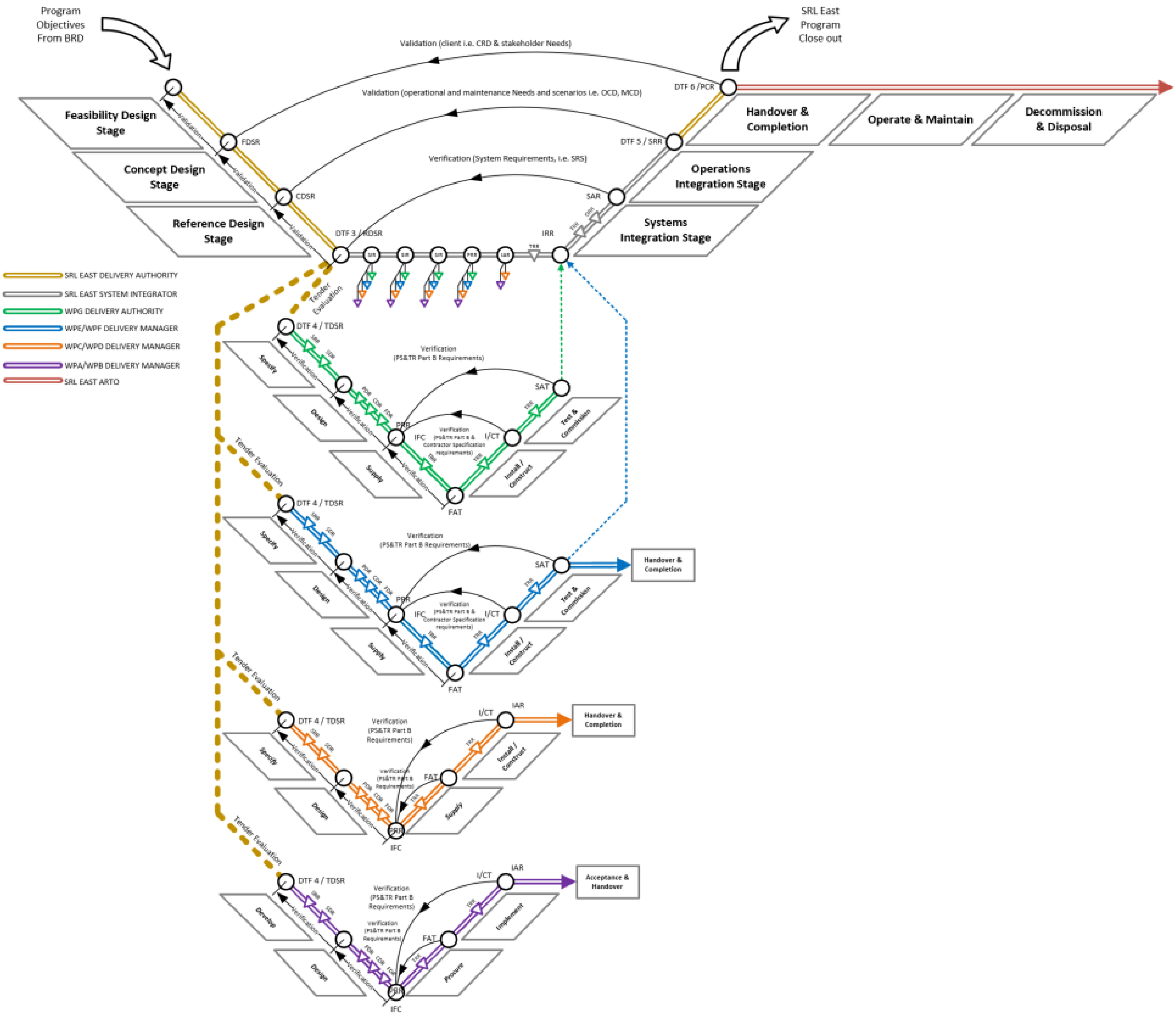
Abbreviation	Definition
MCD	Maintenance Concept Document
NTS	Network Technical Standard
OCD	Operations Concept Document
PCA	Physical Configuration Audit
PHA	Preliminary Hazard Analysis
PS&TR	Project Scope & Technical Requirements
RAM	Reliability, Availability, Maintainability
RIM	Rail Infrastructure Manager
RFP	Request for Proposal
RFT	Request for Tender
RMP	Requirements Management Plan
ROP	Rail Operating Plan
RTO	Rail Transport Operator
SAMP	Systems Assurance Management Plan
SEMP	Systems Engineering Management Plan
SRL	Suburban Rail Loop
SRLA	Suburban Rail Loop Authority
SRS	System Requirements Specification
SoS	System of Systems
TVS	Tunnel Ventilation System
V&V	Verification and Validation
WLC	Whole of Life Cost

14 Related Documents

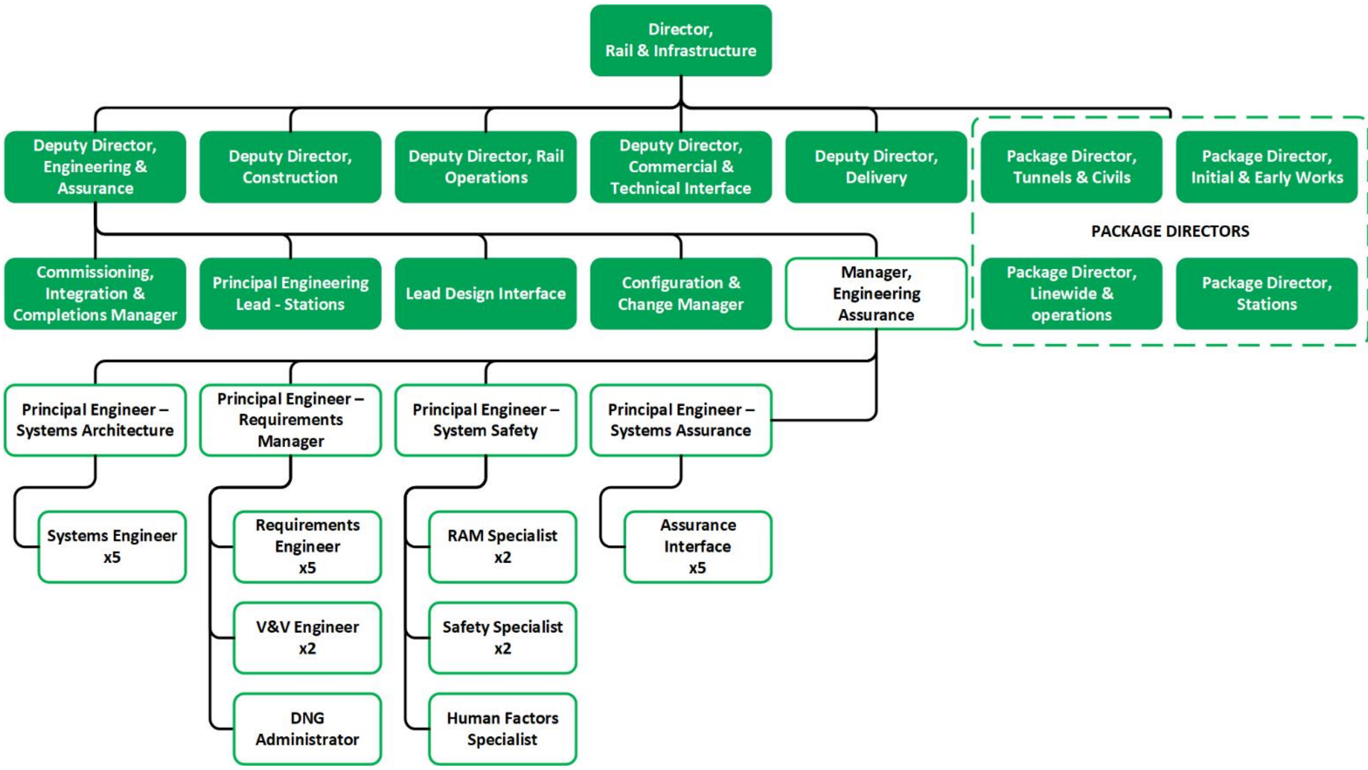
Source	
1	[1] DTF Investment Lifecycle (URL), section 7
2	[2] DoT Network Integrity Assurance Plan – Suburban Rail Loop – East and North – 18 October 2021
3	[3] ISO/IEC/IEEE 15288:2015 (URL), section 7.1
4	[4] Engineering & Design Management Plan (SRL-PWD-SRL-NAP-MPL-XDM-AIM-0001), section 7.1
5	[5] SRLA Procurement & Packaging Strategy
6	[6] SRLA Configuration Management Plan (SRL-PWD-SRL-NAP-MPL-XSE-AIM-0001), section 7.2
7	[7] SRL Stage One Scope Summary Document (TBD), section 9.1.3.4
8	[8] SRLA Quality Management Plan (SRL-PWD-SRL-NAP-MPL-XQA-AIM-0001), section 16
9	[9] SRLA Systems Assurance Management Plan (SRL-PWD-SRL-NAP-MPL-XSE-AIM-0003)
10	[10] SRLA Requirements Management Plan (SRL-PWD-SRL-NAP-MPL-XSE-AIM-0008)
11	[11] SRLA Architecture Management Plan
12	[12] SRLA Interface Management Plan (SRL-PWD-SRL-NAP-MPL-XIF-AIM-0001)
13	[13] Integration Management Team Terms of Reference
14	[14] OAWG Terms of Reference
15	[15] RWG Terms of Reference
16	[16] Not Used
17	[17] ARCADIA Method (URL)
18	[18] SRLA Completions & Handover Management Plan (SRL-PWD-SRL-NAP-MPL-XCP-AIM-0001)
19	[19] SRLA Testing & Commissioning Management Plan (SRL-PWD-SRL-NAP-MPL-XCP-AIM-0002)
20	[20] SRLA Governance Management Plan (SRL-PWD-SRL-NAP-MPL-XPR-AIM-0002)
21	[21] SRLA Change Management Procedure (SRL-PWD-SRL-NAP-PRC-XCM-AIM-0001

15 Appendix

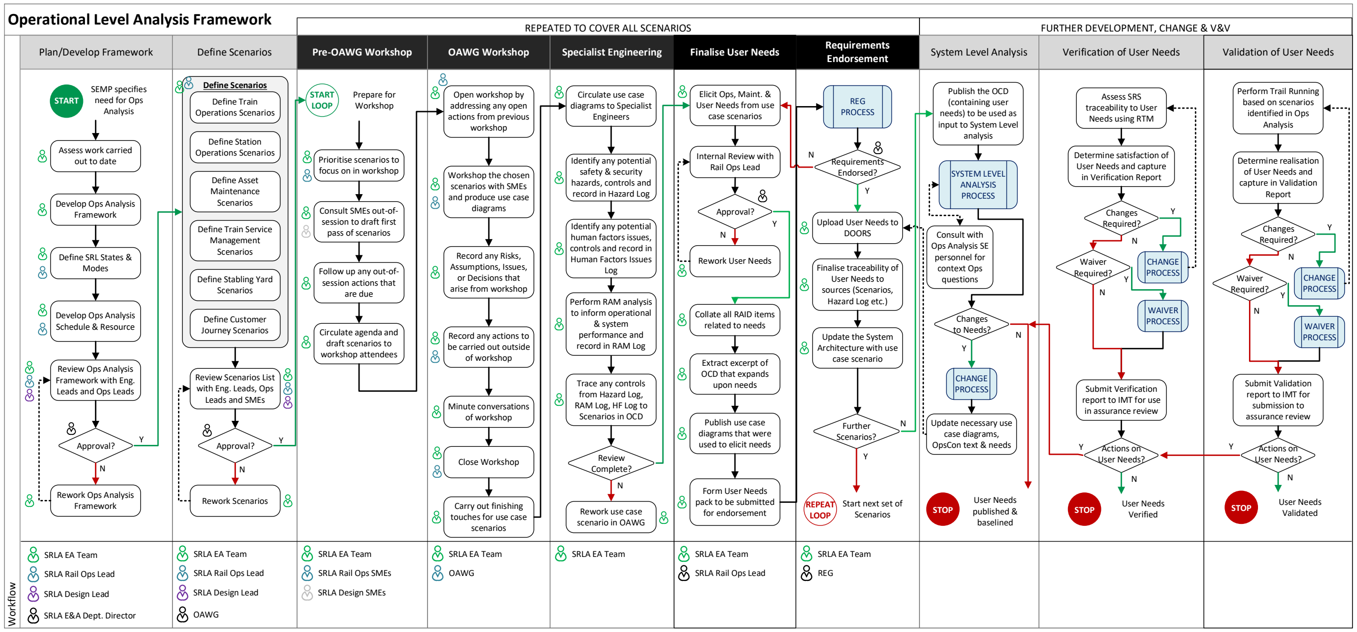
15.1 Appendix A - Program & Project Lifecycles



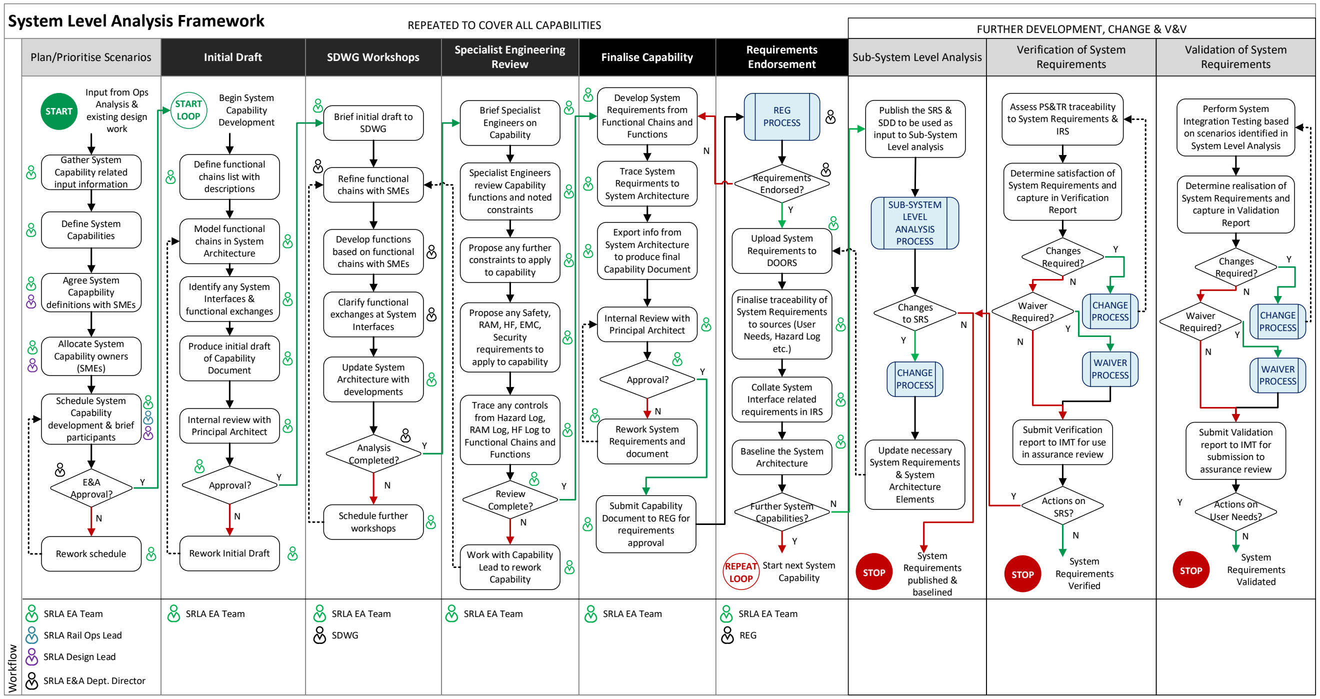
15.2 Appendix B - SRLA Engineering & Assurance Team



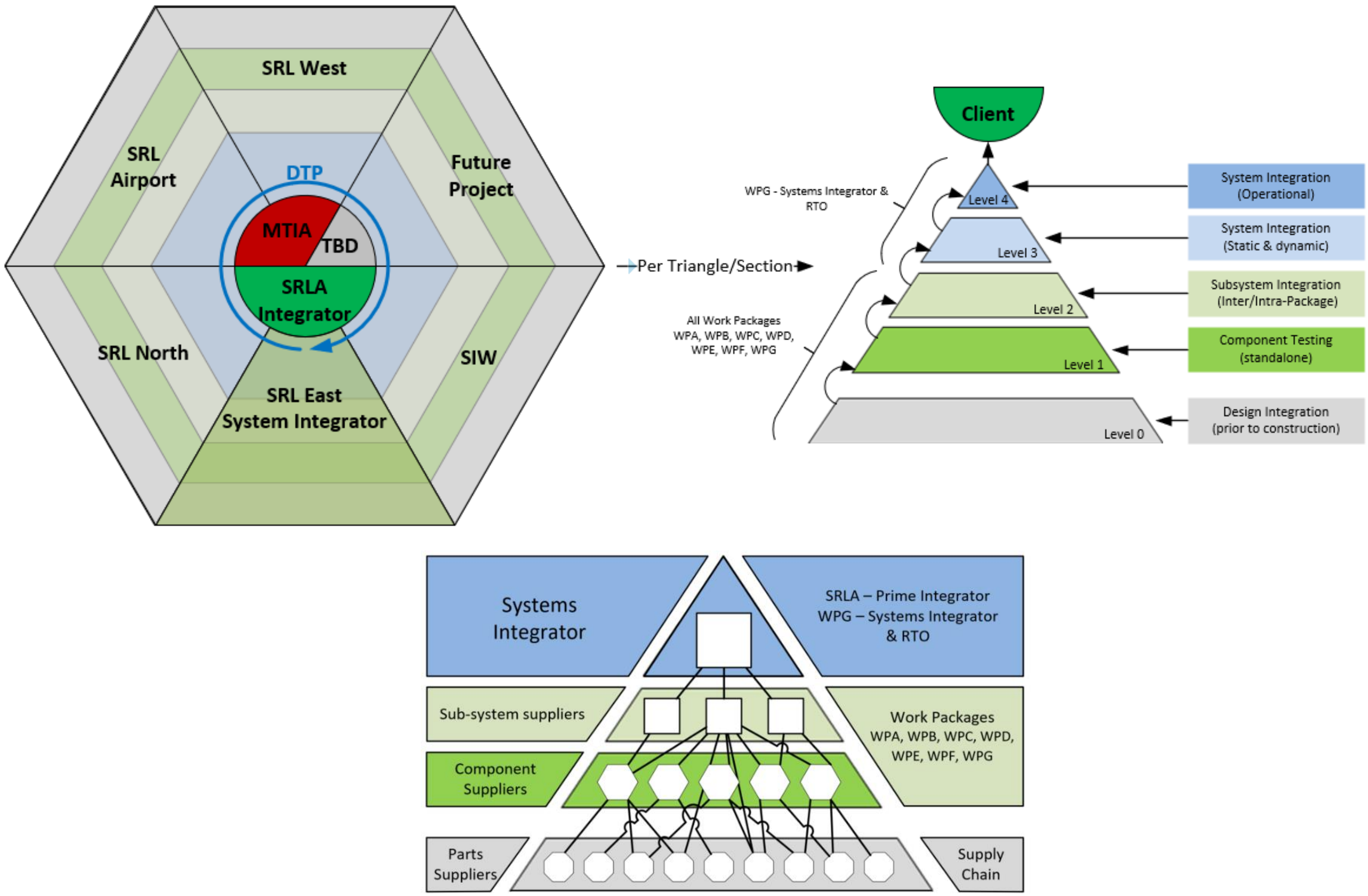
15.3 Appendix C - Operational Level Analysis Framework



15.4 Appendix D – System Level Analysis Framework



15.5 Appendix E – Integration Approach



15.6 Appendix F – SEMP Subordinate Plans

This appendix presents the vision, goals and key principles for each Engineering Assurance discipline. Each section will all be structured in the following way:

- The Inputs – the documentation that has been prepared to date and the relevant national or industry standards or guidance documents
- The Process – an outline of the process focusing on the key outcomes to be provided for each stage in Pathways lifecycle process.
- The Outputs – a list of the main products and program outcomes for each stage in the SRL lifecycle.

The strategies outlined in the SEMP will provide the guide for the more detailed 'management plans' that will be produced and maintained by the appropriate discipline lead (Figure 4).

It is the responsibility of the Engineering Assurance Manager to ensure the SEMP and its subordinate management plans align with the vision, goals and key principles of the SRLA Program.

The Engineering Assurance Manager will be responsible for 'Approving' the management plans.

The Deputy Director of Engineering Assurance is responsible for 'Endorsing' the management plans for implementation.

15.7 Appendix G – Requirements Management

15.7.1 Inputs

Title	Description
Business Requirements Document	The purpose of this business requirements document is to provide a clear specification of the envelope within which the SRLA project must deliver, and to what level of service/performance.
Scope Definition & Staging Document	This document gives high level details on the planned 'migration' of the SRLA project to gradually design, build and integrate the system. This will inform the staging of CRDs and provide prioritisation for any requirements work.
Rail Operations Plan	The Rail Operations Plan provides the SRLA with a concept of how the 'future' railway is predicted to be operated and maintained. The operational activities elicited from this concept document give rise to a set of functions that the SRLA solution will need to perform.
Key Decisions Register	The Key Decision Register provides SRLA with constraints that must be factored into any requirements analysis work.
Preliminary Hazard Analysis	Any preliminary hazard analysis undertaken during feasibility and concept work for the SRL must feed into any requirements analysis work to be developed further and incorporated into the SRLA solution.
INCOSE Guide for writing requirements	This guide has been developed by the International Council of Systems Engineers (INCOSE) to give clear rules on how to write 'good' requirements that should be adhered to when developing requirements at SRLA.
INCOSE Systems Engineering Handbook v4	The INCOSE SE Handbook offers up guidance for undertaking different levels of requirements analysis and eliciting. Chapter 4 covers general

Title	Description
Chapter 4 – Technical Processes	technical processes that should be tailored and expanded further in the Requirements Management Plan.

15.7.2 Process

During the Feasibility Stage, DTP has provided SRLA with a draft issue of the business requirements stating how DTP needs SRL to perform.

It is SRLA's responsibility to demonstrate, throughout the project lifecycle, that the business requirements are being delivered by the SRLA solution. To achieve this, SRLA will implement a robust requirements management process and provide dedicated resources within the Engineering & Assurance Team to manage SRLA's requirements.

During the Development and Delivery phases of the SRL Program the four primary objectives for requirements management are:

- To plan, develop, document and implement the requirements management process
- To select, set up, administer an appropriate requirements management tool, and ensure the necessary team members are given access and training commensurate to the individual's role
- To prepare a Railway-level System Requirements set and demonstrate to DTP's satisfaction that the elicited requirements are compliant with the business requirements
- To support the procurement process to package the Railway-level System Requirements into multiple specifications to define what part of the system each package must deliver.

15.7.3 Requirements Management Planning

The Requirements Manager will develop a Requirements Management Plan (RMP) for SRLA and maintain this plan for the duration of the program.

The RMP will describe how the principles of this strategy will be realised through the life of the project.

The RMP shall tailor the technical and management processes outlined in the INCOSE SE Handbook to ensure requirements practices are in keeping with industry standards.

In addition to producing the RMP, the Requirements Manager shall liaise with the Configuration Manager to ensure that an appropriate change control process is established and that the mechanism for agreed technical changes to be instructed onto SRLA is defined.

The Requirements Manager will ensure all parties involved in the requirements process (including DTP and delivery agencies) have access to and are adequately briefed on the RMP and that any subsequent updates or amendments to the plan are cascaded appropriately in a timely manner.

The Requirements Manager will also be responsible for ensuring that the requirement management tasks are included in any Engineering Assurance schedules/plans and for managing the timely delivery of these activities.

The Requirements Manager will monitor and report progress of the development of the requirement set by maintaining a register of the status, condition and baseline of the full requirements schema.

Assurance of the requirement management process will be undertaken through a combination of internal self-assessment and external peer review:

- The Requirements Manager will undertake regular checks on the project teams and delivery agencies to ensure their application of, and compliance to, the RMP process. Where shortcomings are identified, the

Requirements Manager and the associated System Engineer (from project teams of the delivery agency) shall develop and implement a remedial action plan

- The SRLA Requirements Manager shall invite their equivalents from other projects (that are jointly developing or delivering the SRL) and engage with the DTP Systems Engineering team to undertake peer-review assessments and to disseminate lessons learnt/good practice experience within the wider transport portfolio, effectively creating an open community of practice for requirements management.

15.7.4 Requirements Tool Selection, Setup and Administration

SRLA shall implement an appropriate software solution for recording and managing all requirements, assumptions, verification evidence and non-compliances.

The Requirements Manager will be responsible for evaluating the available requirement management tools and choosing an appropriate solution for SRLA.

It is recognised that the IBM Requirements Engineering DOORS Next (DNG) is the industry standard tool for requirements management and likely to be the preferred solution, but the Requirements Manager shall still ensure that that the tool is capable of meeting the current and future needs of SRLA.

The Requirements Manager is responsible for the setup of the requirements management tool and organising of the data/requirements structure in accordance with the requirement schema defined in the RMP.

Access to the requirements management tool shall be made as widely available to the SRLA as possible, but it is recognised that licensing and other availability limitations may apply. Regardless, the latest baseline of the requirements set shall be made available to all SRLA team members.

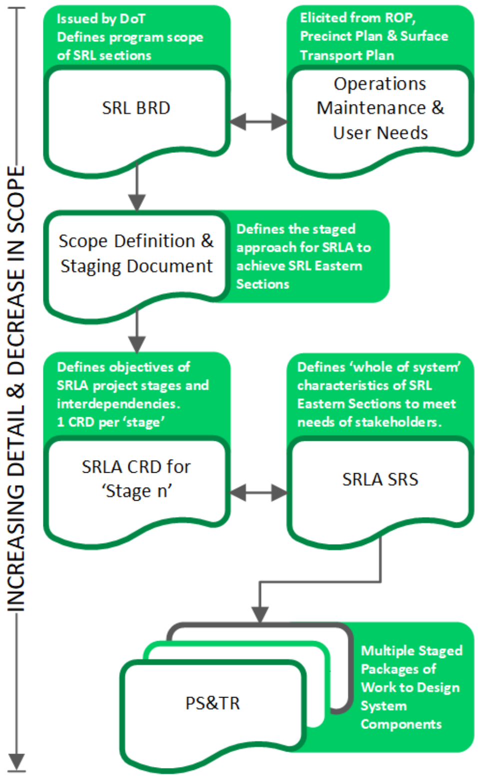
The Requirements Manager will establish appropriate access rights and permissions for staff that need to work in the tooled environment and to mitigate the risk of unauthorised changes and errors occurring.

The Requirements Manager shall provide training to all staff who have access to the requirements management tool and on-going support and advice where necessary.

15.7.5 Railway-Level System Requirements Set

The Requirements Manager, in consultation with the Engineering & Assurance team and DTP, will design a requirements hierarchy structure (the requirements ‘Schema’) to meet the needs of the SRL program.

The Requirement Schema shall tailor the proposed structure documented in the business requirements document, which is:



1.1.1.1. Client Requirements Document (CRD first developed during the concept stage of a project) – Defines the User requirements and constraint that SRLA must deliver and acts as the 'contract' between the SRLA Project and the SRL Program.

1.1.1.2. System Requirements Specification (SRS first developed in design stage of a project) – Defines the technical requirements that need to be delivered by the 'whole' system to satisfy the User Requirements and constraints.

1.1.1.3. Project Scope & Technical Requirements (PS&TR first developed in tender stage of a project) – The PS&TR is to be prepared at the single system, asset or service level and will provide the detailed technical requirements, acceptance criteria, constraints and parameters for the asset to be procured. It must also provide any 'statement of works' (SoW) requirements that instruct a delivery agency to 'do something' that will not be satisfied by the implemented solution. An example of this may be for the delivery agency to produce a set of management plans to show how they will meet the minimum expectations of SRLA's management plans.

1.1.1.4. The Requirements Manager is free to set up any other supporting sets or sources of requirements as they deem fit to best support the Engineering & Assurance activities undertaken on SRLA.

During the design and tender stages, SRLA shall develop the System-level and PS&TR-level requirements in parallel. This will be undertaken through elicitation of requirements from the 'source' documents such as the CRD, ROP, Assumptions and hazard registers. A forum shall be established (either standalone or included as part of the Terms of Reference of an existing forum) through which requirement proposals can be reviewed and agreed while the requirement set is being developed.

The development of the System-level requirements set shall be considerate of the phasing of the works so that the partial or incremental delivery of functionality is planned for. This will result in specific requirements for each migration state (as per the Scoping Definition & Staging Document) and unique requirements that exist for each phase.

The Engineering & Assurance Manager will devolve responsibility for production of the PS&TR-level requirements to the appropriate requirements engineers.

The Requirements Manager is responsible for showing traceability between all levels of requirement contained within the SRLA requirements management tool and must produce requirements traceability matrices and reports to communicate coverage and maturity of requirements sets.

15.7.6 Support procurement and packaging of Requirements

Due to the lawful and contractual nature of the procurement process, any requirements specified within these contracts must be clear, but more importantly a 'complete' set. The opportunity to negotiate and add (or remove) requirements into contracts is usually very slim, so it is important to set the requirements correctly in the first instance.

The Requirements Manager is responsible for supporting the procurement teams in packaging requirements appropriately for procurement purposes. This is expected to be in the form of PS&TRs.

The Requirements Manager shall work with the Configuration Manager to ensure there are clear baselines created of each requirements set delivered to procurement so that any suggested changes can be accurately tracked and communicated to all parties affected.

15.7.7 Outputs

Title	Description
Requirements Management Plan	The management plan that provides detail on how requirements management is undertaken at SRLA in line with the SEMP.
Requirements Management Tool (selected and setup)	The tool that SRLA is to use for managing requirements is set up, in use and being administered by the Engineering & Assurance team.
Requirements Management input to Change Control Process	The requirements manager has collaborated with the configuration manager to ensure there is sufficient controls and process with regards to managing change to individual or sets of requirements and their associated baselines.
Client Requirements Document for SRLA	The high level set of requirements that translates the BRD from DTP into a SRLA specific set of stakeholder type requirements.
Systems Requirements Specification for SRLA	The set of requirements that define the performance characteristics of the SRLA solution.
Project Scope & Technical Requirements for each package of works	Allocated requirements from the SRS appropriately packaged to define what technical solution a delivery agency must deliver and any process requirements they must follow.
Requirements Traceability Matrices	Exports of the traceability between requirements sets to support satisfaction arguments.
Requirements Management Reports	Any reports that need to be published to convey the maturity, status, and coverage of requirements sets.

15.8 Appendix H - Systems Architecture Management

15.8.1 Inputs

Title	Description
The Victorian Transport System Architecture – Metamodel Definition	This DTP document sets the vocabulary to be used for any architecture development undertaken by SRLA to ensure that future integration of SRLA & DTP architectures can be made.
Victorian Digital Asset Strategy	This OPV document offers guidance on implementing digital engineering on high value and high-risk projects. It is necessary to understand the intent of the digital engineering effort of SRLA and ensure any architecture work integrates well with any other digital engineering workstreams.
Rail Operations Plan	The Rail Operations Plan provides the SRLA with a concept of how the 'future' railway is predicted to be operated and maintained. The operational activities elicited from this concept document give rise to a set of functions that the SRLA solution will need to perform and should be captured within the system architecture.
Scope Definition & Staging Document	This document gives high level details on the planned 'migration' of the SRLA project to gradually design, build and integrate the system. This will inform the staging of system components and provide prioritisation for any architecture work.
Key Decisions Register	The Key Decision Register provides SRLA with constraints that must be factored into any architecture work.
INCOSE Systems Engineering Handbook v4 Chapter 4 – Technical Processes	The INCOSE SE Handbook offers up guidance for undertaking different levels of systems analysis and architecture development. Chapter 4 covers general technical processes that should be tailored and expanded further in the Architecture Management Plan.

15.8.2 Process

During the Feasibility Stage, initial system architectures were developed, and a rich-picture view of the systems involved in SRL's final state was created to show an overview of what the final railway would look like once all the upgrades were completed.

It is the program's responsibility to describe the status of the design in a common definition for all stakeholders, throughout the lifetime of the program and particularly during the development & design phases. To achieve this, the program will implement a robust plan for the creation of system description and architecture drawings describing the nature of the system design through the migration stages of SRL (as per the Scoping Definition & Staging Document.

During the development & design phase, the primary objectives for System Definition and Architectures are:

- Creation of an Architecture Management Plan
- Defining the type of views for which architectures are to be created and the configuration points in the program for the different railway states to be captured
- Creation of the defined architectures using the existing understanding of the system design.

15.8.3 Architecture Management Planning

The Systems Engineering Manager is responsible for the development of System Architectures and associated descriptions for the program. An Architecture Management Plan will be created to explain the detail of how the System Architectures and descriptions are to be created, managed and communicated.

The Architecture Management Plan will continue to be updated as the program progresses, but at this point it will capture the major activities to be performed in the development & design phases of SRL. Methodology and tools will be described in the plan to facilitate a common approach to developing and designing the SRLA solution from a configuration managed single source of truth.

The plan must align the architectures and description documents with the configuration points identified in the Scoping Definition & Staging Document, so that the architectures can be used to verify the works necessary to achieve each of the configuration stages.

As a minimum, the Architecture Management Plan must define how to undertake the following:

- Stakeholder/Operational Needs Analysis
- System/Functional Analysis
- Logical Architecture Design
- Physical Architecture Design
- Contracts Allocation

This will facilitate SRLA to gradually and logically breakdown SRL into its smaller components, attribute them requirements, allocate the design and construction of those components into contracts, validate lower level designs against and eventually show how the implemented solution has been derived from and meets the needs of the program stakeholders.

15.8.4 View Definition

System Architecture is System Engineering toolset, whereby a system is represented in terms of its fundamental structure.

The process of system architecting typically involves examining a system from various different contextual viewpoints, breaking the system into its fundamental component parts and then creating a model of these ‘building blocks’ showing their dependencies.

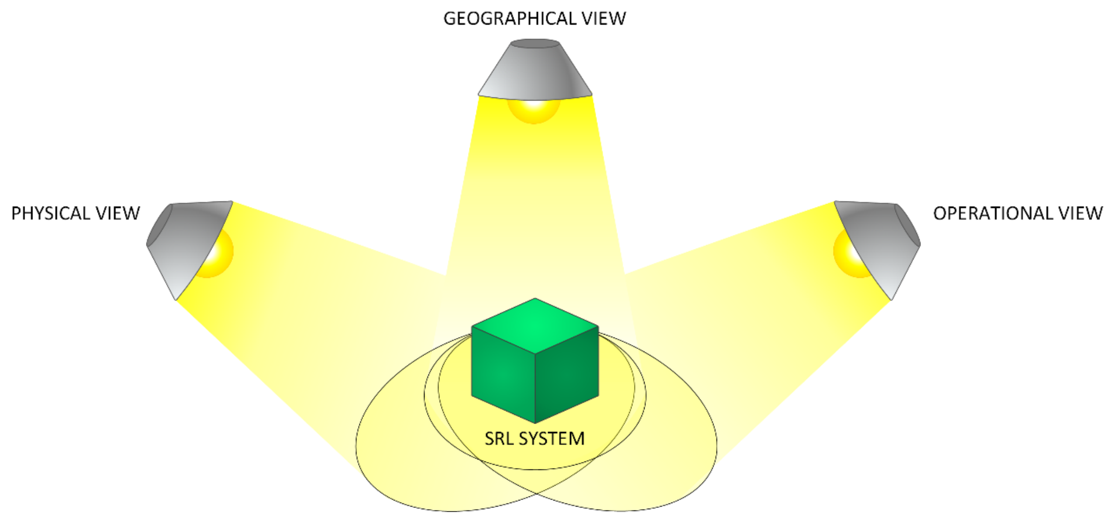


Figure 15 - Architecture Views

Architectures do not describe the detailed system design nor should they be considered as a constraint on the designs. They will primarily be used as a communication and collaboration tool to show higher level views capturing a variety of system assets, they can also be used for gap analysis and aid in the elicitation of requirements.

Due to the complexity of the program, a wide variety of architectures can be created, e.g., physical, geographical, schematic, etc.

During the development and design phases, a series of views will be selected that are to be drawn against each configuration point. These 'viewpoints' of the system are able to be shown graphically, but also defined textually in the relevant System Definition Document.

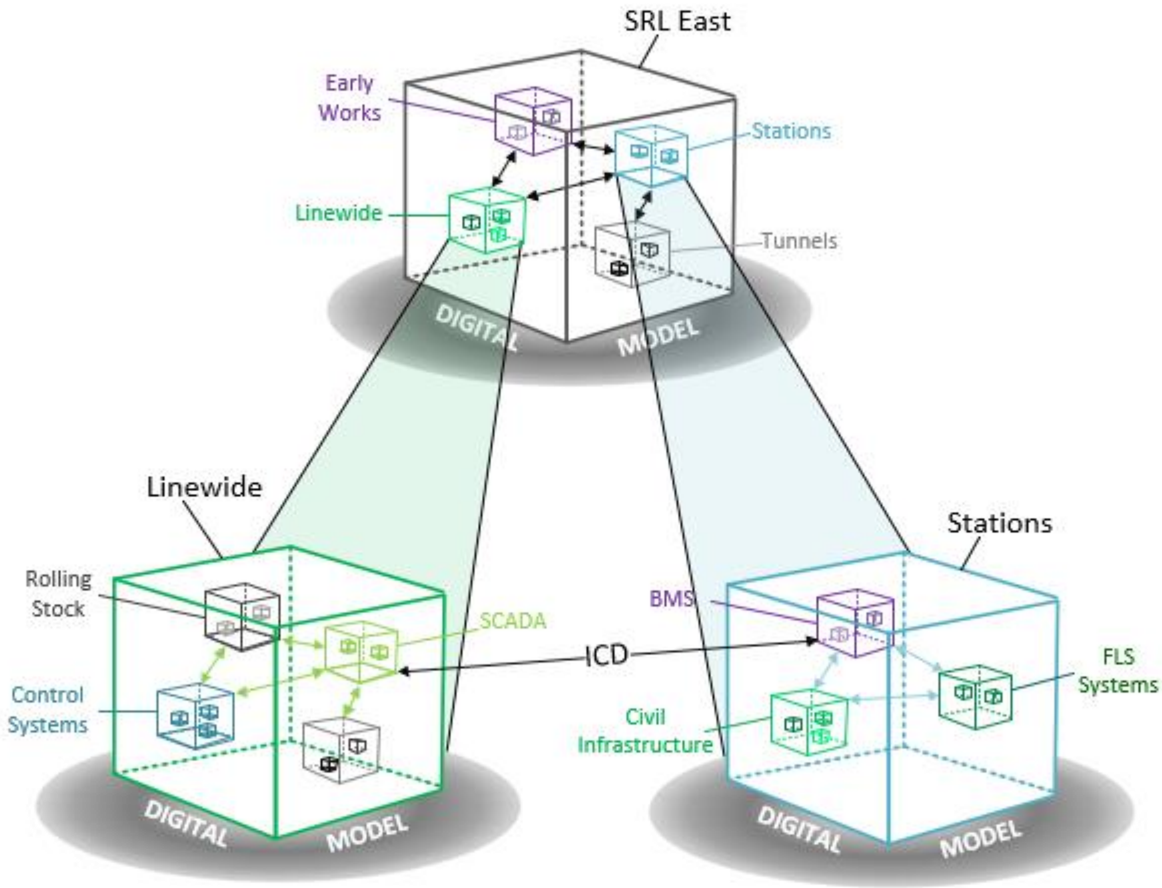
15.8.5 Architecture Creation

Architectures are conceptual views of a system, with different views having specific value to certain roles in the program.

The architectures will be produced with an input and review process with the relevant disciplines for that architecture type to ensure that all of the expected assets/attributes are included on the architecture.

During the development & design phases, the baseline versions of each architecture view will be created for their expected configuration points. They will be drawn using the current level of technical specification the program can provide and will continue to be updated as the design of each system becomes clearer until they show a final representation of the actual systems deployed.

The system and its components will be defined in System Definition Documents, with interfaces between the SRL system and other systems being defined in Interface Control Documents, as depicted below.



15.8.6 Outputs

Title	Description
Architecture Management Plan	The management plan that provides detail on how architecture management is undertaken at SRLA in line with the SEAMP.
SRLA System Architecture Model	The digital model that holds the system architecture and all of its associated information.
SRLA System Architecture diagrams/viewpoints	Visual and textual outputs from the digital model to be analysed by stakeholders.
SRLA System Definition Document	A document that defines all the system components existing within the system architecture. This document defines each component, describes its functions, interfaces and any other specific characteristics.
Interface Control Documents	Multiple documents that define technical information/data exchanges across interfaces identified within the system architecture. These documents typically define the type of interface, the performance it must possess, the information/data that is to be exchanged across it, and other characteristics, such as fixings and fittings.

15.9 Appendix I - Interface Management

15.9.1 Inputs

Title	Description
Business Requirements Document	The purpose of this business requirements document is to provide a clear specification of the envelope within which the SRLA project must deliver and what interfaces the SOI may have with stakeholders and existing systems
Rail Operations Plan	The Rail Operations Plan provides the SRLA with a concept of how the future railway is predicted to be operated and maintained. The operational activities elicited from this concept document give rise to interfaces between systems and its users
Key Decisions Register	The Key Decision Register provides SRLA with constraints that must be factored into any interface definition work
Preliminary Hazard Analysis	Any preliminary hazard analysis that has been undertaken during feasibility and concept work for the SRL must feed into any interface management work to ensure safety critical interfaces are identified and appropriately managed.
System Architecture	The system architecture is the tool that houses the technical interfaces and should be used for interface verification and gap analysis
INCOSE Systems Engineering Handbook v4 Chapter 4 – Technical Processes	The INCOSE SE Handbook offers up guidance for interface management. Chapter 4.4 covers general technical processes for documenting interfaces that should be tailored and expanded further in the Interface Management Plan

15.9.2 Process

Through the development and design phases, the SRLA program shall develop Interface definition documents to manage all identified interfaces.

The Interface Manager or the Engineering Assurance team has the responsibility to ensure the collection, definition and documentation all of the technical interfaces in the program.

For the development and design phases of the SRLA program, the primary objectives for Interface Management are:

- To create an interface management plan that describes how technical interfaces will be captured and documented for the program
- To capture the system interfaces and their high-level definitions (interface matrix)
- To assign responsibility for the management of the identified interfaces.

15.9.3 Creation of Interface Management Plan

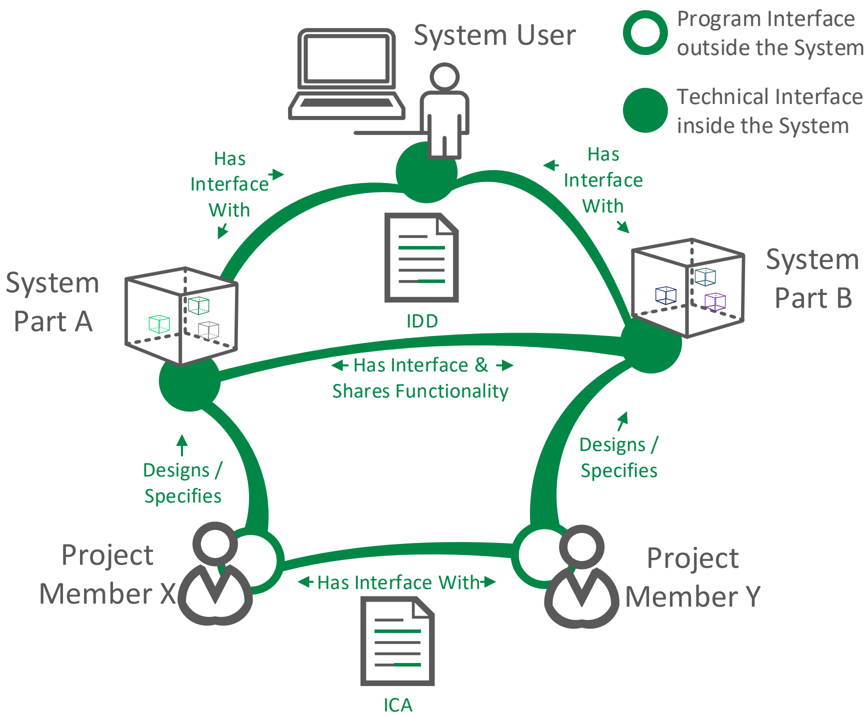
The Interface Manager will create an Interface Management Plan describing how the interfaces will be broken down, and how the interfaces between systems will be defined and managed.

The plan will include a template for Interface Control Documents, a description of the interface capture methodology and the relevant processes that will be needed to capture them.

15.9.4 Capture Program Interfaces

The Interface Manager will be responsible for gathering and defining the system, program and operational interfaces for SRLA.

These interface definitions will be updated regularly, with major changes reviewed at each design review stage.



Interface Control Documents will be created to describe the interfaces between each of the systems that comprise the overall architecture of the railway.

Documentation will continue to be developed as the detailed design is realised, so the lower level interfaces, and the contract interfaces can be identified and defined.

External interfaces to interdependent programs will be captured through regular meetings with interfacing program and the high level governance boards of both the program and DTP. Any agreements made between these interfaces must be captured in an Interface Control Agreement.

15.9.5 Assign Responsibility for Interfaces

Interface definition documents will be created for each identified interface. The responsible party for creating and maintaining these documents will depend on the level of integration required for that interface. It is expected that:

- Where the interface is entirely within one project/contract, a responsible engineer from within that project will be responsible for it
- Where the interface is between two systems/projects/contracts, a responsible engineer will be nominated by SRLA. This engineer is expected to be part of the leading system in the interface where there is one
- Where the interface is between multiple systems/projects/contracts, the responsible engineer will be nominated by SRLA who will coordinate all systems/projects/contracts impacted by the interface.

The Interface Manager will be responsible for reviewing each of the interfaces and their associated documentation, including those describing the interfaces between contractors.

15.9.6 Outputs

Title	Description
Interface Management Plan	The management plan that provides detail on how interface management is undertaken at SRLA in line with the SEAMP.
Interface Matrix	A matrix that shows interfaces between system components, their type, criticality, and ownership.
Interface Control Agreements	Multiple documents that specify a formal agreement between two projects, defining which party is the lead, what information must be exchanged, and how the interface between the project must be managed.
Interface Control Documents	Multiple documents that define technical information/data exchanges across interfaces identified within the system architecture. These documents typically define the type of interface, the performance it must possess, the information/data that is to be exchanged across it, and other characteristics, such as fixings and fittings.

15.10 Appendix J - System Safety Management

15.10.1 Inputs

Title	Description
ONRSR Guidelines	Guidelines that must be adhered to in order to qualify the railway as safe to operate
Rail Safety National Law	Laws that must be adhered to in order to qualify the railway as safe to operate
INCOSE Systems Engineering Handbook v4 Chapter 10 – specialty Engineering Activities	The INCOSE SE Handbook offers up guidance for undertaking system safety activities. Chapter 10 covers general Specialty Engineering Activities that should be tailored and expanded further in the System Safety Management Plan

15.10.2 Process

System Safety is primary concern for SRLA and therefore this discipline needs to underpin all engineering activities so that SRLA is designing a safe solution.

Responsibility for Engineering Safety Management for the program will lie with the Manager, Engineering Assurance and enabled by the System Safety Lead.

- The Systems Safety Lead is responsible for maintaining an overall safety hazard log for the integrated safety risks of the SRLA program and its subsequent projects
- The individual design authorities will be responsible for creating and maintaining the safety hazard log for their systems. These will be reviewed with the System Safety Lead
- Each safety hazard log will identify possible hazards, likelihood of occurrence, and severity along with any mitigation to be implemented
- An Engineering Safety and Assurance Case will be created by the System Safety Lead and updated as the detail of the design increases
- Individual System Safety Cases will be created by the system designers and reviewed by the Systems Safety Lead

- The System Safety Lead will review all supplier contracts to ensure safety requirements are accounted for and can be assured upon delivery
- A System Safety Management plan will be created with details of how the safety processes will be followed for the program.

15.10.3 Outputs

Title	Description
System Safety Management Plan	The management plan that provides detail on how system safety management is undertaken at SRLA in line with the SEAMP
Safety Hazard Log	A digital log containing all hazards identified, their mitigating actions and resolutions
Engineering & Assurance Safety Cases	Safety cases developed to assure SRLA that what is being designed and what has been built can be qualified as safe to operate

15.11 Appendix K - Reliability, Availability and Maintainability (RAM) Management

15.11.1 Inputs

Title	Description
Business Requirements Document	The purpose of this business requirements document is to provide a clear specification of the envelope within which the SRLA project must deliver, and to what level of service/performance.
BS-EN-50126 – Railway Applications – The specification of Reliability, Availability, Maintainability and Safety (RAMS)	The industry wide standard for the application of RAMS on railway programs. This standard offers guidance on how RAMS activities should be applied with methodologies tailored for the rail environment.
Key Decisions Register	The Key Decision Register provides SRLA with constraints that must be factored into any RAMS analysis work
Rail Operations Plan	Any preliminary system performance analysis that has been undertaken during feasibility and concept work for the SRL must feed into any RAMS analysis work to be developed further and incorporated into the SRLA solution.

15.11.2 Process

During the Feasibility phase, DTP undertook an activity to establish top-level reliability targets for SRL based on benchmarking against comparable high capacity metros. The result of this exercise is a set of requirements included in the BRD.

For the Development & Design phases of the SRL program, the primary objectives of the RAM workstream are to:

- Plan, develop, document and implement the RAM management process
- Verify the feasibility of the top-level reliability targets,
- Apportion the R&M performance through to system/sub-systems to ensure appropriate requirements are available for the procurement process
- Evaluate tender responses

- Oversee the RAM assessments undertaken by the delivery agencies

15.11.3 RAM Planning

The RAM Lead will develop a Reliability, Availability & Maintainability Plan (RAM Plan) for SRLA to describe the way in which RAM tasks are to be applied for the duration of the program.

The RAM Plan is a living document and will be continually updated and maintained by the RAM Lead.

The RAM Plan shall tailor the generic processes outlined in BS-EN-50126 to ensure RAM practices are in keeping with industry standard.

In addition to producing the RAM Plan, the RAM Lead shall liaise with the Requirements Manager to ensure an appropriate relationship is established between RAM analysis (and any resulting requirements) and the requirements management tool.

The RAM Lead will ensure all parties involved in the RAM process (including DTP and delivery agencies) have access to and are adequately briefed on the RAM Plan and that any subsequent updates or amendments to the plan are cascaded appropriately in a timely manner.

The RAM Lead will also be responsible for ensuring that the RAM tasks are communicated to the Systems Engineering Manager (SEM) to be included in any Engineering Assurance schedules/plans and for managing the timely delivery of these activities.

15.11.4 Verify the top-level reliability targets

The Requirements Manager and RAM Lead will coordinate an exercise to validate the top-level reliability requirements in the BRD. The purpose of the activity is to demonstrate that the expected system architecture is capable of delivering the required reliability performance to prove that the BRD requirement is achievable and realistic.

15.11.5 Apportion the RAM Performance Requirements

Once the top-level requirements have been validated and any necessary change assessment/authority has been gained, then the RAM Lead will undertake an apportionment activity to split the RAM targets at the top-level into rail system and asset reliability requirements and define the contribution from Staff and Passenger causes.

The RAM Lead will work closely with the Operations & Maintenance representatives to ensure that the Operational & Maintenance analysis workstreams result in appropriate system design requirements to mitigate the likelihood and consequence of events that could result in service affecting failures. Additionally; the Operations & Maintenance representatives will be involved in all future reliability performance trade-off discussions and Whole of Life Cost analysis.

15.11.6 Evaluation of Tender Responses

The output of the apportionment exercise will be a set of RAM requirements that can be included in the PS&TRs for procurement of delivery agencies. The RAM Lead will be involved in the evaluation of the bidder's response to the PS&TRs and assess supplier predictions for reliability against a whole of life cost model.

15.11.7 Oversee the RAM assessments undertaken by Delivery Agencies

Detailed RAM modelling and assessments are expected to be the responsibility of the delivery agency. It is expected that the supplier will propose RAM modelling and analysis approaches commensurate to the system risk profile and technology maturity levels of the particular system/asset.

The RAM Lead will oversee the RAM activities and review the verification evidence presented by the delivery agency against the RAM targets.

The RAM Lead will be responsible for ensuring the WLC models are kept up to date and for assessing and providing recommendations on trade-off, should RAM performance not be achievable.

15.11.8 Outputs

Title	Description
RAM Management Plan	The management plan that provides detail on how RAM management is undertaken at SRLA in line with the SEAMP
Validated BRD reliability targets	The service performance requirements that have been given to the project have been validated as feasible and allocated to appropriate components of the proposed solution
RAM related System Requirements	The output of any RAM analysis will result in a set of non-functional requirements that specify the performance of particular components of the proposed solution
RAM Targets apportioned to Railway System Components & Design Packages	This will show clear linkage & traceability between high level RAM Targets and the lower level system components that contribute to them.
Critical Items Log	This register captures definitions and mitigating actions of any system components considered as 'critical items' in the successful functioning of the SRL system. Critical Items are uncovered as part of undergoing any RAM analysis activities.

15.12 Appendix L - Electro-Magnetic Compatibility (EMC) / Interference (EMI) Management

15.12.1 Inputs

Title	Description
IEC EN 61000 & CISPR 16	Two generic standards that give guidance on minimum application of EMC analysis, measurement, and testing
INCOSE Systems Engineering Handbook v4 Chapter 10 – Speciality Engineering Activities	The INCOSE SE Handbook offers up guidance for undertaking EMC analysis. Chapter 10.2 specifically covers application of EMC activities that should be tailored and expanded further in the EMC Management Plan

15.12.2 Process

It is the program's responsibility to ensure full EMC compatibility of both existing legacy and newly installed systems that are part of the SRLA solution.

There will be resources allocated with the direct responsibility for Electromagnetic Compatibility related activities within the Engineering Assurance team.

For the development and design phases of the SRL program the primary objectives for Electromagnetic Compatibility are:

- To plan ongoing EMC activities for the design and delivery phase
- To create and EMC Risk Register and begin the hazard identification for the program and contributing projects
- To provide EMC support to the project teams to assist them in developing EMC requirements for their contracts.

15.12.3 Planning EMC Activities

The allocated resource for EMC management will be responsible for creating a list of EMC activities to be undertaken during the development & design phases of SRLA. These activities will be tailored for the specific needs of the program and will take particular note of interfacing systems where EMC issues are likely to occur.

EMC specialists will be expected to support the Engineering Assurance team for these EMC related activities where necessary as their expert knowledge will be required to achieve compliance, and the commonality in approach will enable the program safety case to be more easily signed off.

Some EMC related works will be delegated to design teams or contractors, where the EMC issue is seen to be internal to a single system or demonstration of compliance to standards can be shown through testing of that system alone. The Engineering Assurance team will only be required to review the process and results in this case.

In the case of integrated EMC issues, the Engineering Assurance team will be responsible for ensuring compliance to standards and conducting the required work to do so.

15.12.4 Creation of EMC Risk Register

The Engineering Assurance team has the overall responsibility to ensure EMC compliance for the program and is therefore responsible for collecting and considering each of the potential EMC risks, their level of criticality and mitigation steps. These risks will be collected through a series of workshops with relevant members of the program team, led by the allocated resource for EMC management. The risks, severity and mitigation steps will be gathered together in an EMC Risk Register for the program. If necessary, additional risks with greater levels of detail will be created for specific systems that contribute to the SRLA program.

In the design and specification stage, the EMC Risk Register will be developed with the intention of creating formal hazard identification documents once the design of various systems is complete. The program team holds responsibility for Rail-System level EMC hazard identification, but system internal EMC hazard identification is the responsibility of the various contractors.

15.12.5 Development of EMC requirements

There are standards and legislation to be adhered to for EMC for both design and delivery of components and systems, the allocated resource for EMC management and supporting specialist(s) are expected to ensure compliance to these standards. This will be achieved in the design phase by providing input to the design teams to include EMC requirements in their specifications and contracts.

15.12.6 Outputs

Title	Description
EMC Management Plan	The management plan that provides detail on how Electromagnetic Compatibility is managed at SRLA in line with the SEAMP
EMC Risk Register	A digital register that captures any EMC related risks that require mitigating either through design or by procedure
EMC Requirements	Requirements specifying particular electrical/electronic characteristics of the system and its components

15.13 Appendix M - Human Factors Management

15.13.1 Inputs

Title	Description
NTS-013-Human Factors Integration for Public Transport Network Projects	Network Technical Standard that describes how the scientific discipline of human factors should be applied throughout a project's lifecycle to understand the interactions amongst humans and the elements of a system.
ONRSR Guideline – Safety Management System	The ONRSR guidelines offer specifics about human factors analysis for safe rail systems.
INCOSE Systems Engineering Handbook v4 Chapter 10 – Speciality Engineering Activities	The INCOSE SE Handbook offers guidance for undertaking human system integration. Chapter 10.13 specifically covers application of human system integration & human factors engineering that should be tailored and expanded further in the Human Factors Integration Plan
Rail Operations Plan	Any preliminary operational analysis that has been undertaken during feasibility and concept work for the SRL must feed into any human factors integration work to be developed further and incorporated into the SRLA solution.

15.13.2 Process

It is the Human Factors Lead's responsibility to ensure that all Human Factors requirements and potential issues are captured for inclusion in the design of the SRLA solution, and where necessary, mitigation plans are developed and implemented.

For the Development & Design phases of the SRL program, the primary objectives of the Human Factors workstream are to:

- Plan, develop, document and implement the Human Factors Integration process
- Create and manage a human factors issues log to capture all program human factors and ergonomic issues and required actions for their mitigation
- Provide expert knowledge and input to interfacing disciplines, and complete HF specific pieces of work for said disciplines.

15.13.3 Human Factors Integration Planning

The Human Factors Lead will develop a Human Factors Integration Plan (HFIP) for SRLA to describe the way in which Human Factors analysis is to be undertaken throughout the program.

The HFIP is a living document and will be continually updated and maintained by the Human Factors Lead.

The HFIP shall elaborate on the processes laid out in NTS-013-Human Factors Integration for Public Transport Network Project, to ensure Human Factors practices are in keeping with guidelines established by DTP. The Human Factors Lead should also refer to Section 10.3 of the INCOSE Systems Engineering handbook for guidance on application of human systems integration and human factors engineering processes.

In addition to producing the HFIP, the Human Factors Lead shall liaise with the Requirements Manager to ensure that an appropriate relationship is established between Human Factors analysis (and any resulting requirements) and the requirements management tool.

The Human Factors Lead will ensure all parties involved in the Human Factors process (including DTP and delivery agencies) have access to and are adequately briefed on the HFIP and that any subsequent updates or amendments to the plan are cascaded appropriately in a timely manner.

The Human Factors Lead will also be responsible for ensuring that the Human Factors tasks are communicated to the Systems Engineering Manager (SEM) to be included in any Engineering Assurance schedules/plans and for managing the timely delivery of these activities.

15.13.4 Human Factors Issues Log

The Human Factors lead is responsible for overall human factors compliance and validation. This will be achieved through the inclusion of specialist Human Factors expertise that will generate HF requirements for the program assets that have a human interface.

The Human Factors Lead will hold workshops with relevant discipline engineers to generate a Human Factors Issues Log for the program, and the individual component assets, which will be maintained throughout the life of the program.

Each issue on the HFIL will have action(s) generated to progress towards resolution within a given stage of the program. They will also generate Human Factors requirements to be included in the system requirements specification.

15.13.5 Provide Expert Human Factors Knowledge

The Human Factors Lead will also be responsible for providing expert knowledge to interfacing disciplines, this is of particular importance in the detailed design phase to ensure that Human Factors considerations and requirements are captured and included in the system requirements specification.

The Human Factors Lead will provide Human Factors input to other work streams, including activities such as task role analysis, human error analysis, equipment use analysis, and design development.

The Human Factors Lead will be responsible for development of any emergency response and alarms strategies and creating the working group to facilitate this, as it feeds into many of the design disciplines and their requirements.

The Human Factors Lead must develop human factors related system requirements, evaluate tender responses against these requirements, and ensure that they are included (and tested) in accordance with the Testing & Evaluation Plan.

15.13.6 Output

Title	Description
Human Factors Integration Plan	The management plan that provides detail on how Human Factors Integration is managed at SRLA in line with the SEAMP
Human Factors Issues Log	A digital register that captures any Human Factors related risks or issues that require mitigating either through design or by procedure.
Human Factors related system requirements	Requirements specifying particular characteristics of the system and its components in order to ensure human & machine integration